



有机化学

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教学安排

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第2章	烷烃和环烷烃
第3章	不饱和烃和共轭效应
第4章	芳烃
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第九章 羧酸及其衍生物

9.1 羧酸的分类和命名

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9.3 羧酸的物理性质

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9.5 羧酸的制法

9.5 羟基酸

9.6 羧酸衍生物的分类和命名

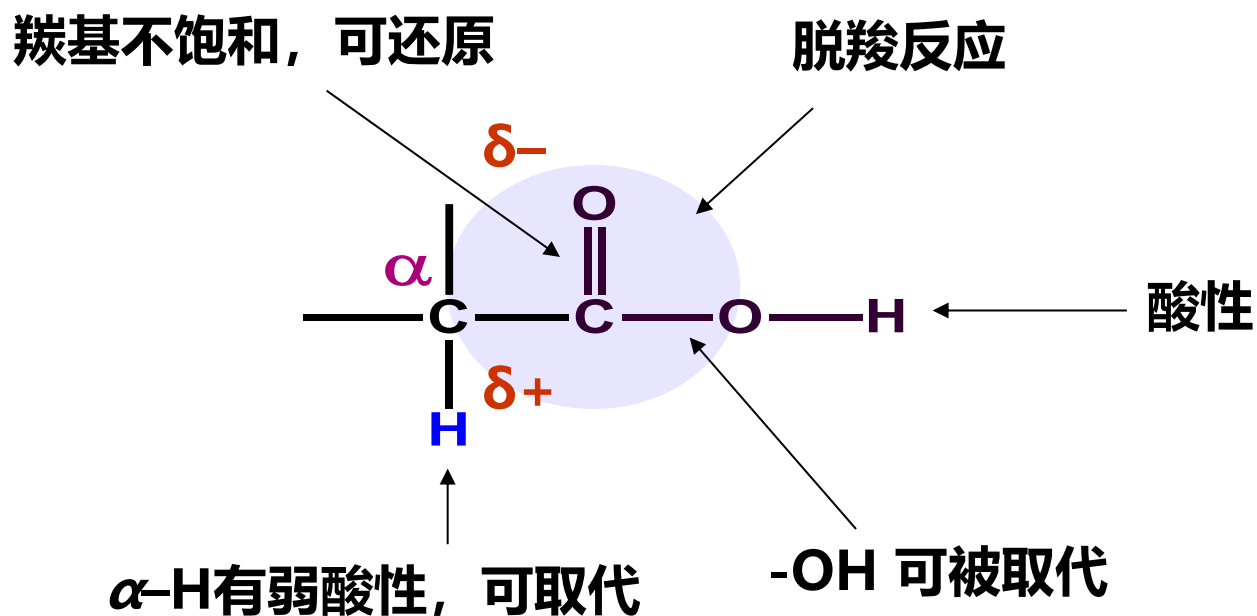
9.7 羧酸衍生物的物理性质

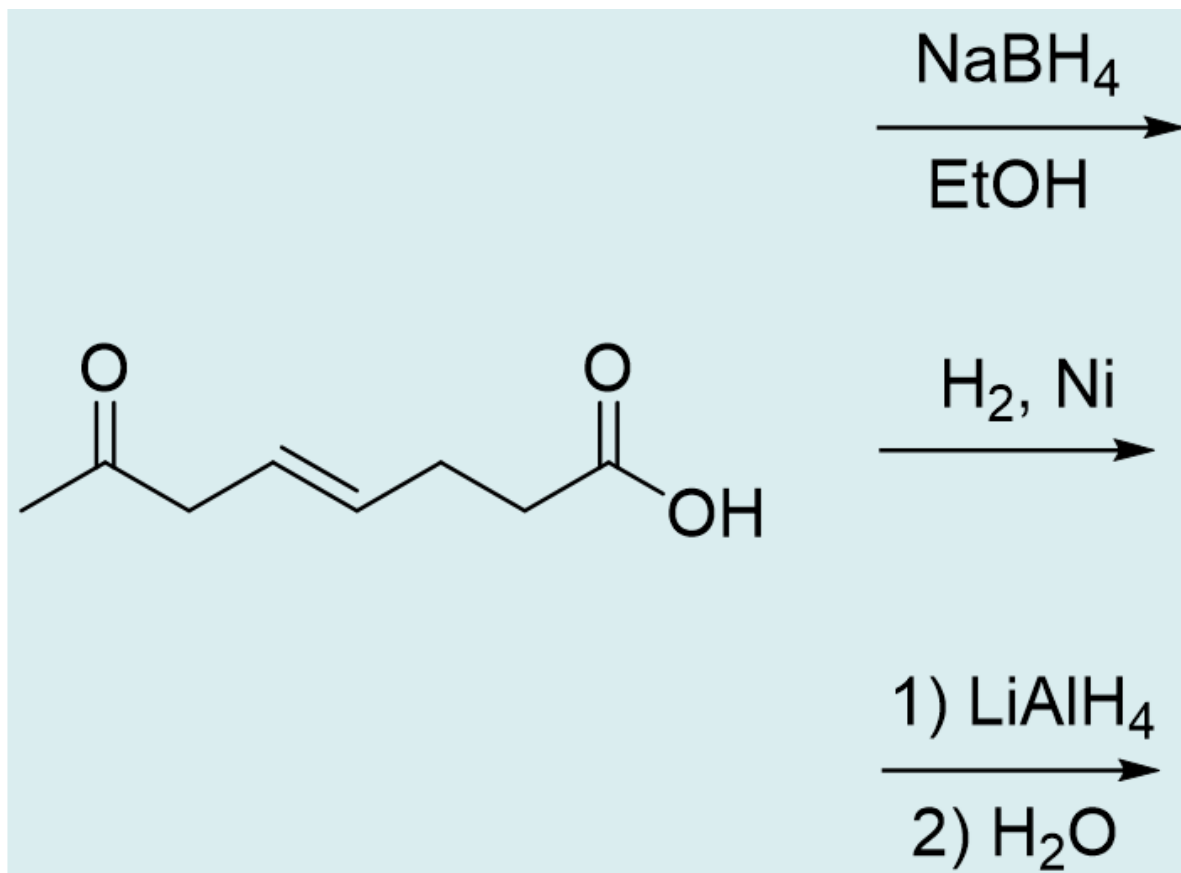
9.8 羧酸衍生物的化学性质

第九章 羧酸及其衍生物



■ 羧酸的结构及化学性质分析

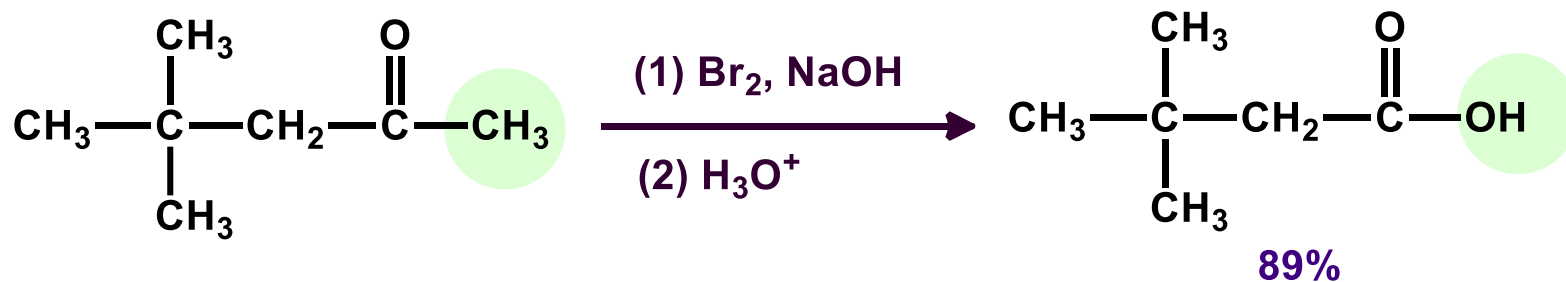
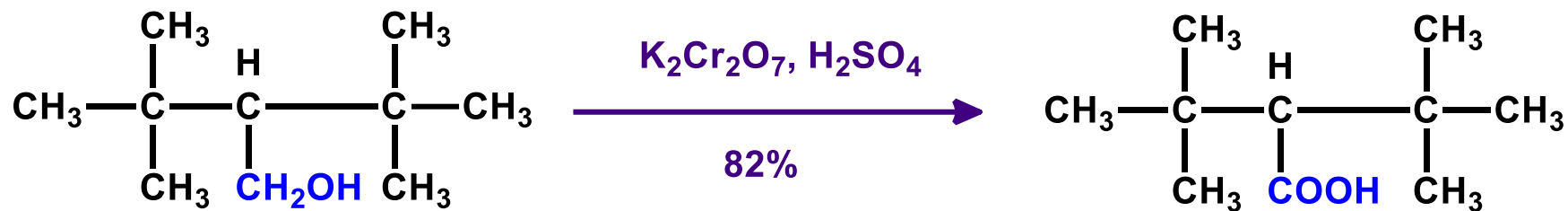




9.5 羧酸的制备



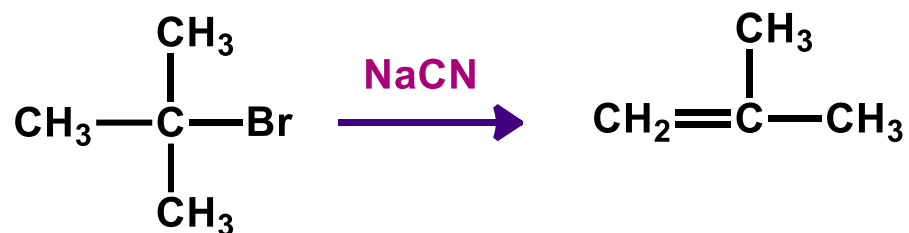
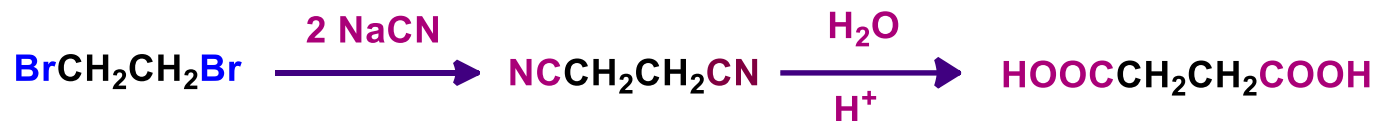
9.5.2 伯醇和醛的氧化



9.5 羧酸的制备



9.5.3 腈的水解

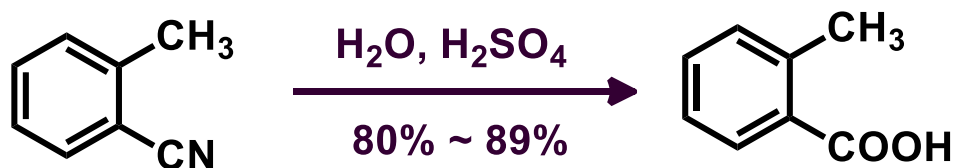
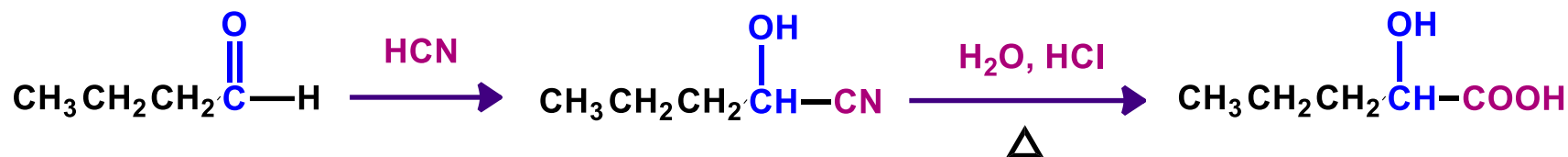


叔卤代烃一般不用于此反应。

9.5 羧酸的制备



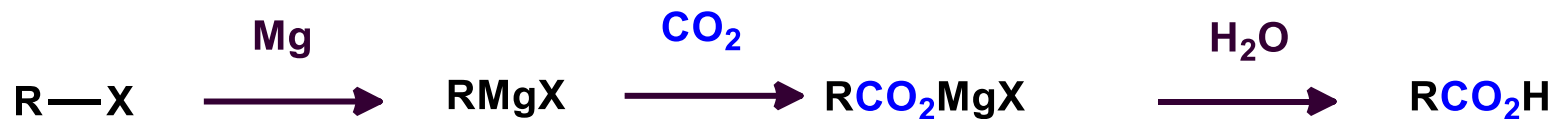
9.5.3 腈的水解



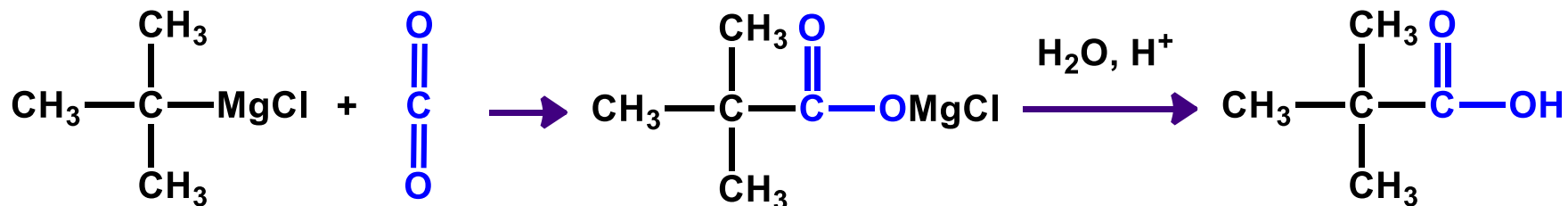
9.5 羧酸的制备



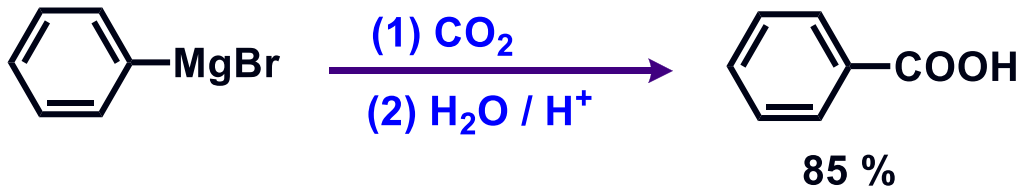
9.5.4 Grignard试剂与二氧化碳作用



例:



79% ~ 80%

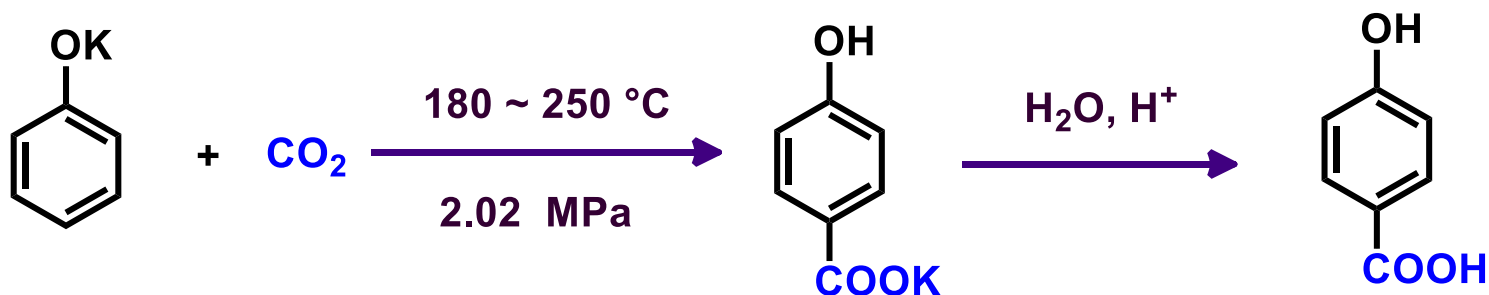
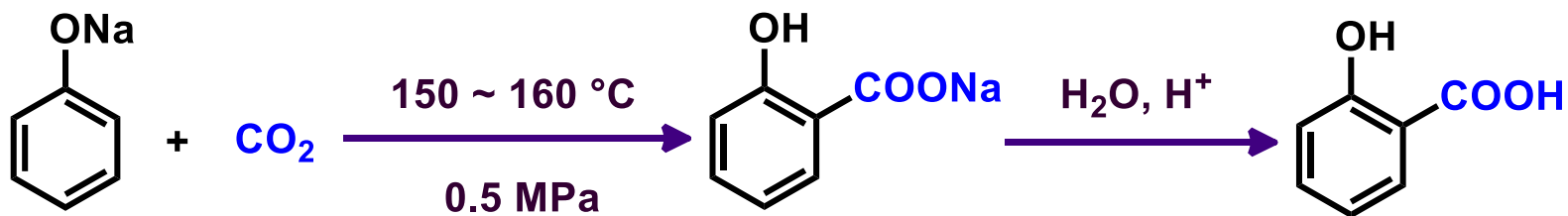


9.5 羧酸的制备



9.5.5 酚酸的合成

Kolbe-Schmitt反应

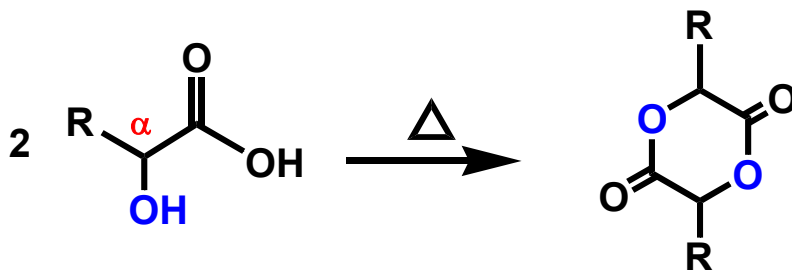


9.6 羧酸



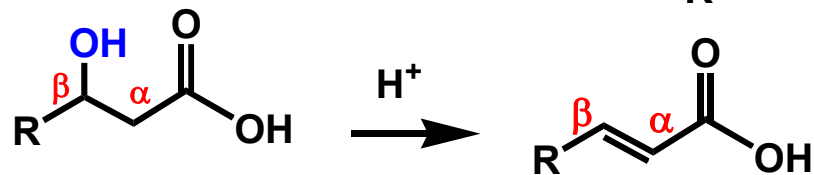
9.6.1 羧酸的脱水

α -羟基酸



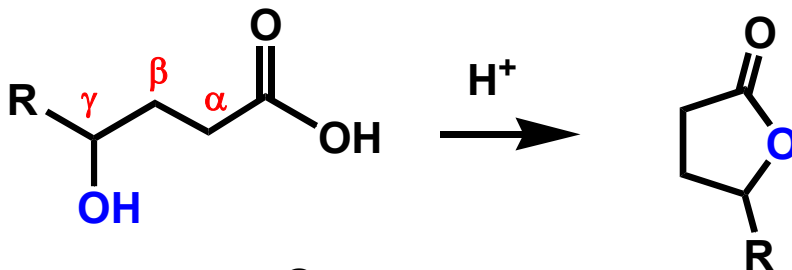
交酯

β -羟基酸



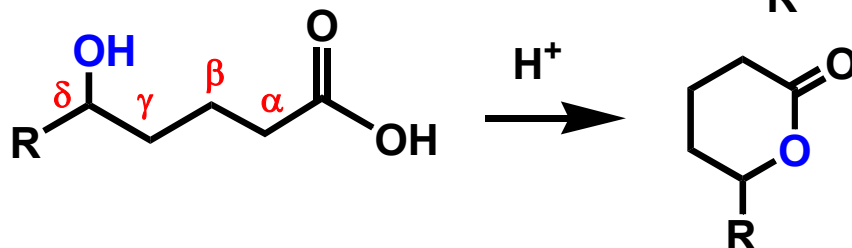
α, β -不饱和酸

γ -羟基酸



γ -内酯

δ -羟基酸



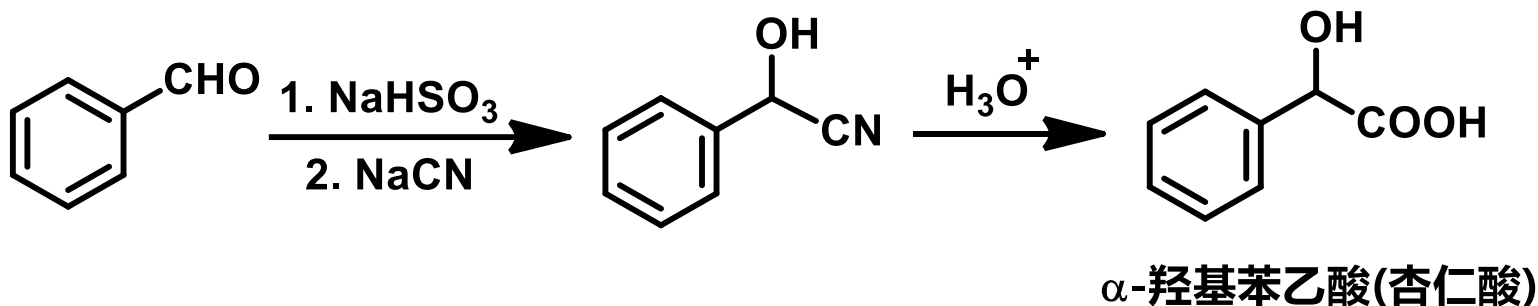
δ -内酯

9.6 羧酸



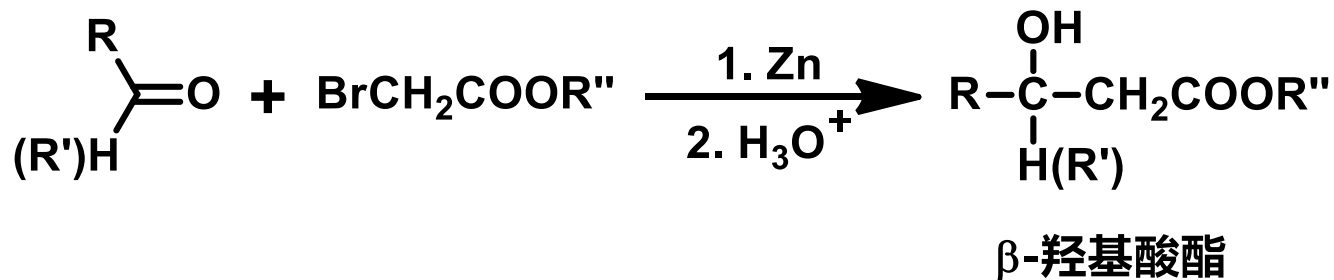
9.6.3 羧酸的制备

□ 水解法： α -卤代酸或 α -羟基腈



□ 瑞佛马斯基(Reformatsky)反应

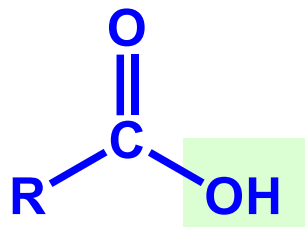
➤ 醛或酮与 α -溴代酸酯和锌在惰性溶剂中相互作用



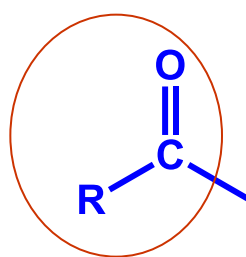
羧酸衍生物



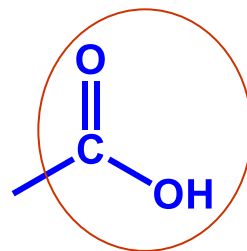
■ 羧酸衍生物 —— 羧酸分子中羧基上的羟基(-OH)被卤原子、酰氧基、烷氧基和氨基替代后的化合物,分别被称为**酰卤**、**酸酐**、**酯**和**酰胺**; 它们均可由羧酸制得, 经水解反应即可转变为羧酸, 因而称为**羧酸衍生物**。



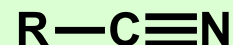
羧酸



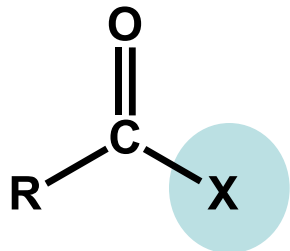
酰基



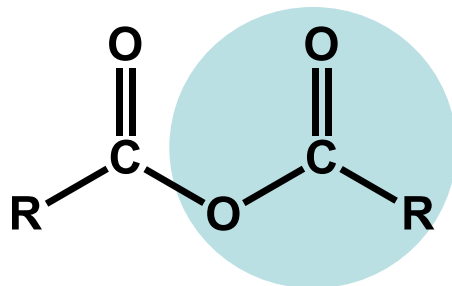
羧基



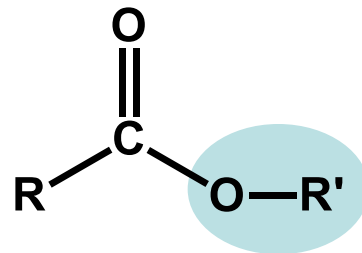
腈



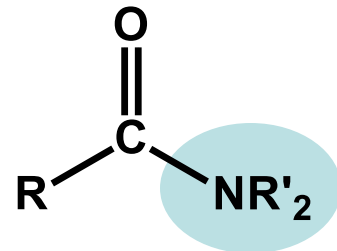
酰卤



酸酐



酯



酰胺

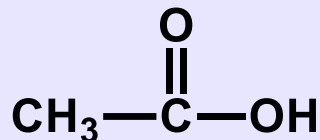
9.7 羧酸衍生物的命名



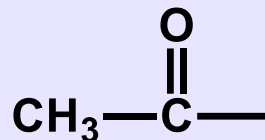
酰卤和酰胺:

根据相应的酰基命名。

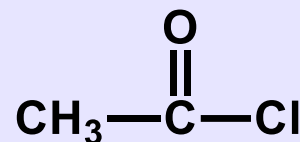
把“酸”改为“酰氯”、“酰胺”即可。



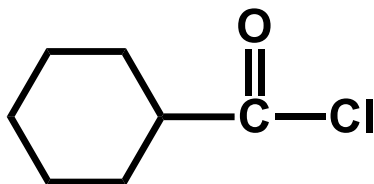
乙酸



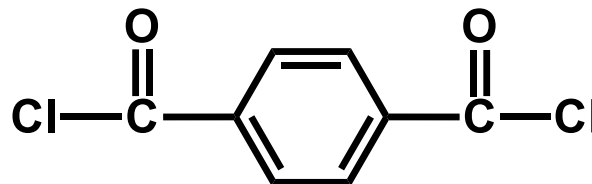
乙酰基



乙酰氯

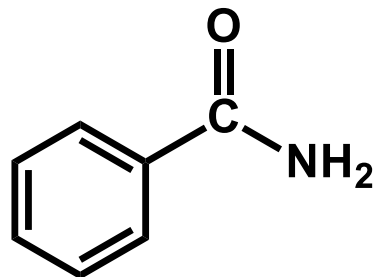


环己烷甲酰氯

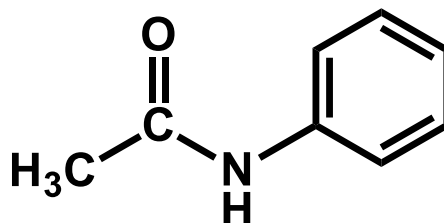


对苯二甲酰氯

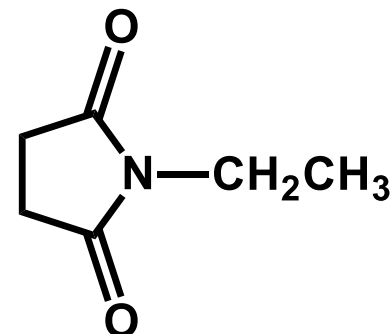
9.7 羧酸衍生物的命名



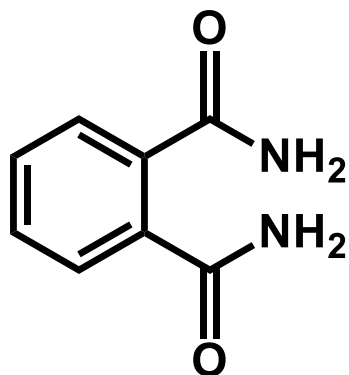
苯甲酰胺



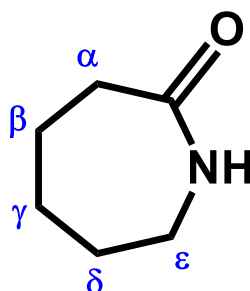
乙酰苯胺



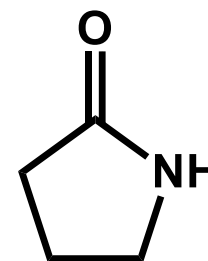
N-乙基丁二酰亚胺



邻苯二甲酰胺



ϵ -己内酰胺

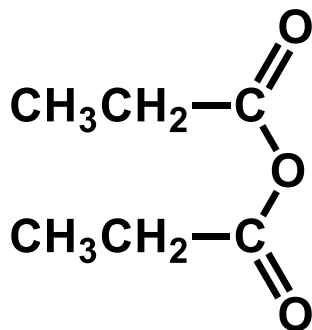


γ -丁内酰胺

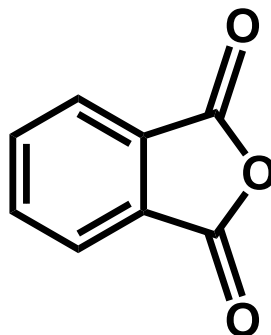
9.7 羧酸衍生物的命名



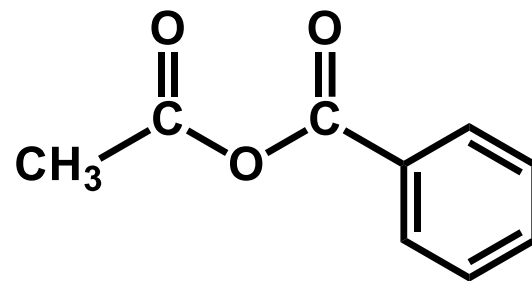
酸酐： 相应酸名+酐



丙(酸)酐

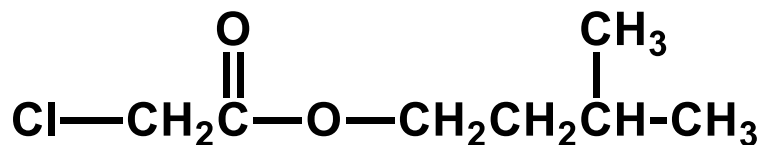


邻苯二甲酸酐

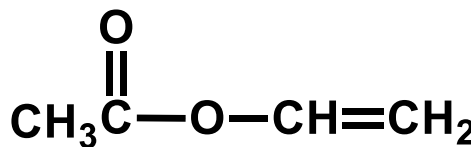


乙酸苯甲酸酐

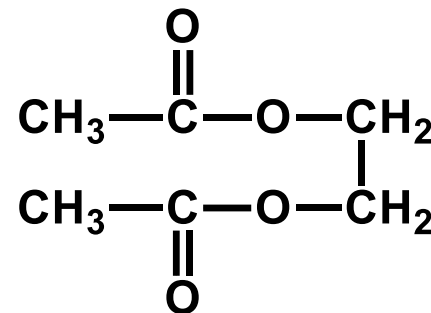
酯： 酸名+醇烃基名+酯



氯乙酸异戊酯



醋酸乙烯酯

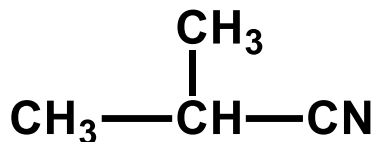


乙二醇二乙酸酯

9.7 羧酸衍生物的命名



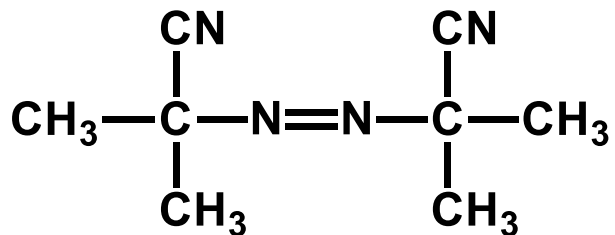
腈: 与同碳数的羧酸相似, 把“酸”字换成“腈”字(有些俗名亦然).



异丁腈



丙烯腈



偶氮二异丁腈



1,4-丁二腈

9.7 羧酸衍生物的命名



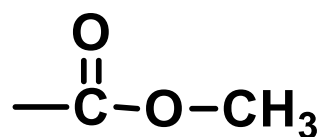
- 含有多个官能团的化合物
- 选择一个官能团为母体化合物，其他官能团作为取代基
- 官能团的优先顺序：



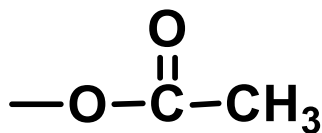
9.7 羧酸衍生物的命名



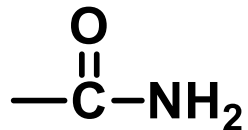
□ 羧酸衍生物的官能团作为取代基时的名称



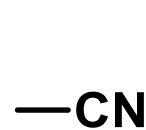
甲氧甲酰基



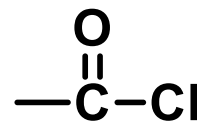
乙酰氧基



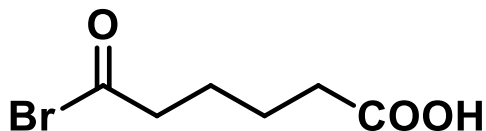
氨基甲酰基



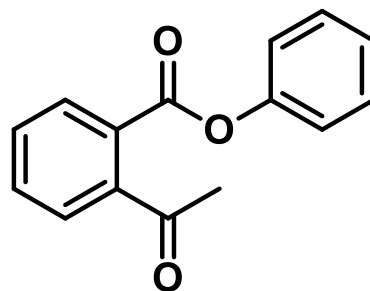
氰基



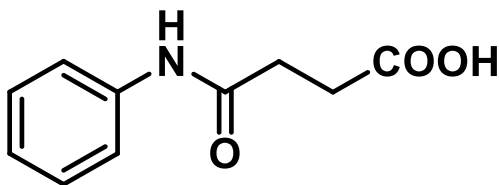
氯甲酰基



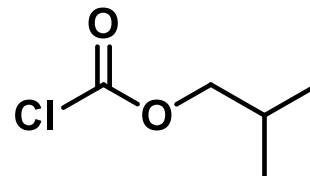
5-溴甲酰基戊酸



2-乙酰基苯甲酸苯酯



3-(苯胺基甲酰基)丙酸

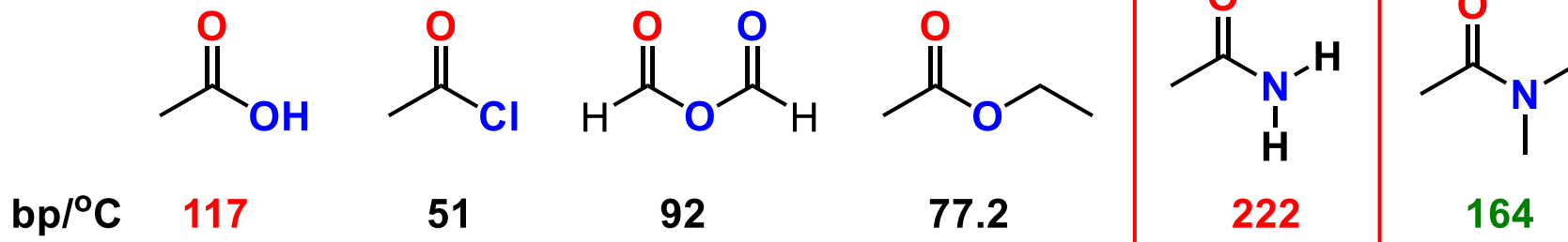


氯甲酸异丁酯

9.7 羧酸衍生物的物理性质



1. 沸点



(1) 酰氯、酸酐、酯不能形成分子间的氢键，沸点比相应的羧酸要低。

(2) 酰胺分子间可形成氢键，缔合作用比羧酸还强，所以沸点比羧酸要高，室温下除甲酰胺一般为固体。当酰胺氮上的氢被烃基取代，沸点降低。

9.7 羧酸衍生物的物理性质



2. 水溶性

酸酐、酰氯不溶于水（低级酰氯和酸酐遇水分解），酯难溶于水；
低级酰胺易溶于水，液体酰胺是良好的有机溶剂。

3. 气味：低级酰氯和酸酐都对粘膜有刺激性。

低级的酯有水果香味，在香料行业有重要的应用：

乙酸乙酯

梨香

乙酸异戊酯

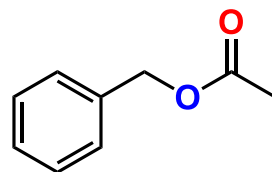
香蕉香

戊酸异戊酯

苹果香

异戊酸戊酯

桔子香



乙酸苄酯

茉莉花香气

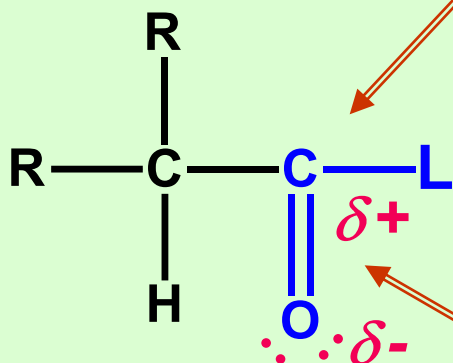
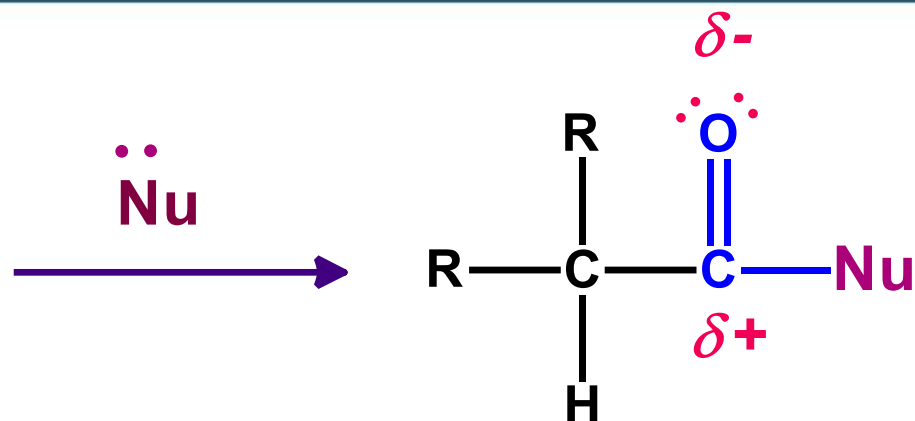
需求量最大的合成香料



9.8 羧酸衍生物的化学性质

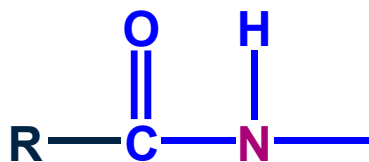
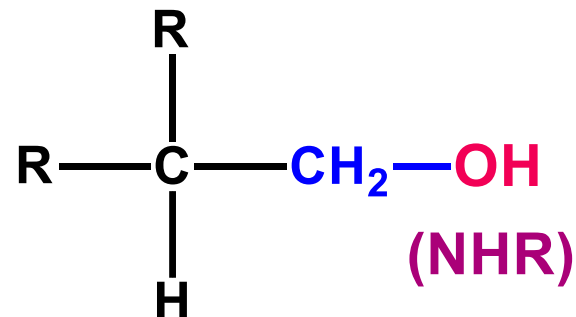


羰基碳有亲电性
可发生亲核取代



羰基被还原

还原

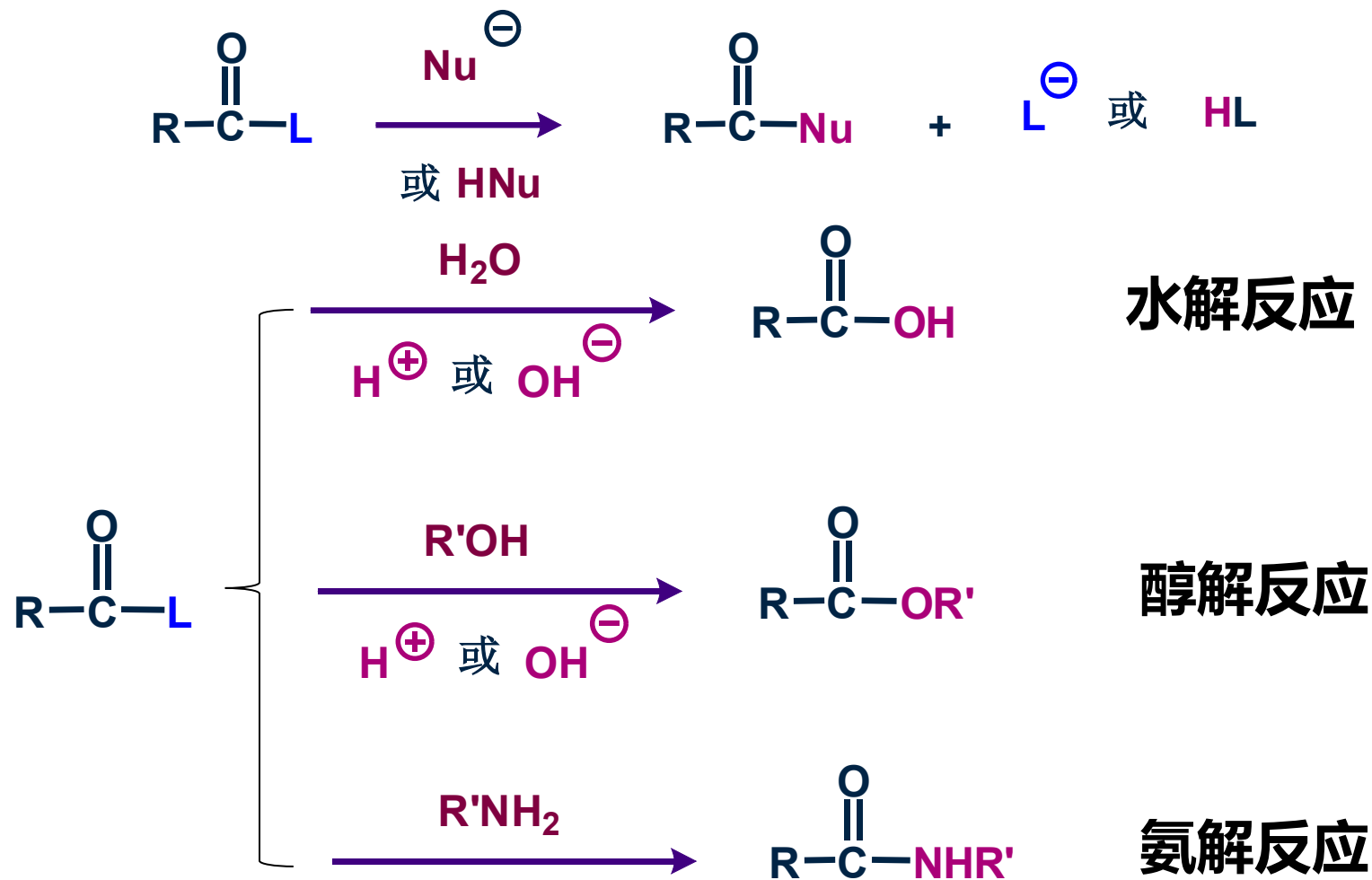


酰胺的特性

9.8 羧酸衍生物的化学性质



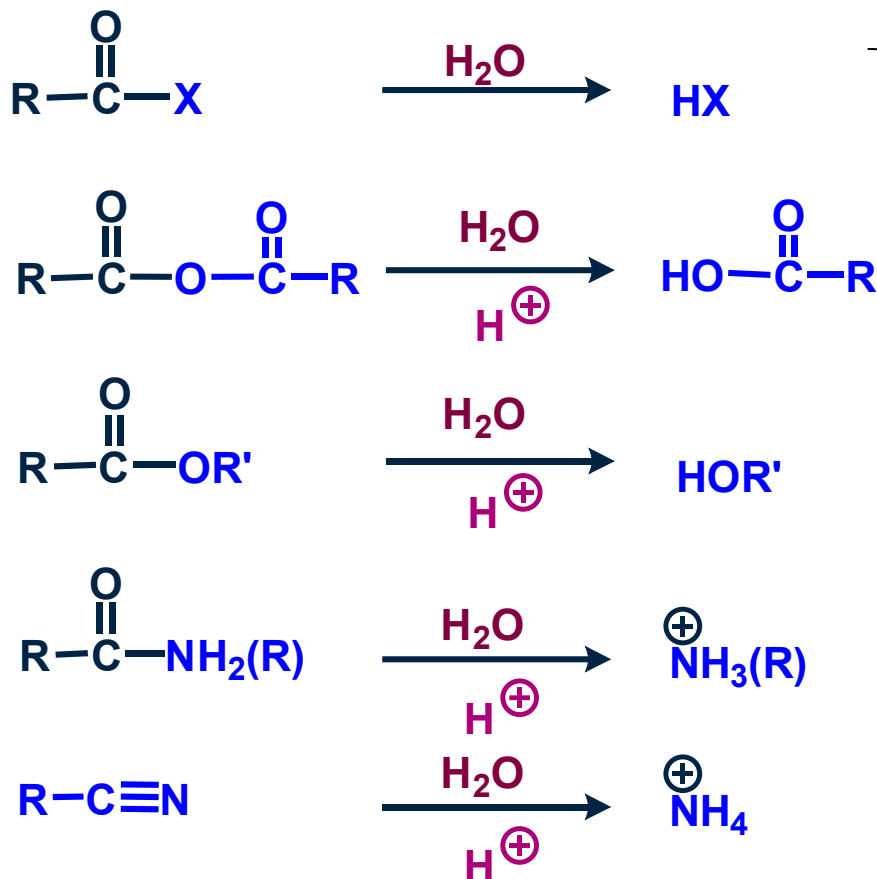
9.8.1 酰基上的亲核取代反应





9.8 羧酸衍生物的化学性质

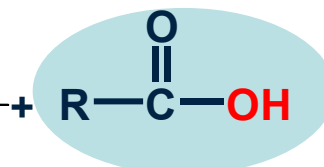
水解反应



反应条件

与水猛烈反应并放热

加热下易反应



需酸或碱催化

酸或碱催化并长时间回流

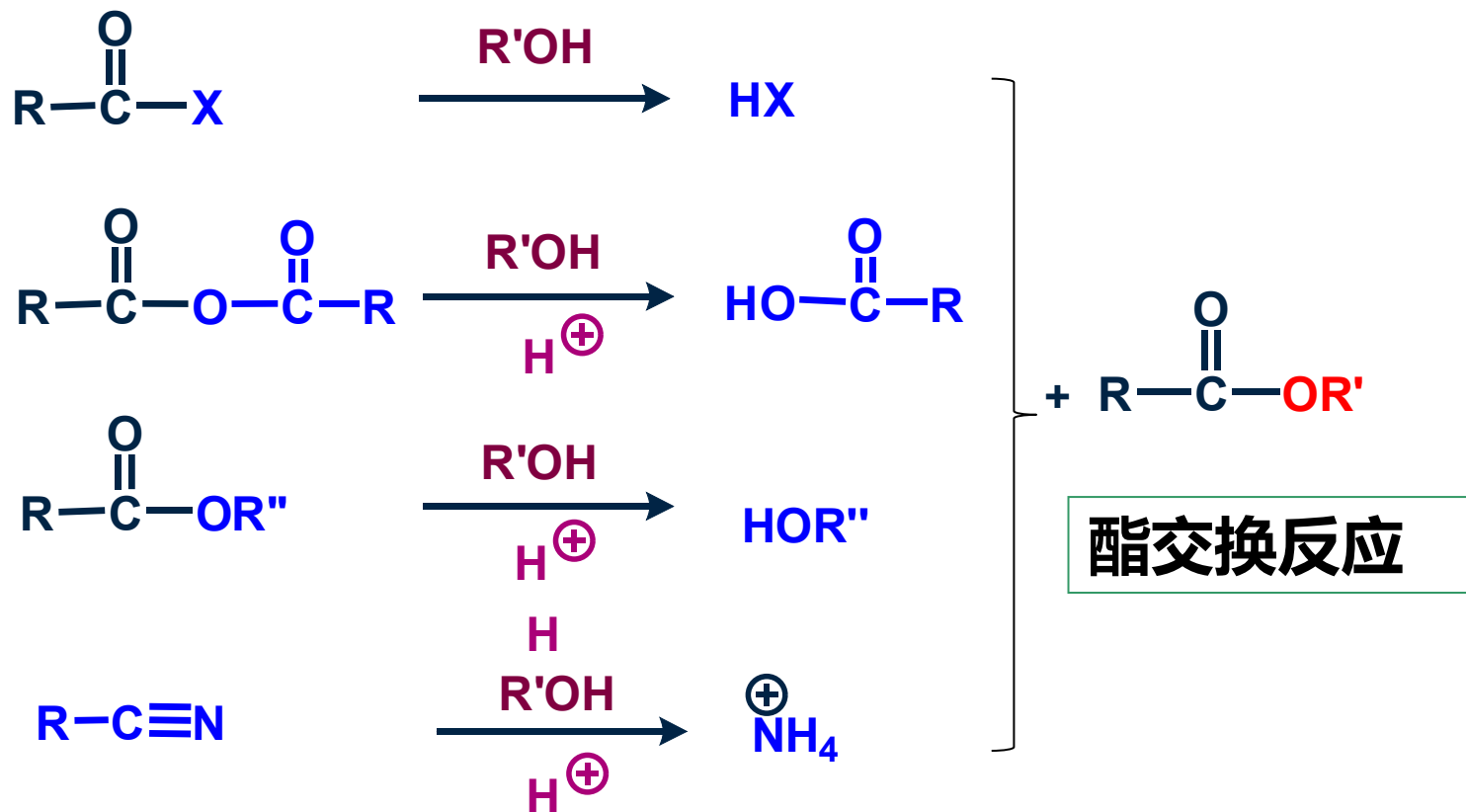
酸或碱催化并长时间回流

碱性条件下，产物生成的酸转变成相应的盐。

9.8 羧酸衍生物的化学性质



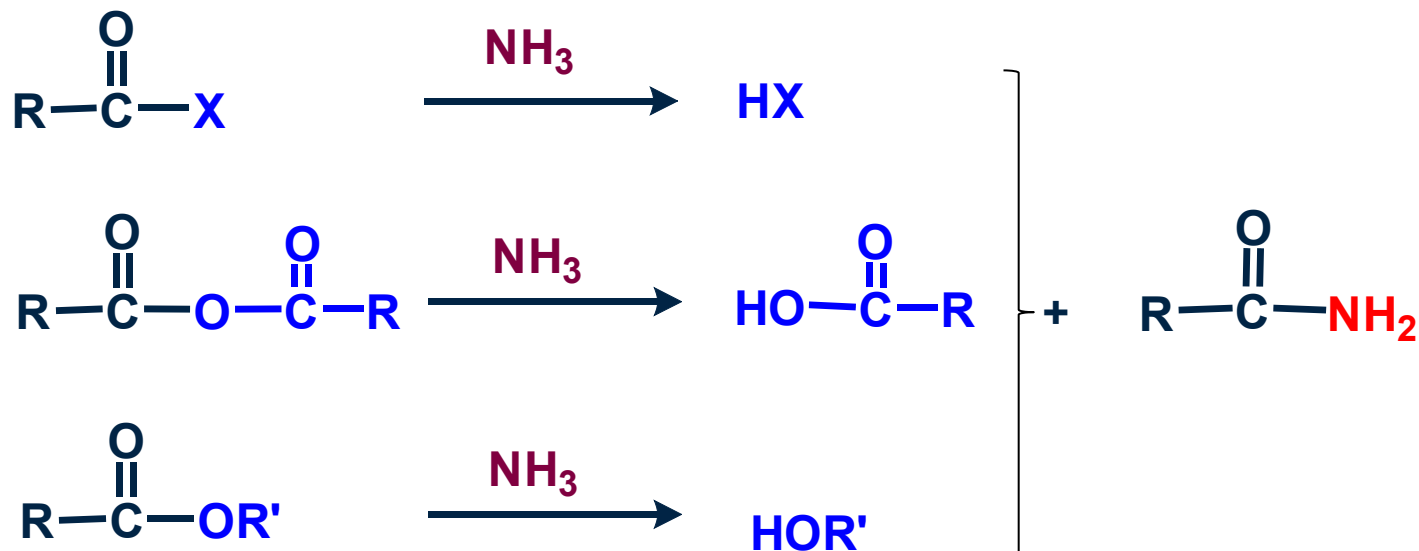
醇解反应



9.8 羧酸衍生物的化学性质



氨解反应

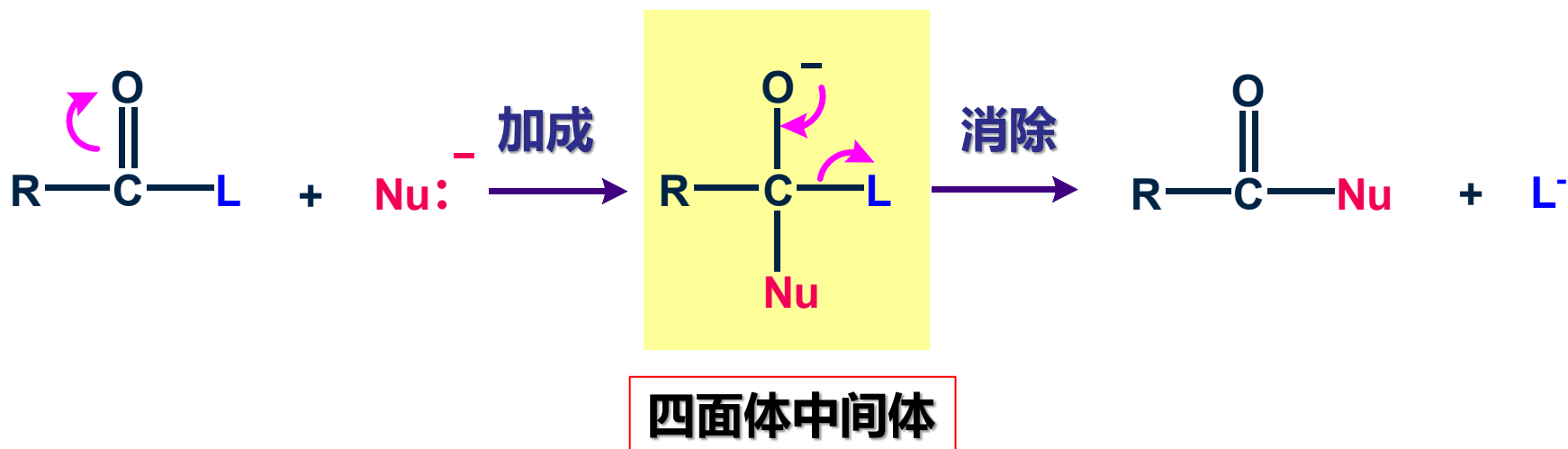


9.8 羧酸衍生物的化学性质



9.8.2 酰基上的亲核取代反应机理及相对反应活性

(1) 带负电的亲核试剂



9.8 羧酸衍生物的化学性质



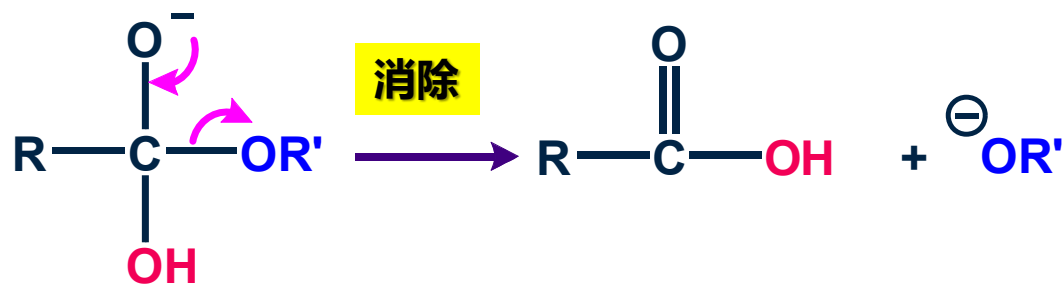
9.8.2 酰基上的亲核取代反应机理及相对反应活性

(1) 反应机理 (以酯的碱性水解反应为例)

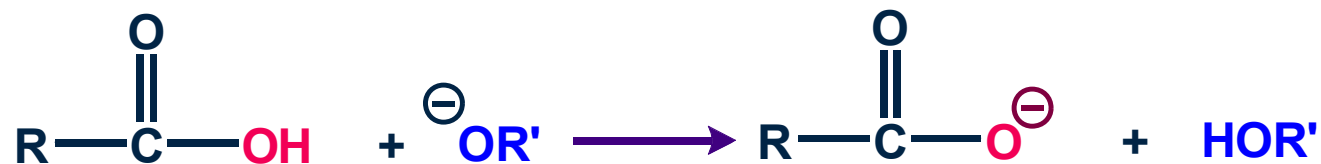
第一步：氢氧根负离子与羰基的加成



第二步：离去基团离去，恢复羰基结构



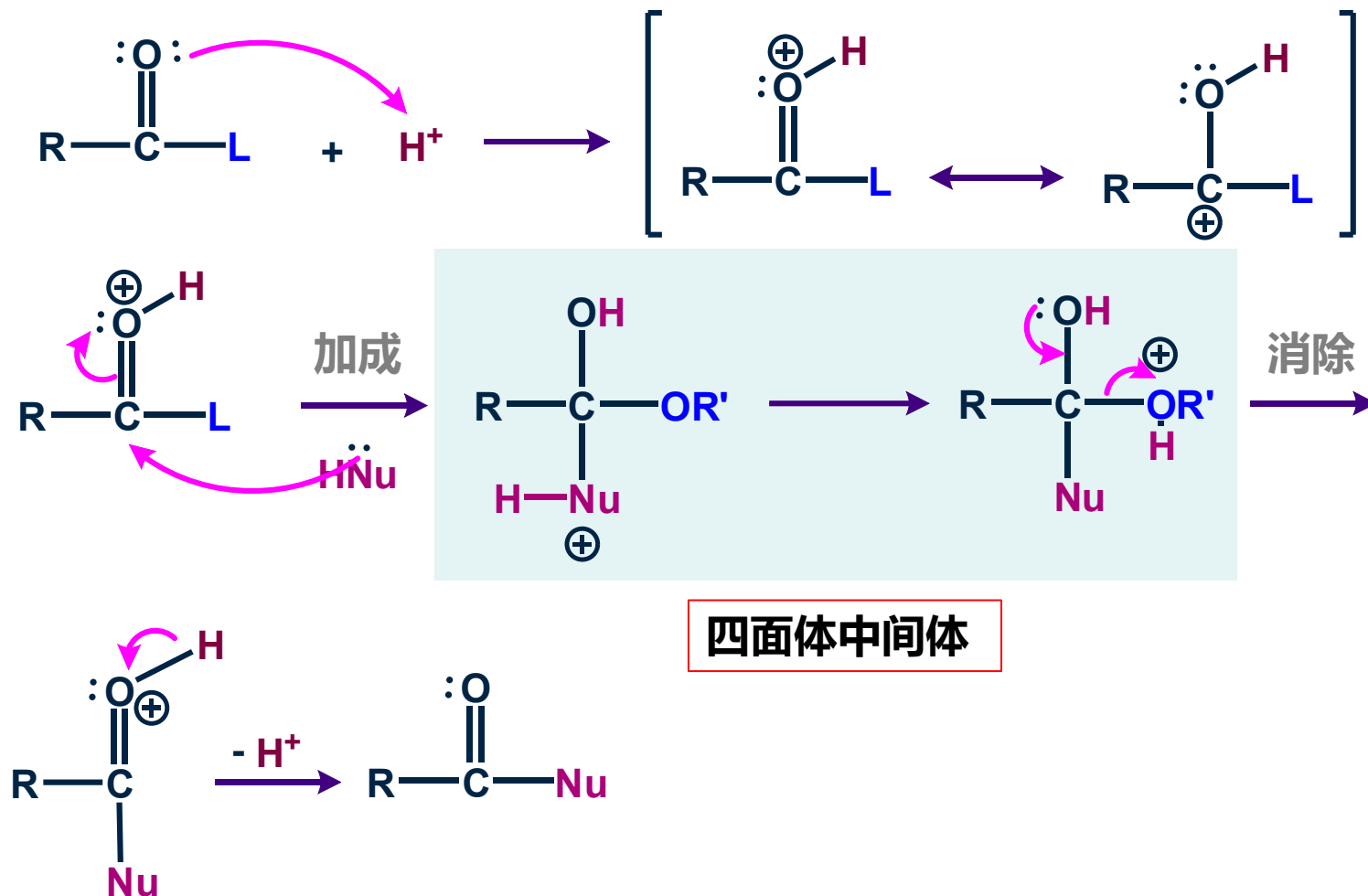
第三步：质子转移，生成醇和羧酸盐



9.8 羧酸衍生物的化学性质



(2)中性的亲核试剂——加成机理

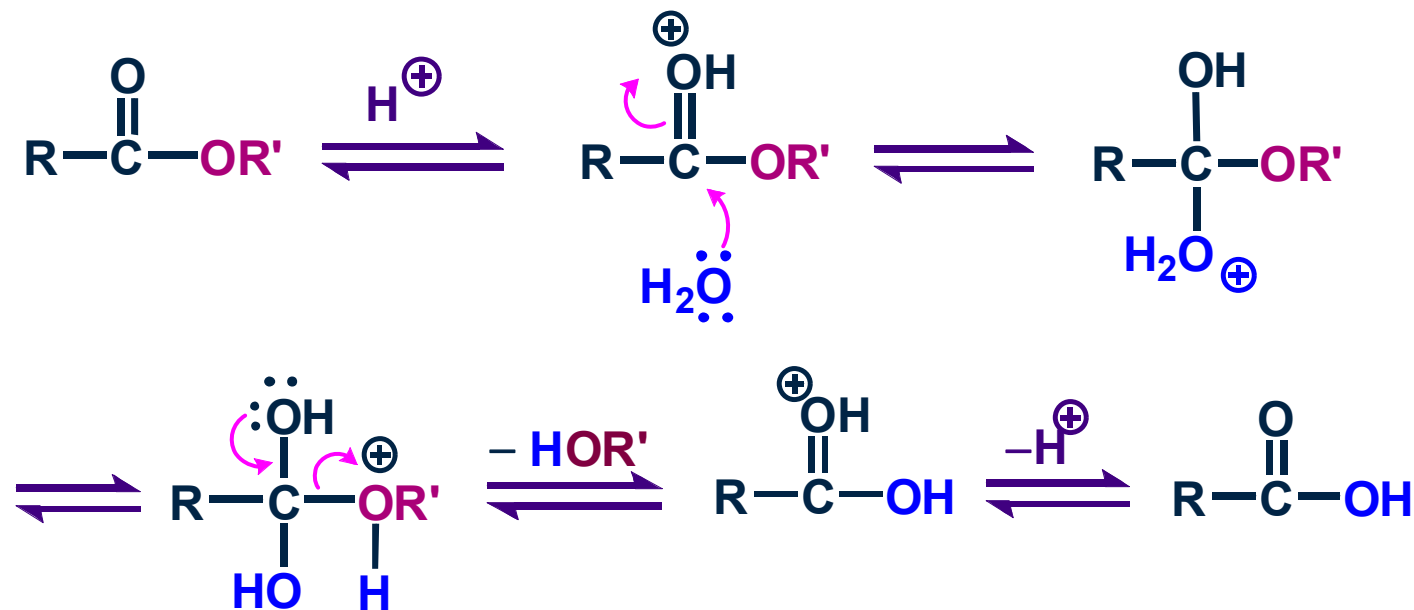


四面体中间体

9.8 羧酸衍生物的化学性质



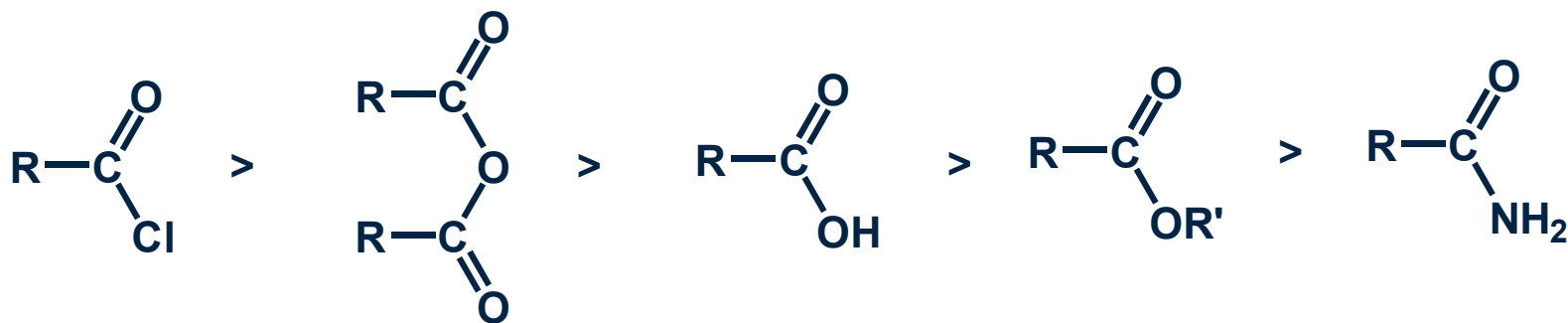
(3) 举例：酯的酸性水解机理





9.8 羧酸衍生物的化学性质

酰基上的亲核取代相对反应活性

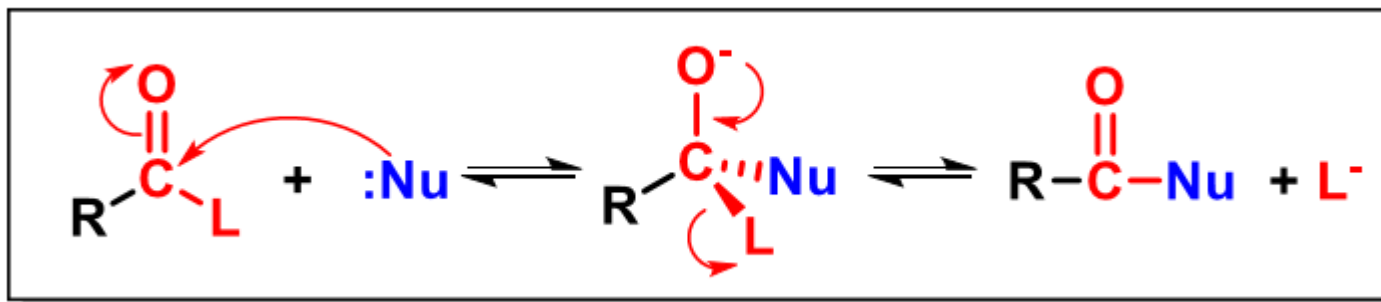


反应活依次性降低

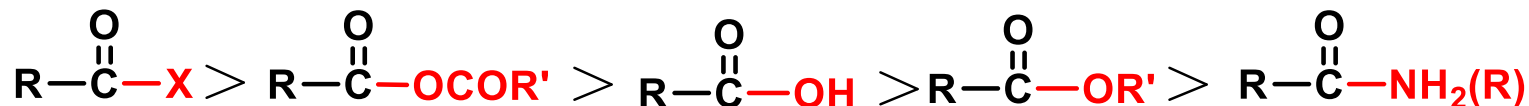
9.8 羧酸衍生物的化学性质



(1) 诱导效应解释相对反应活性



离去基团吸电子诱导效应越强，羰基碳上电子云密度越低，亲核试剂进攻越有利。

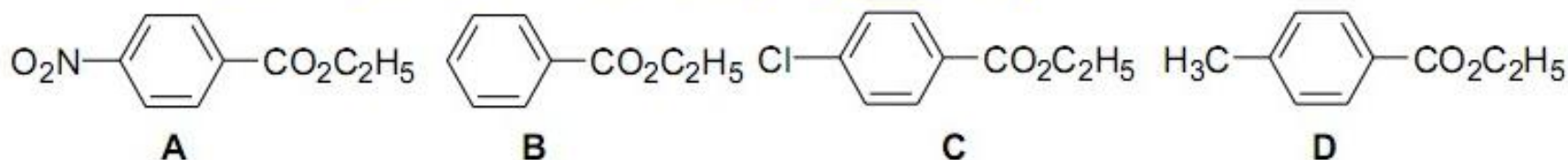


9.8 羧酸衍生物的化学性质



(1) 诱导效应解释相对反应活性

比较如下化合物碱性水解的速度顺序

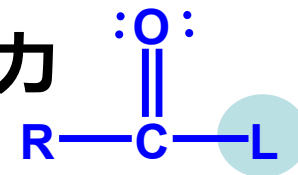


电子效应的影响 $A > C > B > D$

9.8 羧酸衍生物的化学性质



(2) 酸碱理论解释相对反应活性-离去基团离去能力

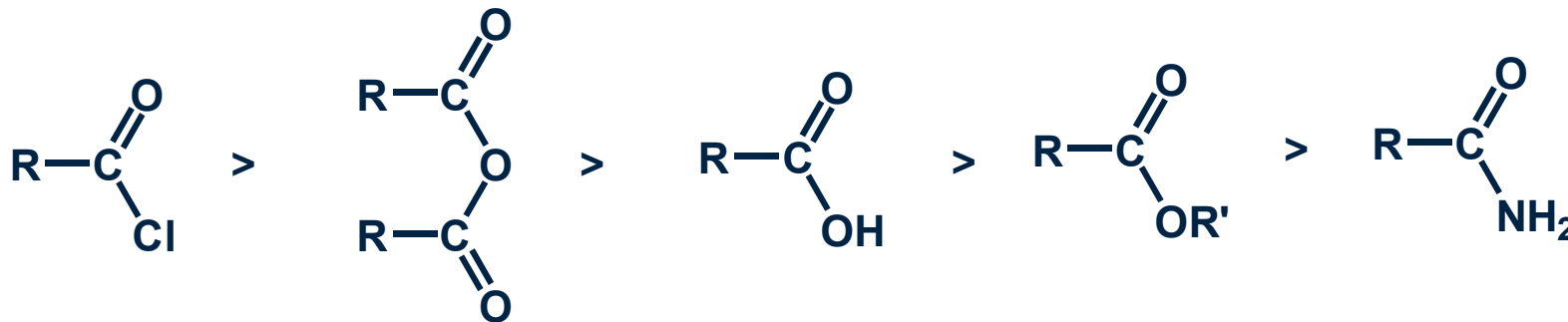


酸性: $\text{HCl} > \text{RCOOH} > \text{H}_2\text{O} > \text{ROH} > \text{NH}_3$

$\text{p}K_a$ -2.2 4~5 15.7 16~19 34

共轭碱碱性: $\text{H}_2\text{N}^- > \text{RO}^- > \text{HO}^- > \text{RCOO}^- > \text{Cl}^-$

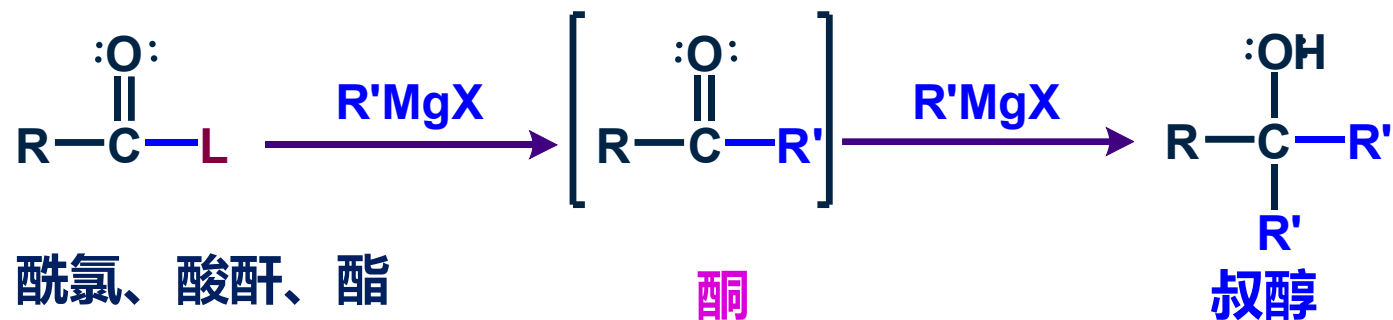
- 碱性越弱, 越容易离去, 相应酰基化试剂的活性越大。



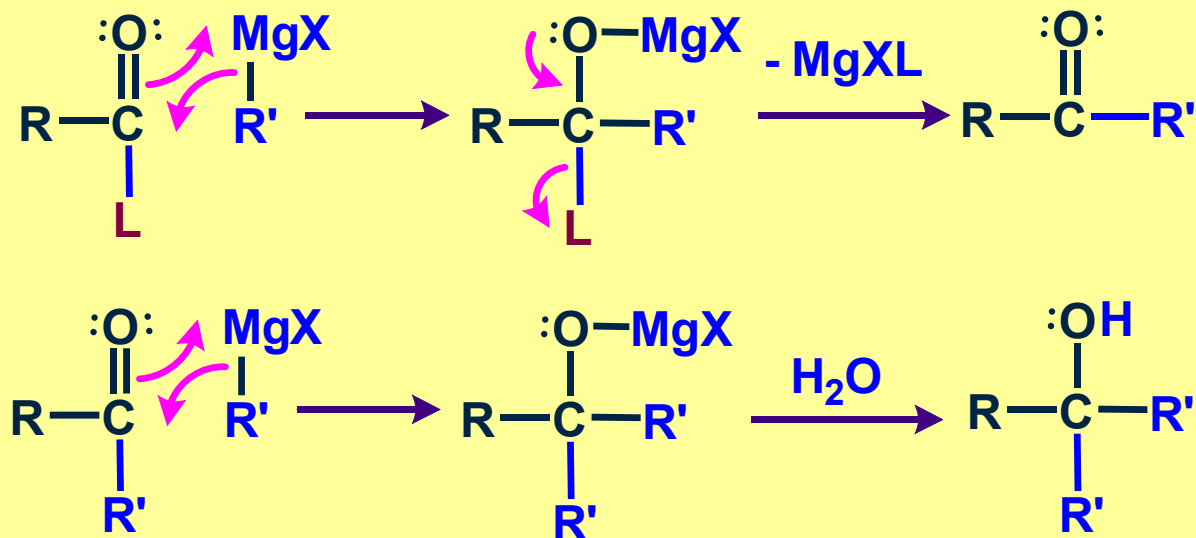
9.8 羧酸衍生物的化学性质



9.8.3 与金属有机试剂的反应



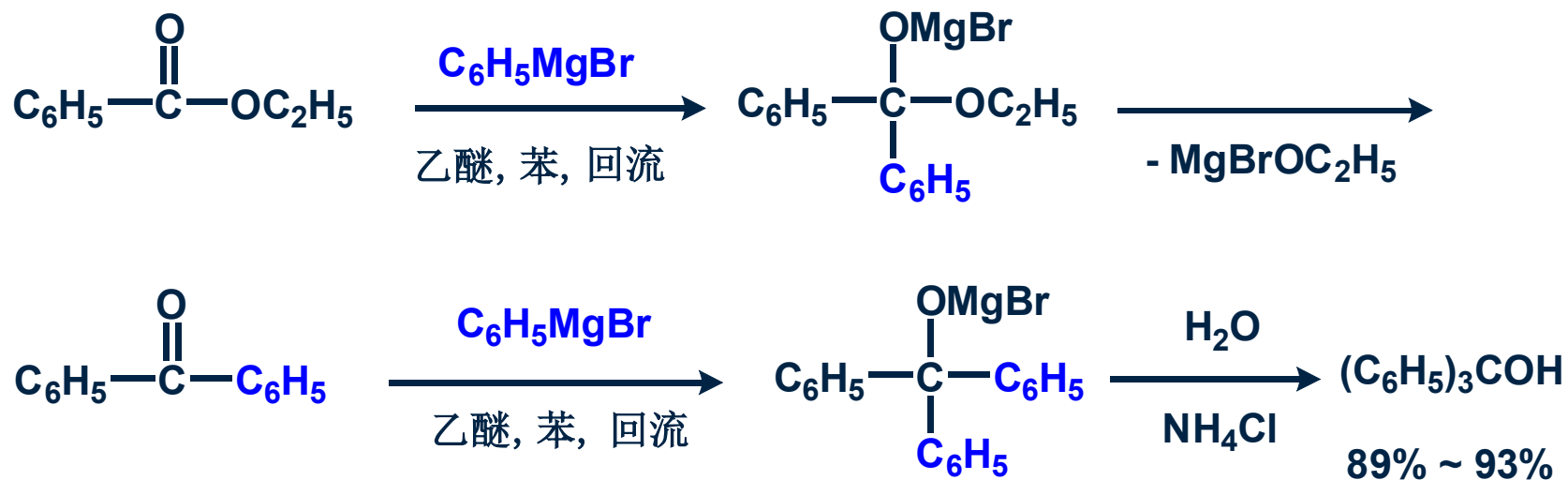
机理:



9.8 羧酸衍生物的化学性质



9.8.3 与金属有机试剂的反应

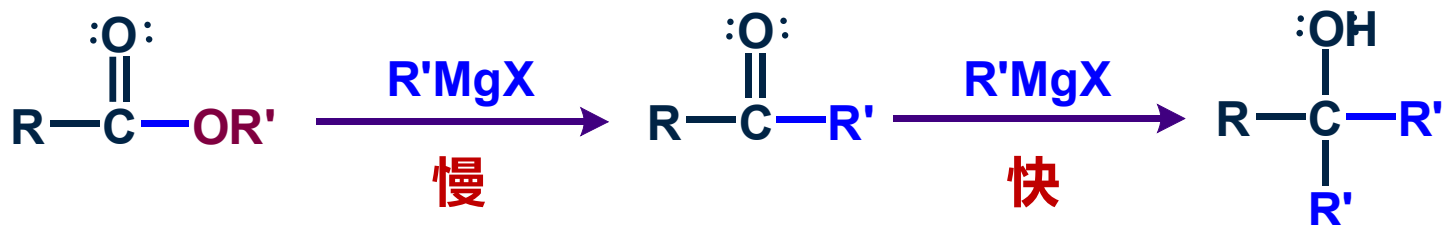
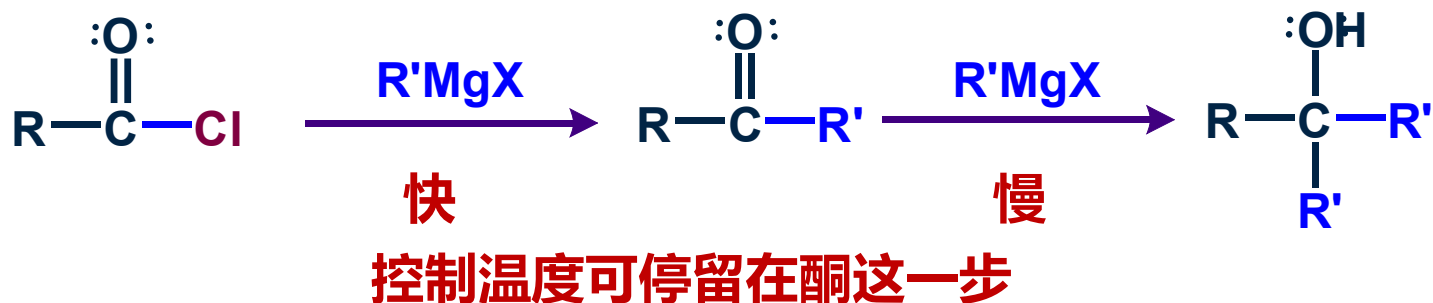
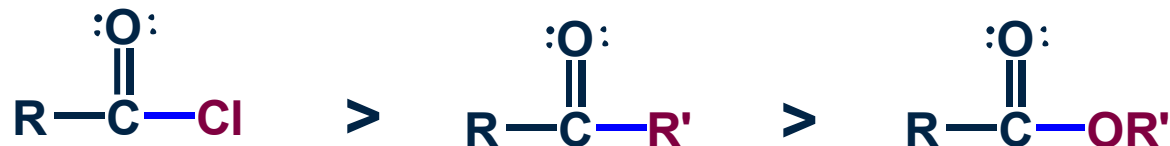


9.8 羧酸衍生物的化学性质



9.8.3 与金属有机试剂的反应

羰基的活性（亲电性）：



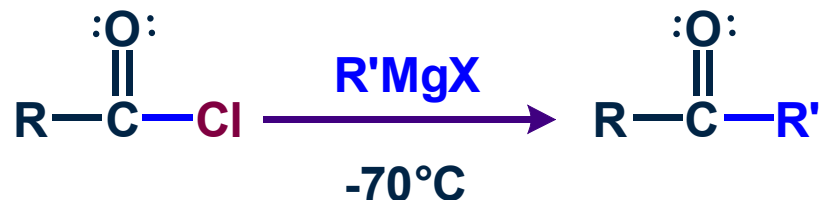
9.8 羧酸衍生物的化学性质



9.8.3 与金属有机试剂的反应

利用羰基活性差别由酰氯制备酮

羰基的活性（亲电性）：



例如：

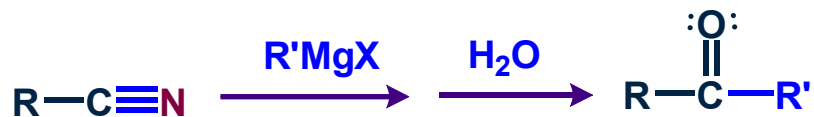


9.8 羧酸衍生物的化学性质

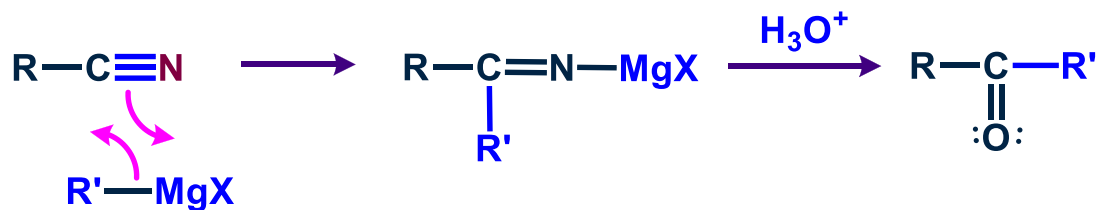


9.8.3 与金属有机试剂的反应

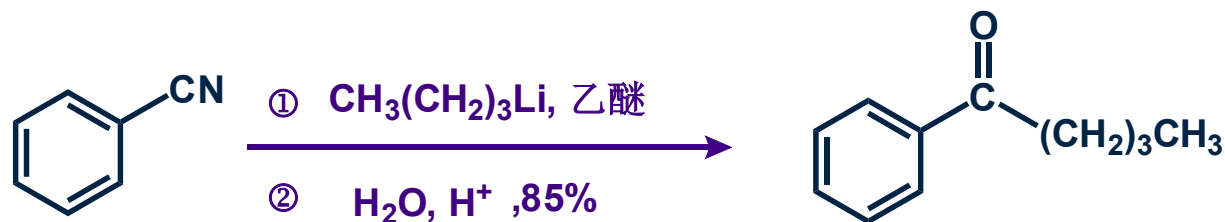
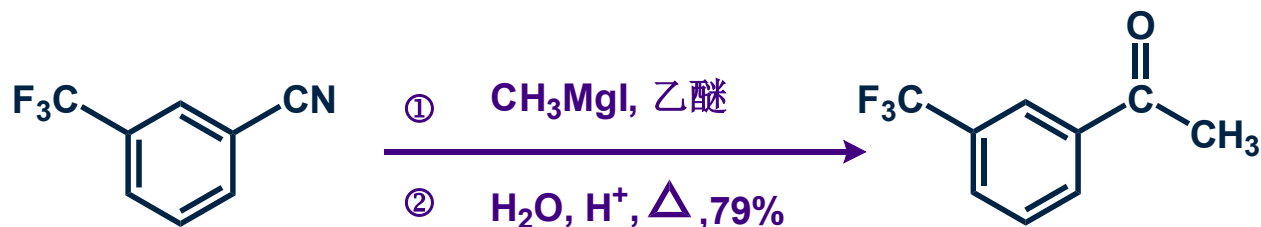
通式:



过程:



示例:

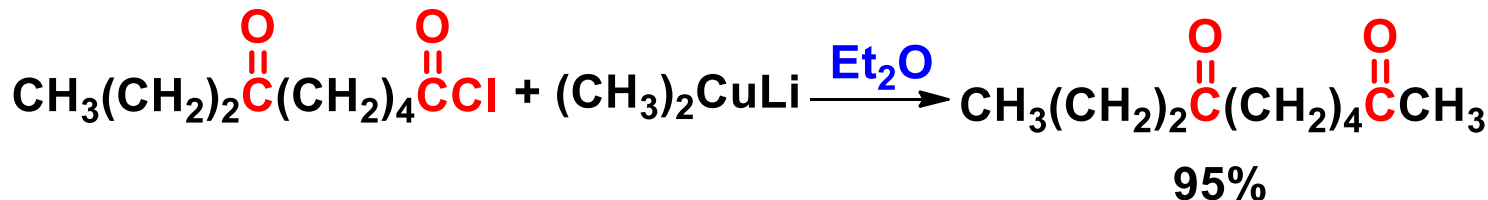
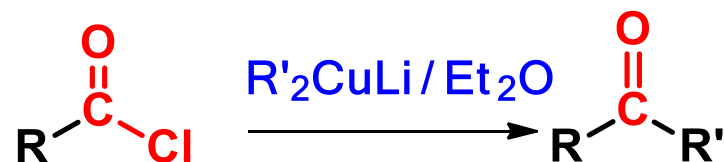




9.8 羧酸衍生物的化学性质

9.8.3 与金属有机试剂的反应-二烷基铜锂 (R_2CuLi)

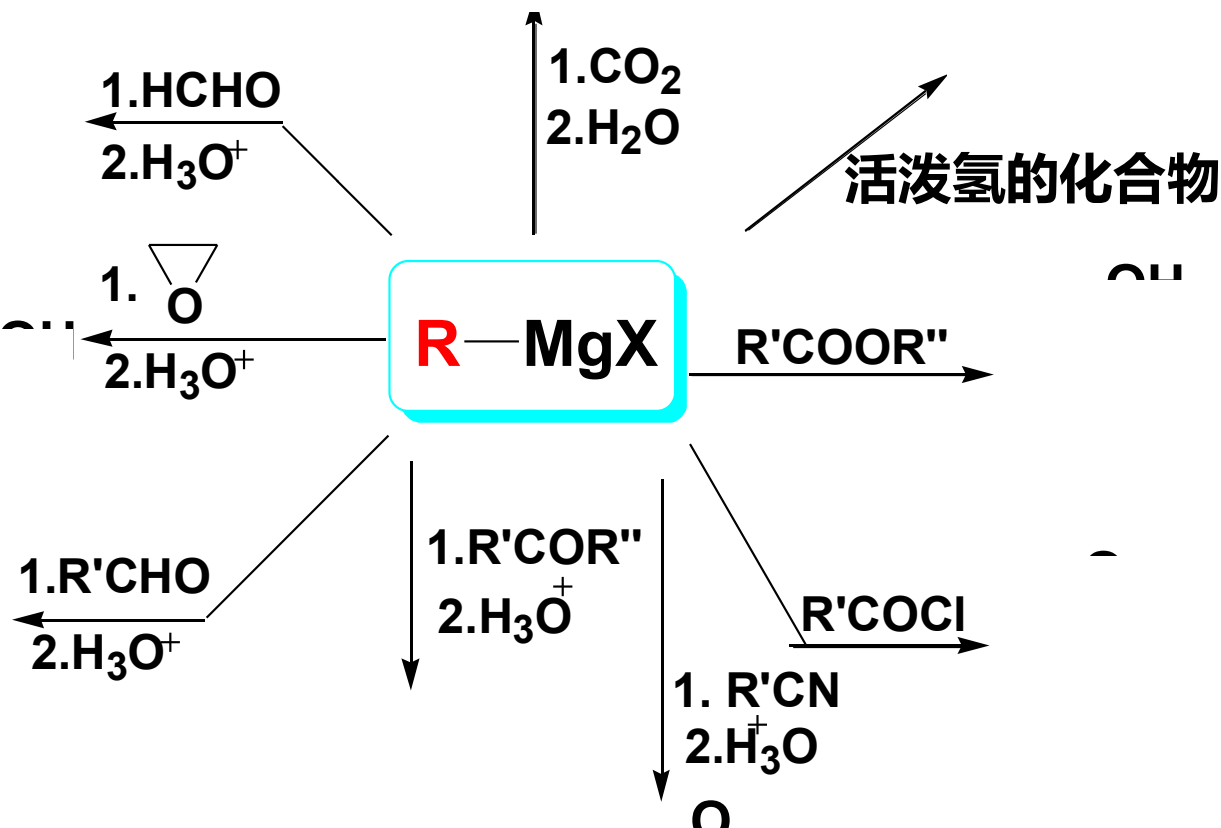
- 二烷基铜锂活性比较低, 可与酰氯、醛反应, 与酮反应很慢, 低温下不与酯、酰胺和腈反应。
- 常用于从酰氯制备酮, 选择性好, 产率高



9.8 羧酸衍生物的化学性质

格氏试剂的应用（总结）

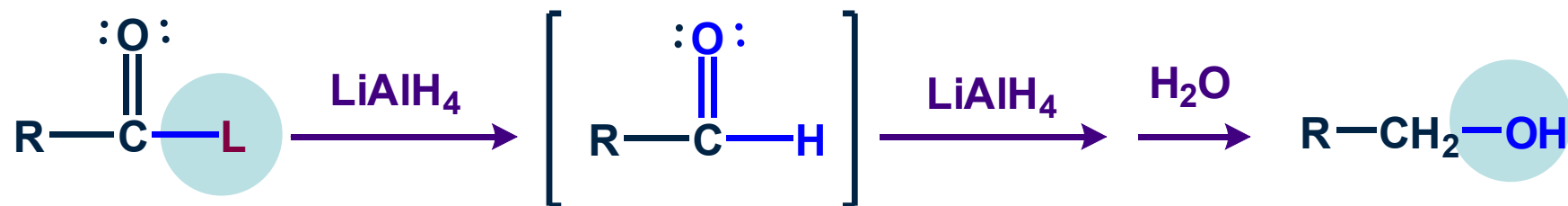
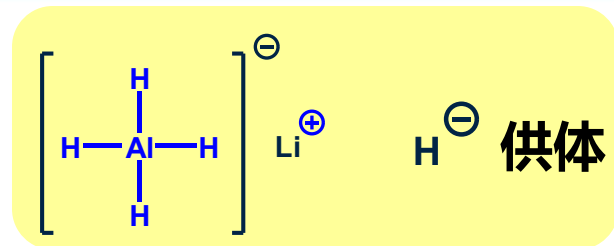
进攻正电中心



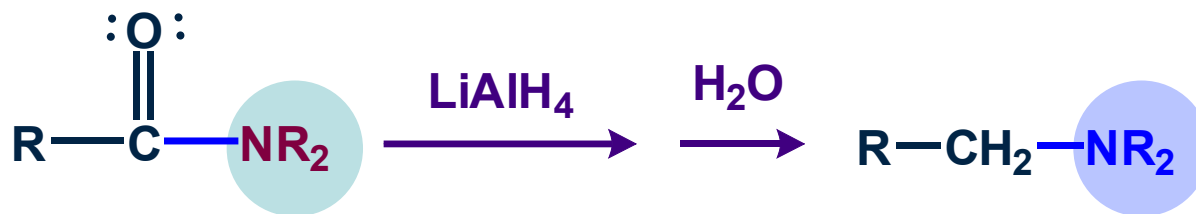
9.8 羧酸衍生物的化学性质



9.8.4 还原反应-氢化铝锂



L = 卤素, 酰氧基, 烷氧基



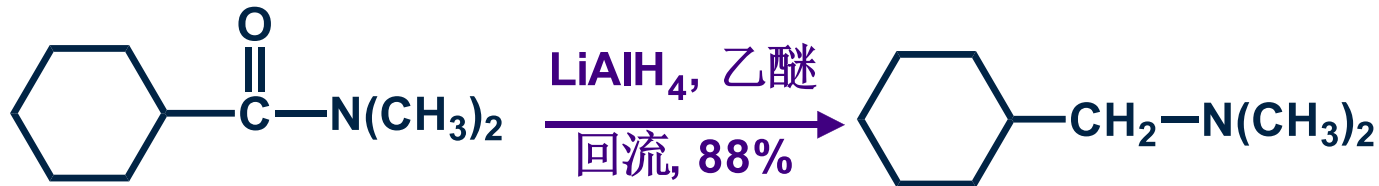
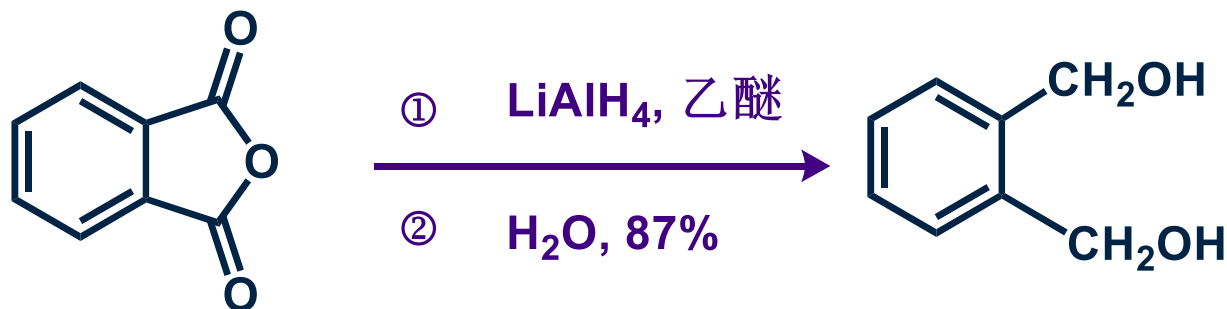
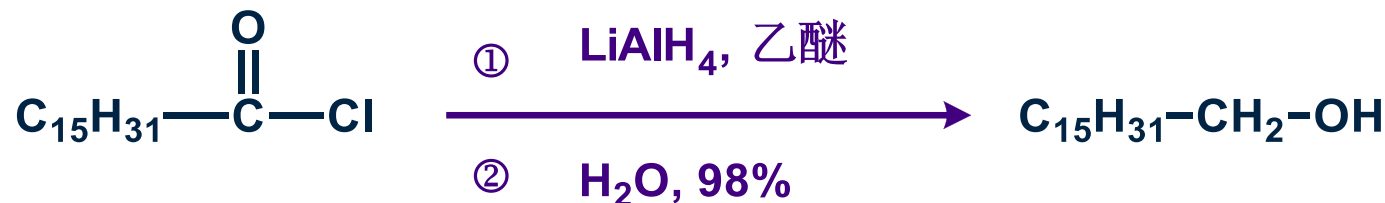
L = NH₂或NHR或NR₂,

腈使用LiAlH₄还原也得胺。

9.8 羧酸衍生物的化学性质



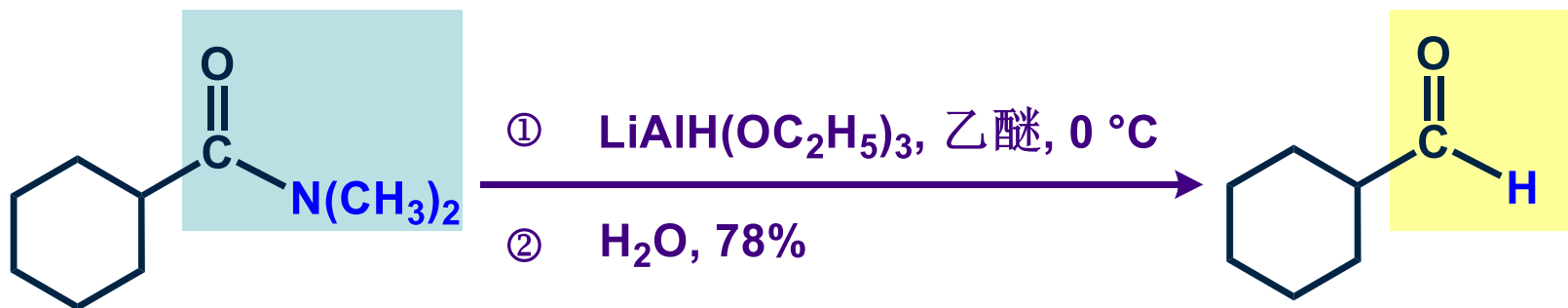
9.8.4 还原反应-氢化铝锂



9.8 羧酸衍生物的化学性质



9.8.4 还原反应-烷氧基氢化铝锂

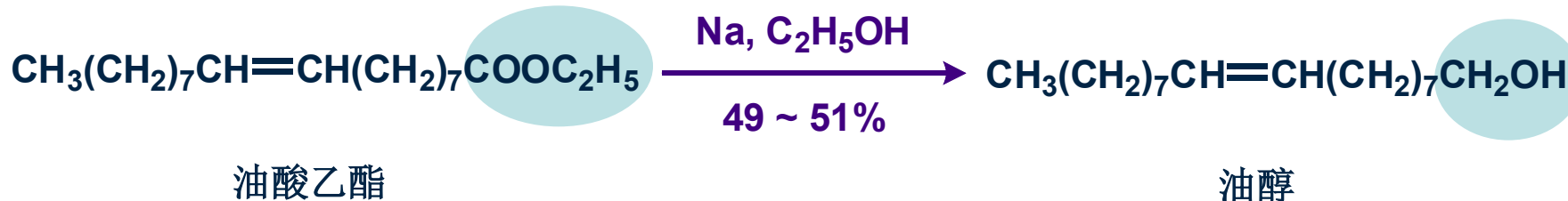




9.8 羧酸衍生物的化学性质

9.8.4 还原反应-金属钠-醇

酯与金属钠在醇(常用乙醇、丁醇或戊醇等)溶液中加热回流,可被还原成相应的伯醇,此反应称为Bouveault-Blanc反应。

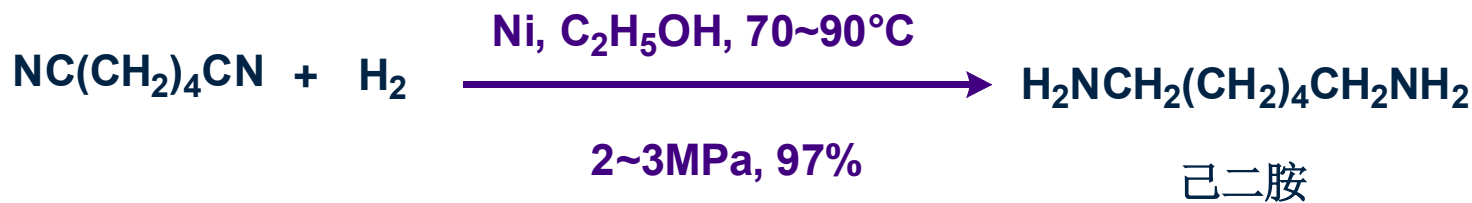
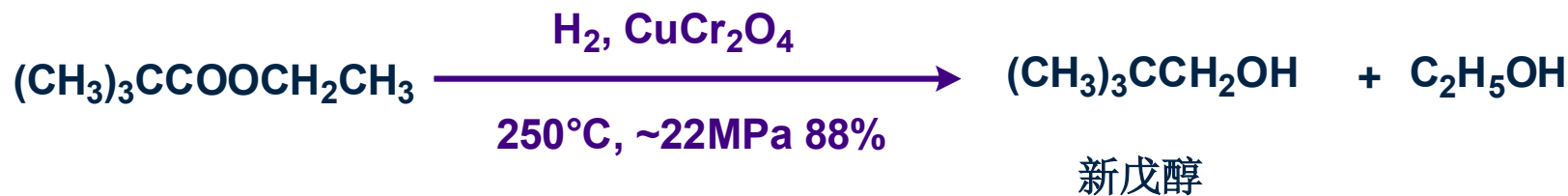


工业上生产不饱和醇的方法



9.8 羧酸衍生物的化学性质

9.8.4 还原反应-催化氢化-了解



9.8 羧酸衍生物的化学性质



9.8.4 还原反应-Rosenmund还原



使Pd催化剂活性减弱
(催化剂中毒)

例:

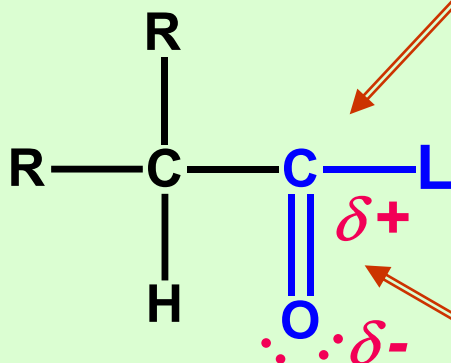
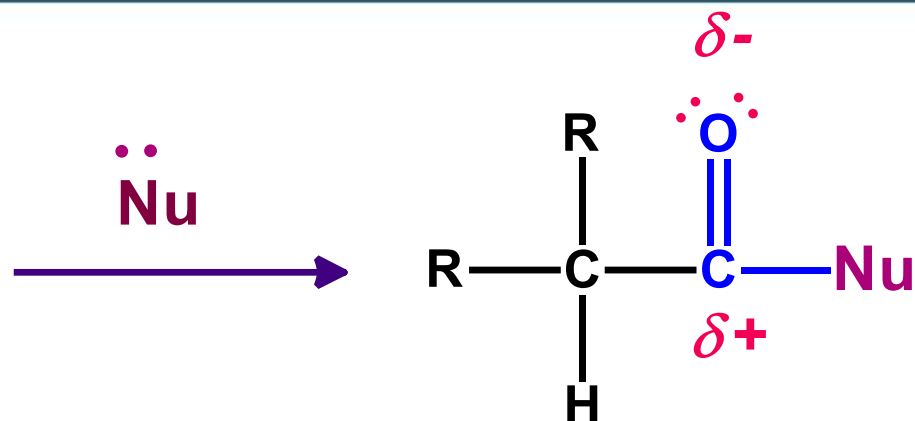


该反应是制备醛的一种好方法.

9.8 羧酸衍生物的化学性质

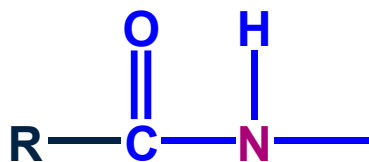
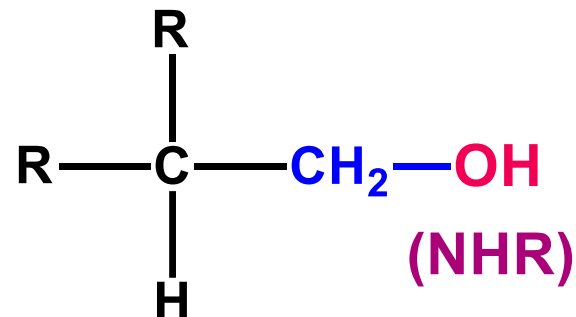


羰基碳有亲电性
可发生亲核取代



羰基被还原

还原

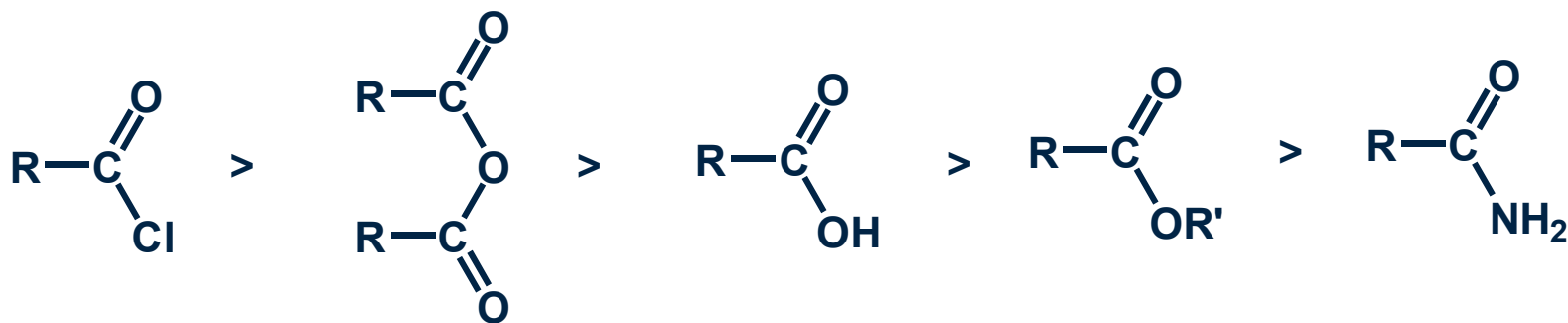


酰胺的特性

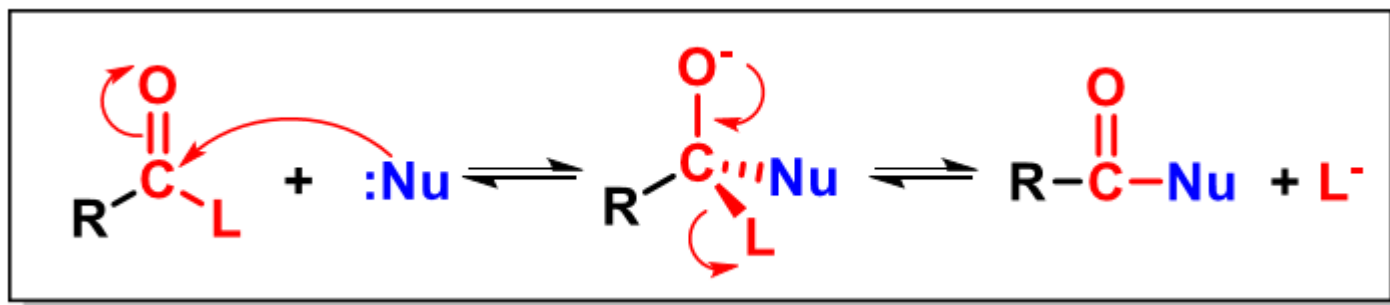
9.8 羧酸衍生物的化学性质

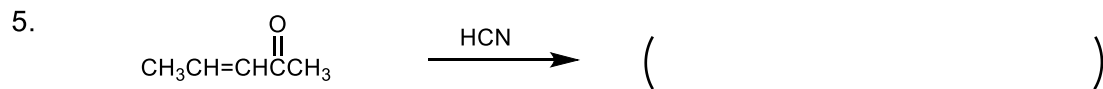
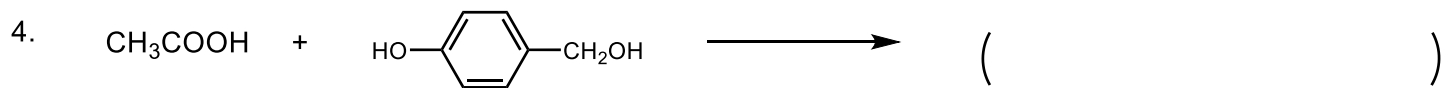
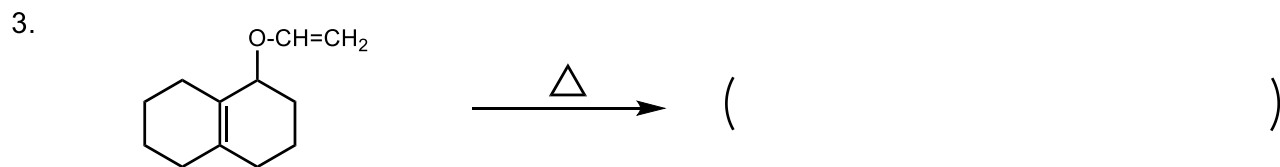
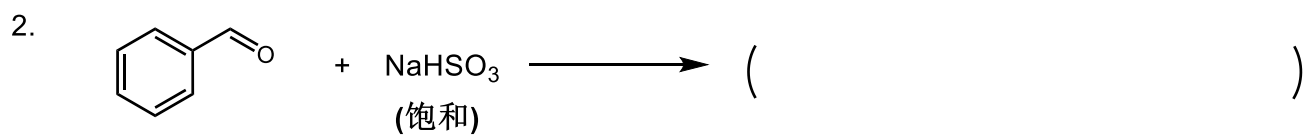
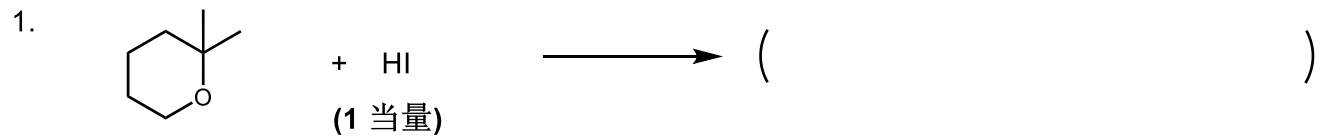


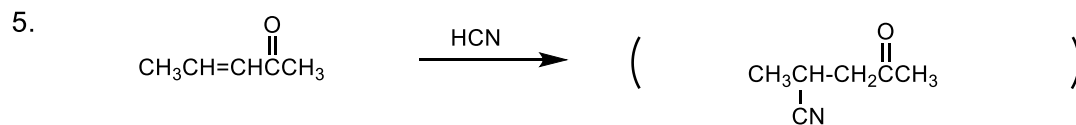
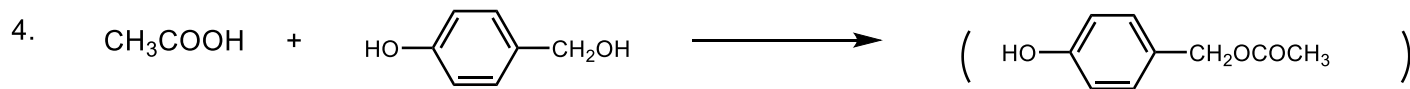
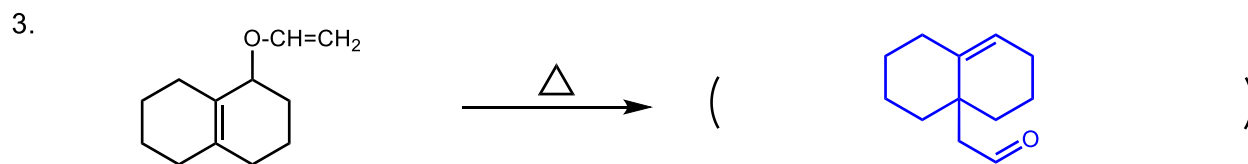
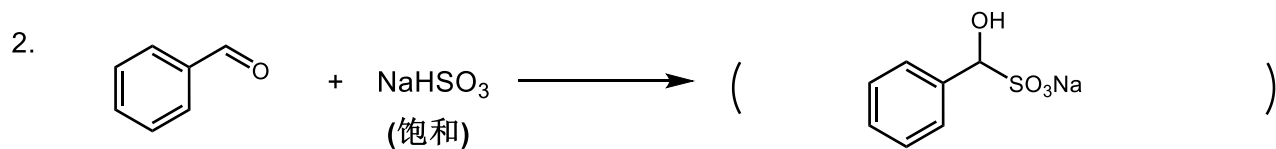
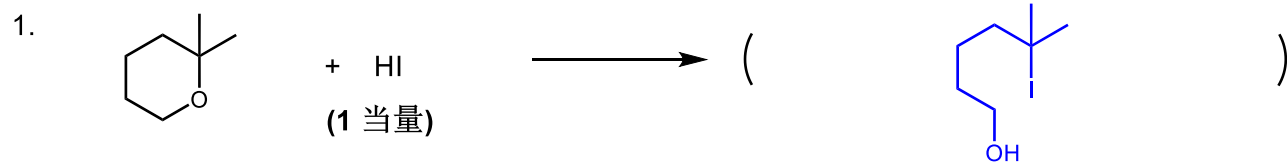
酰基上的亲核取代相对反应活性

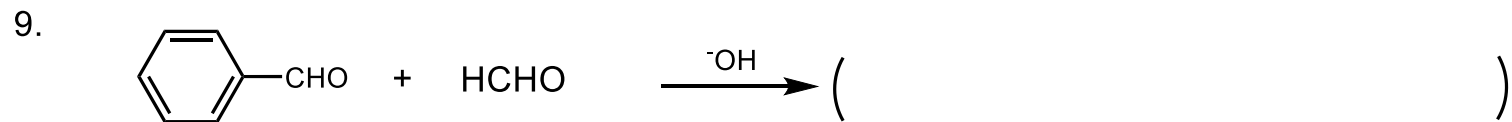
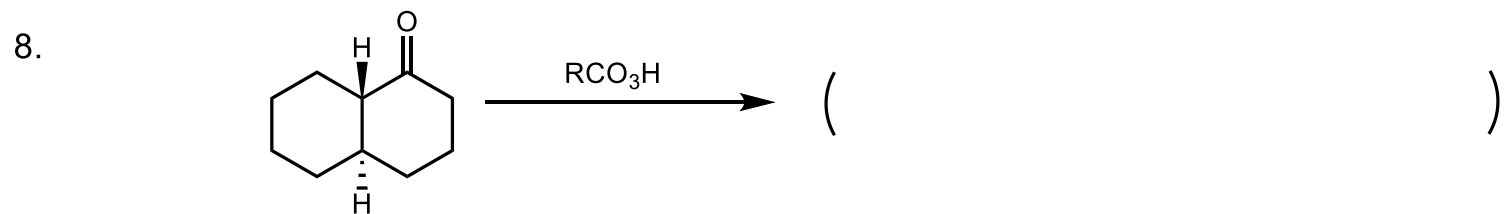
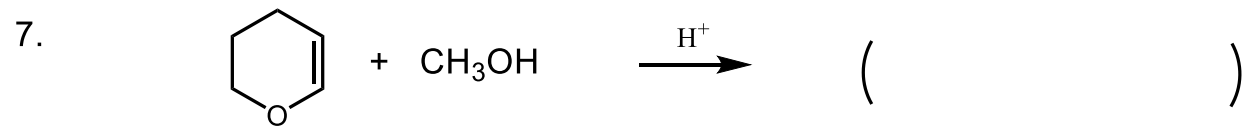
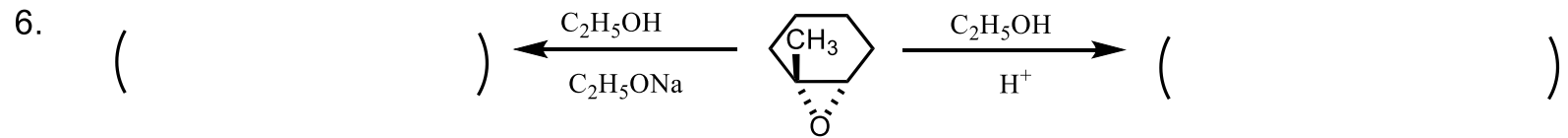


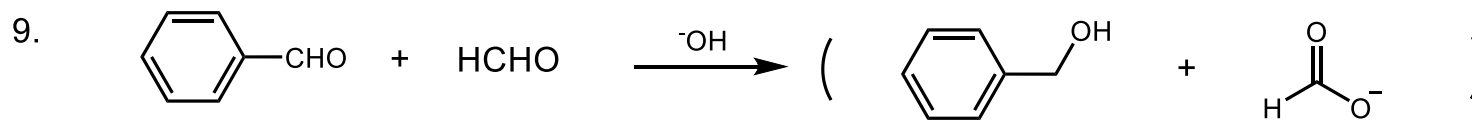
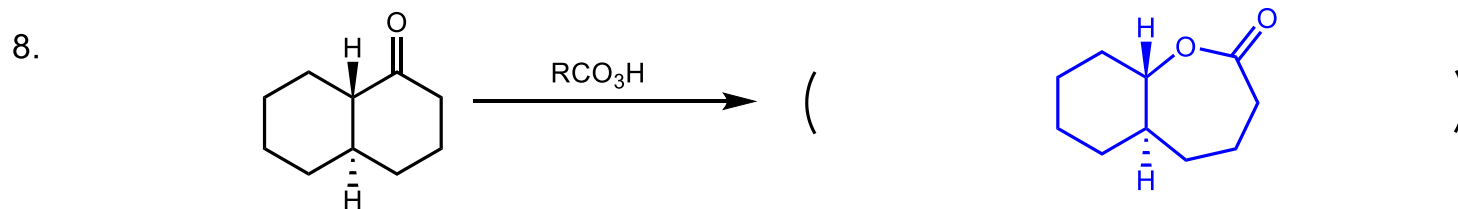
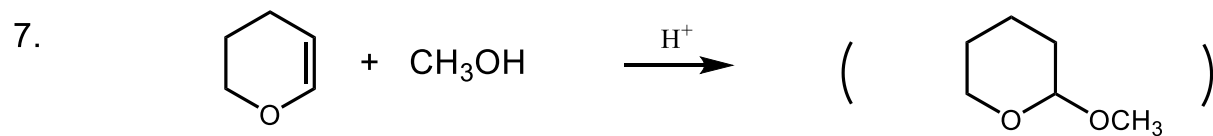
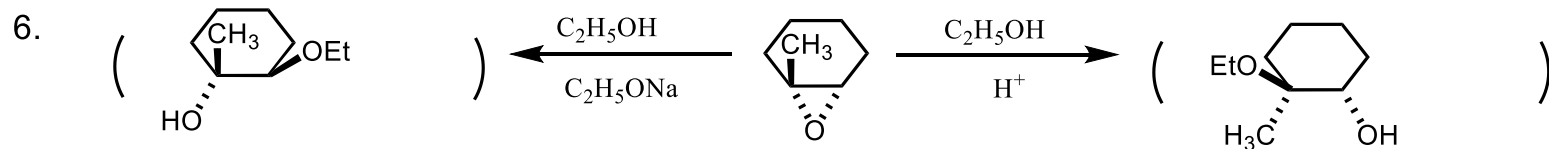
反应活依次性降低

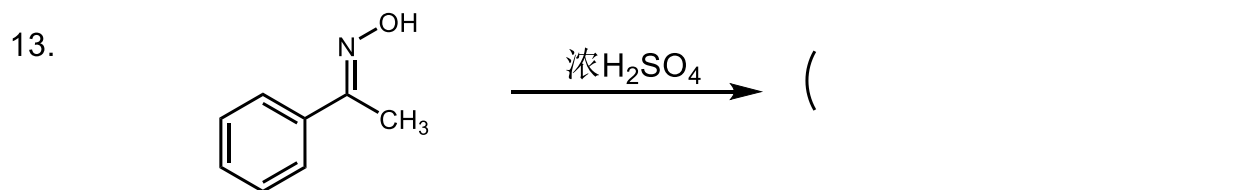
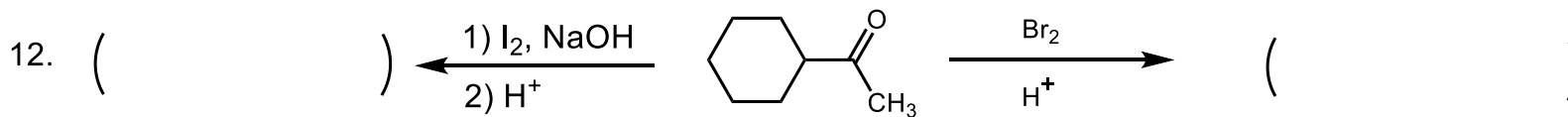
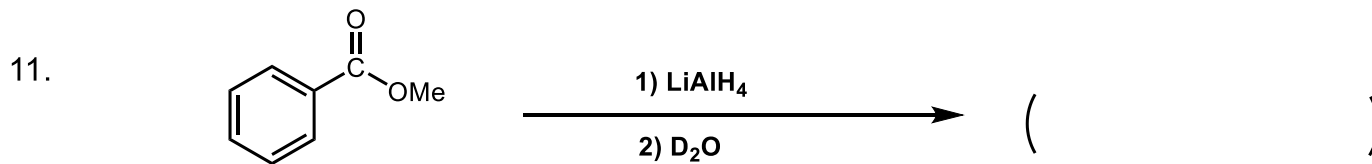






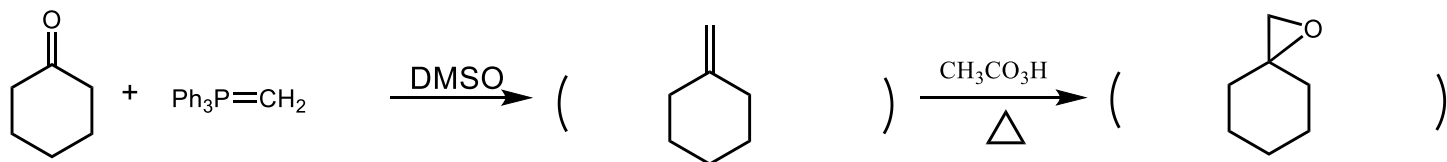




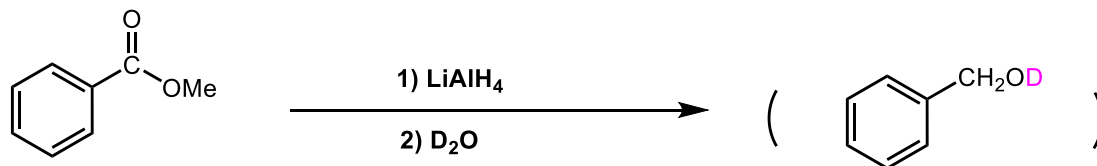




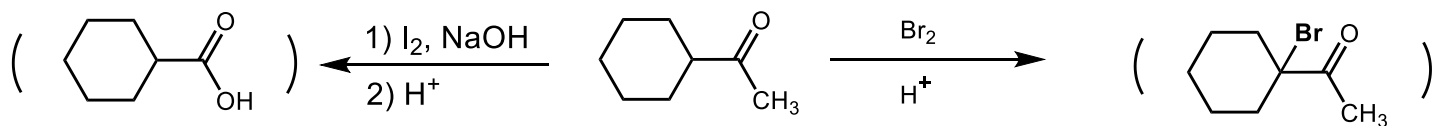
10.



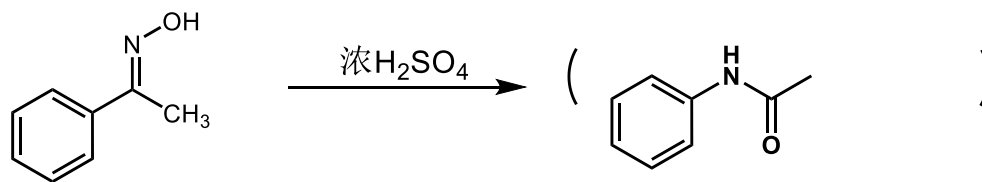
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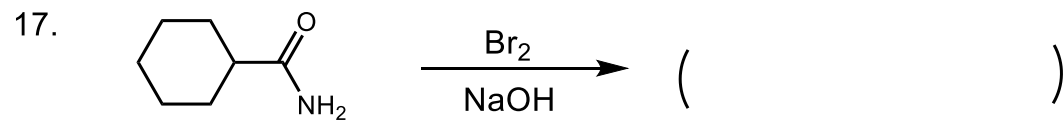
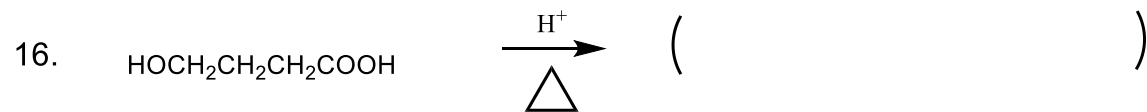
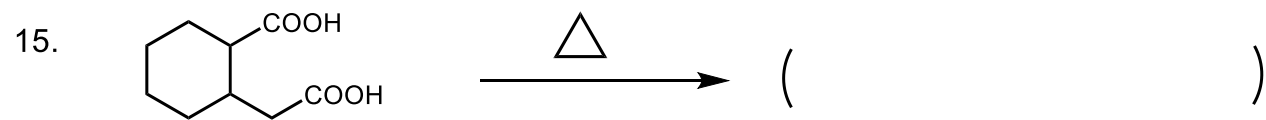


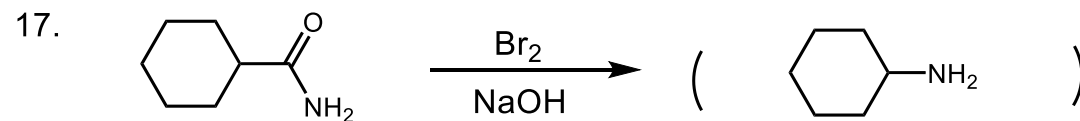
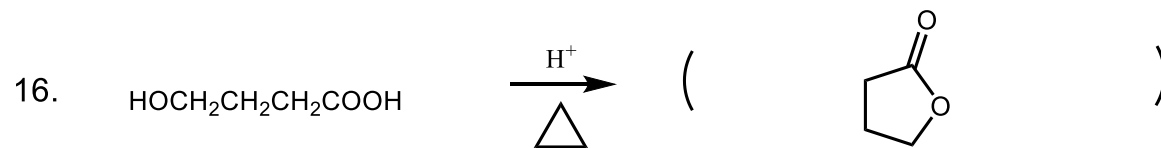
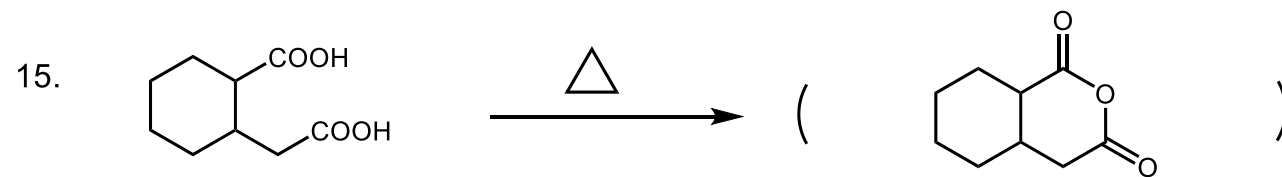
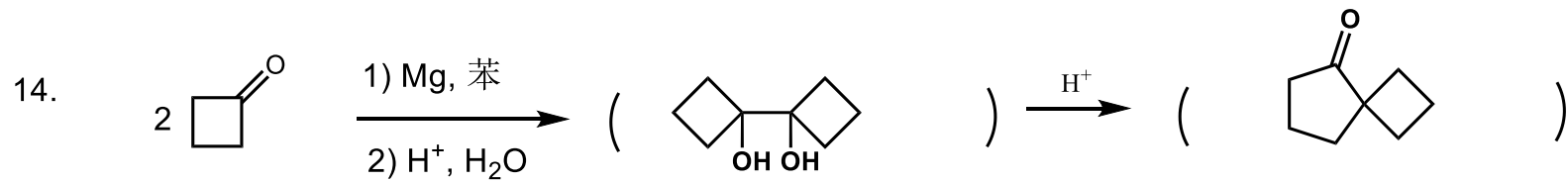
12.



13.







9.8 羧酸衍生物的化学性质

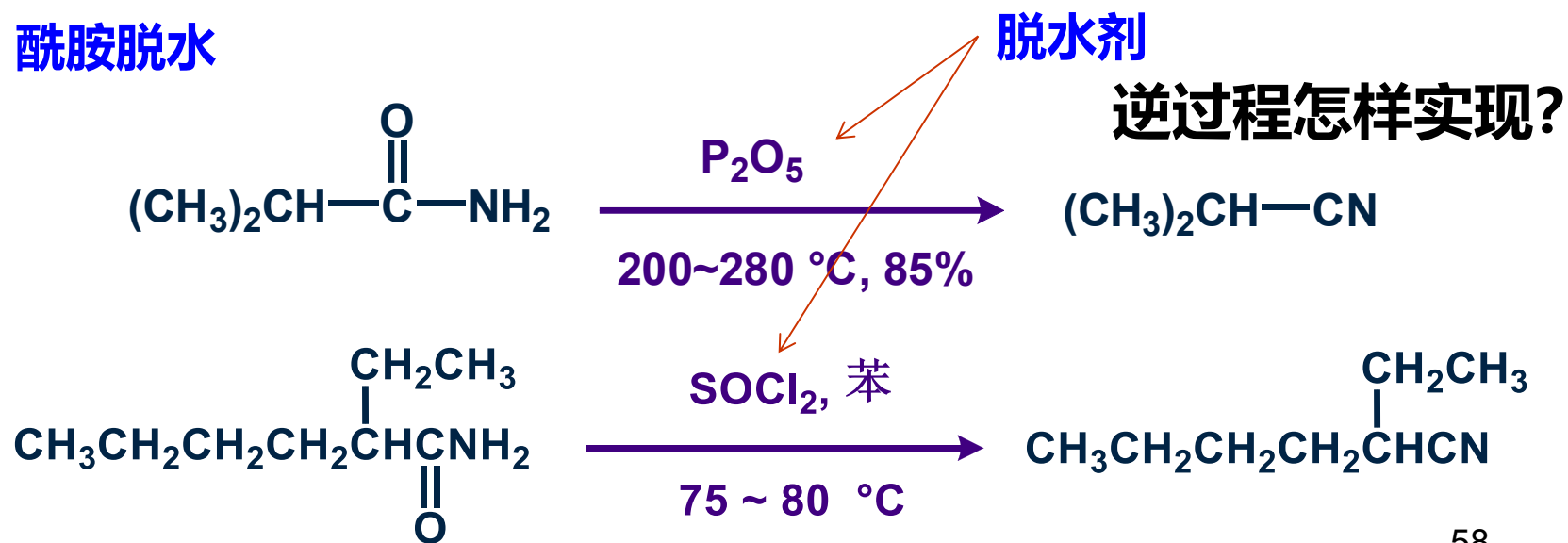


9.8.5 酰胺的特性

(1) 酰胺的酸碱性



(2) 酰胺脱水



9.8 羧酸衍生物的化学性质



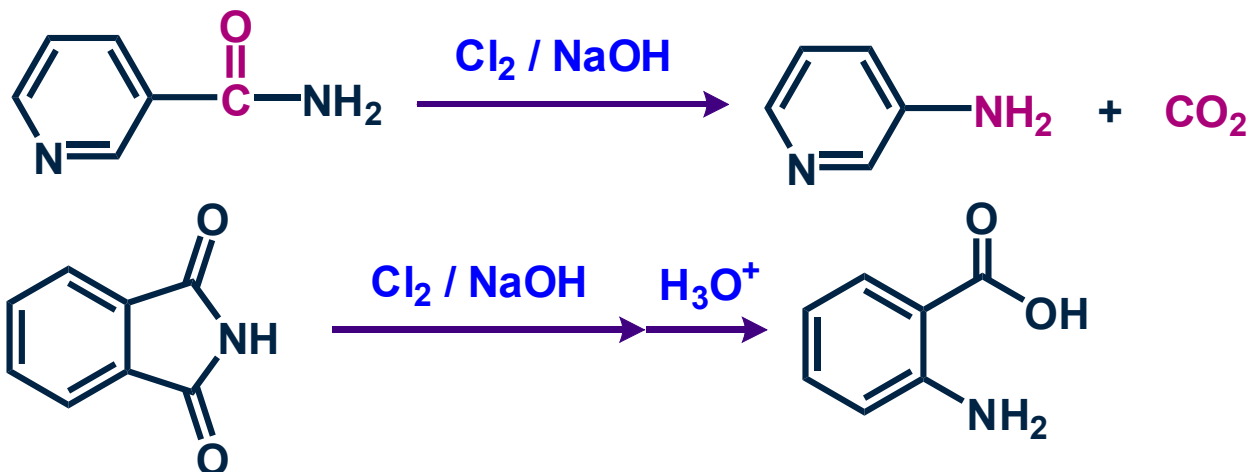
9.8.5 酰胺的特性

(3) Hofmann降解反应



少一个C原子的伯胺

示例:

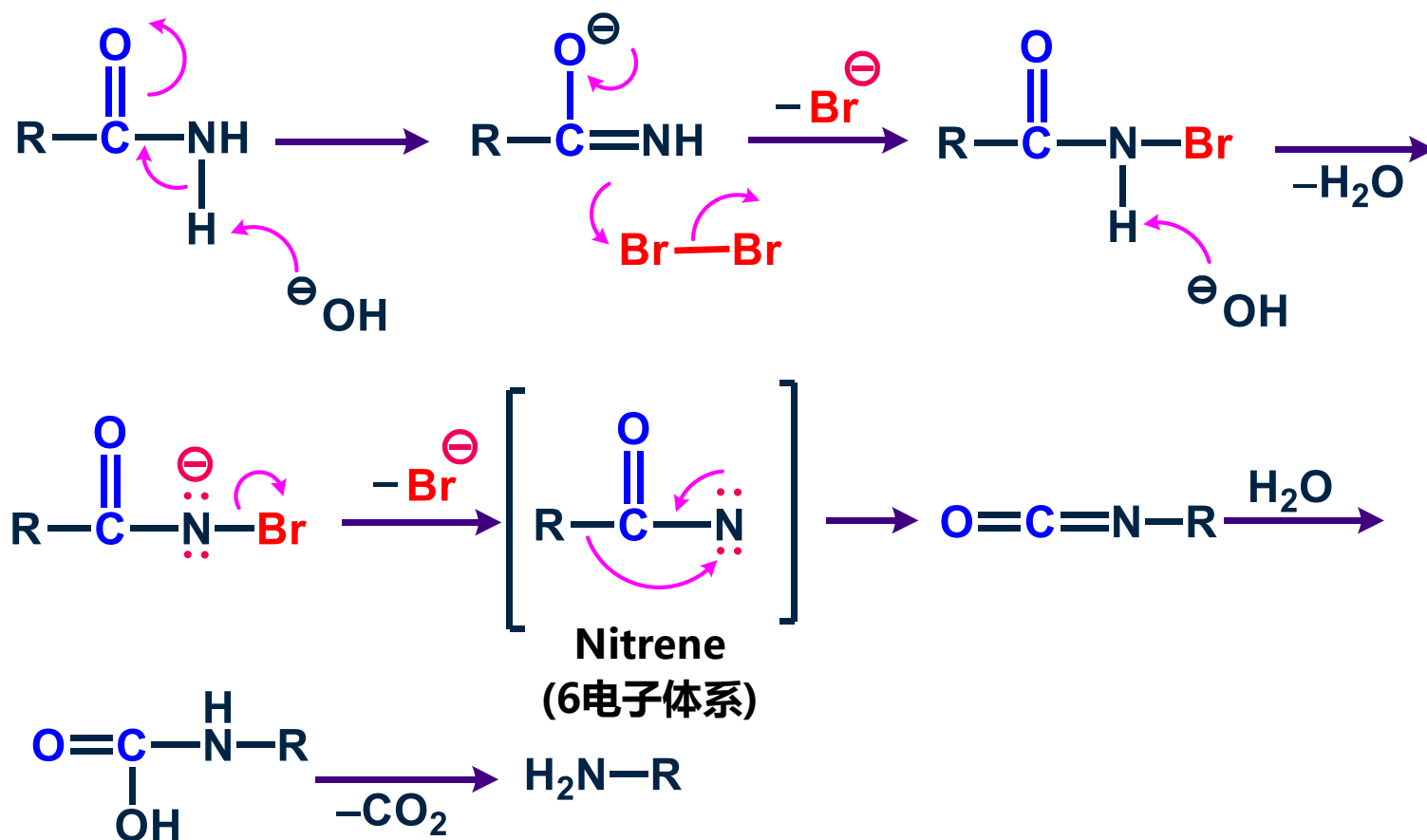


9.8 羧酸衍生物的化学性质



9.8.5 酰胺的特性

(3) Hofmann降解反应机理

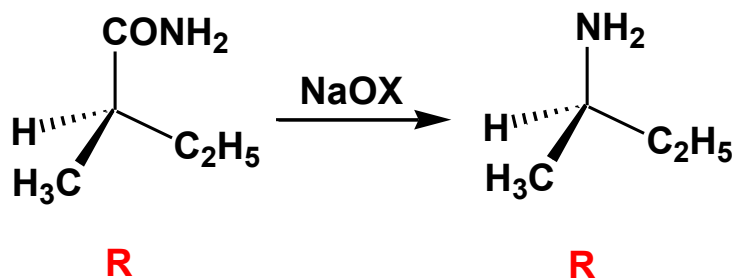


9.8 羧酸衍生物的化学性质



9.8.5 酰胺的特性

(3) Hofmann降解反应-立体化学

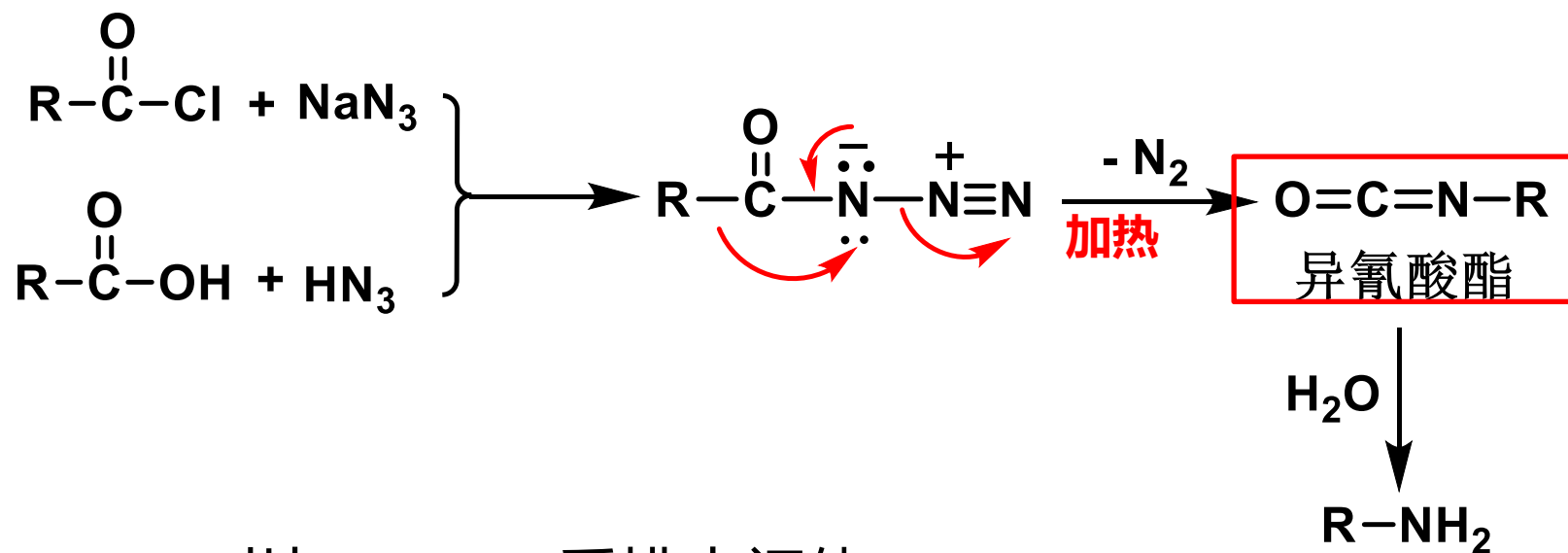


当手性碳原子与酰基碳相连时，Hofmann重排后，手性碳原子的构型保持不变。

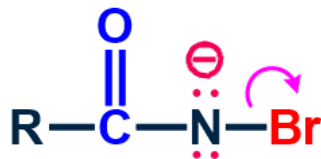
9.8 羧酸衍生物的化学性质



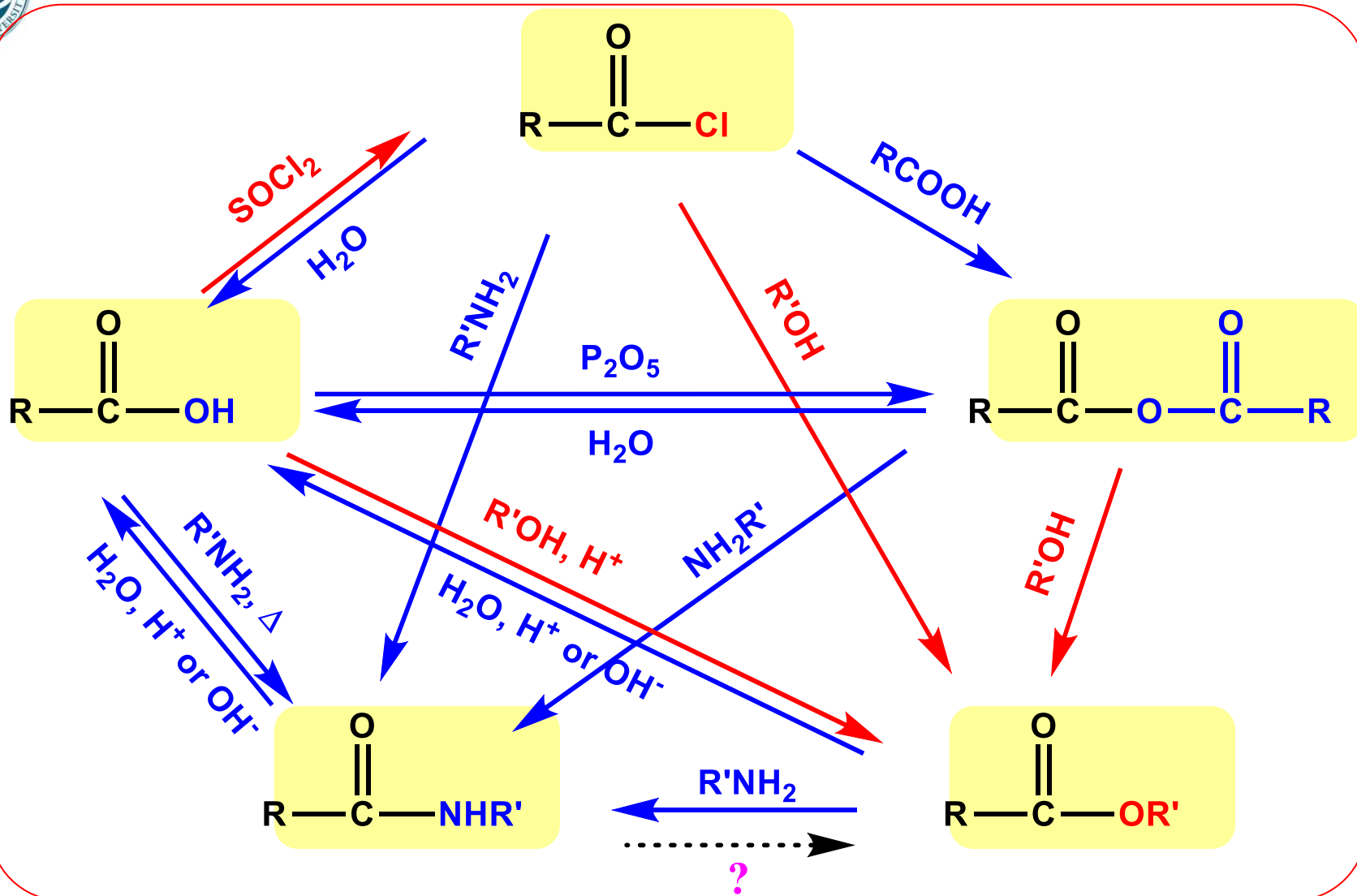
类似的反应—克尔蒂斯(Curtius)重排与施密特(Schmidt)重排



对比Hofmann重排中间体



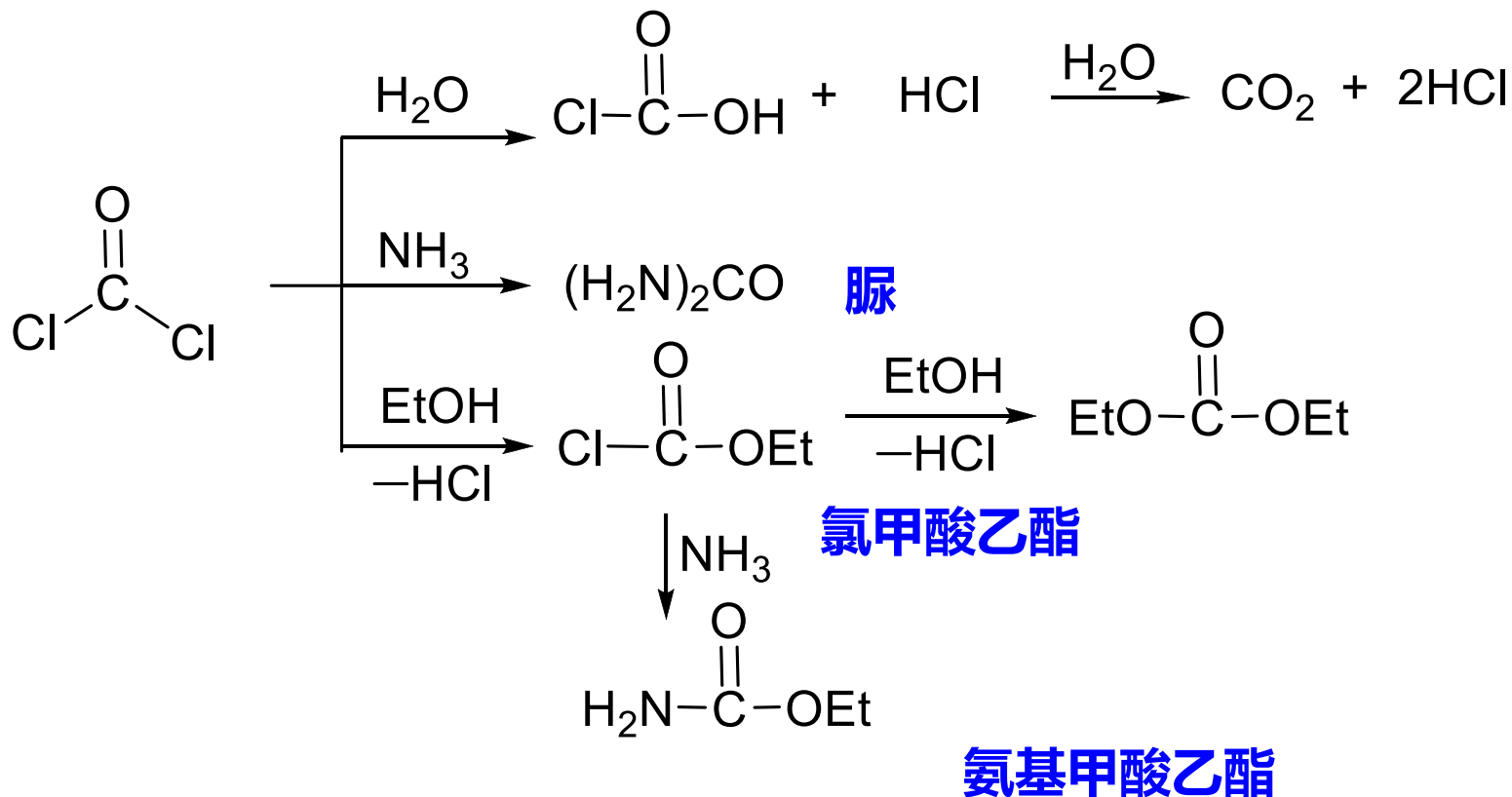
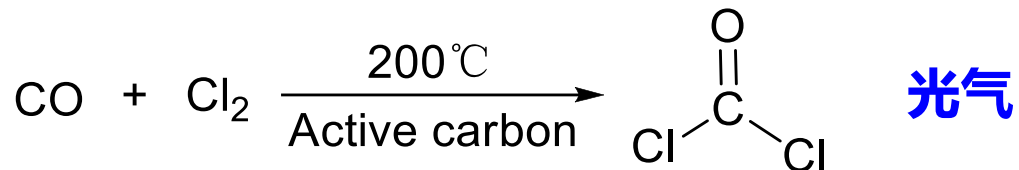
总结-羧酸及其衍生物互相转换





9.9 碳酸衍生物 (了解)

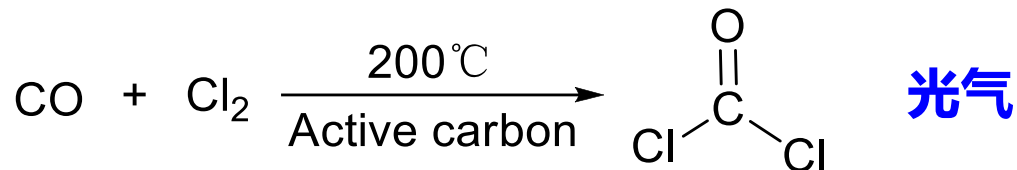
(1) 碳酰氯





9.9 碳酸衍生物（了解）

(1) 碳酰氯



第一次世界大战，哈伯担任化学兵工厂厂长时负责研制、生产毒气并用于战争，造成近百万人伤亡，尽管1907年的海牙公约取缔毒气武器。

有人称他为“天使”，有人称他为“魔鬼”

弗里茨·哈伯
Fritz Haber



1918年--诺贝尔化学奖

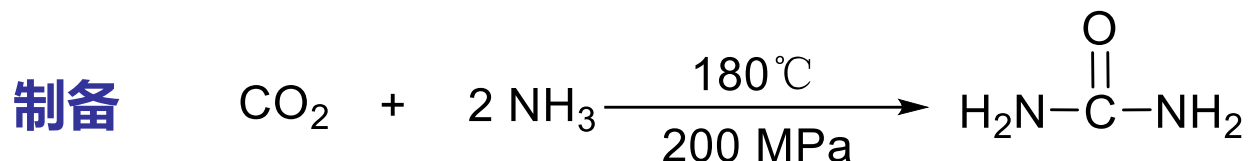


使人类从此摆脱了只能依靠天然氮肥矿产资源，加速了世界农业的发展，粮食产量因此大幅增加。

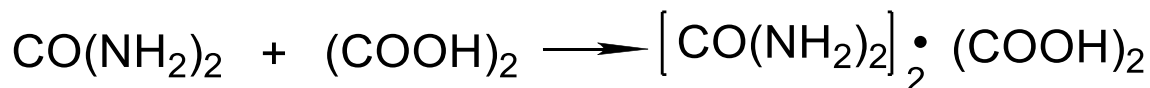
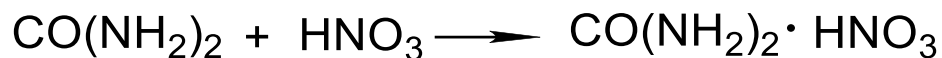
9.9 碳酸衍生物



(2) 碳酰胺(脲 or 尿素)



脲与硝酸/草酸成盐, 可以从水中析出。

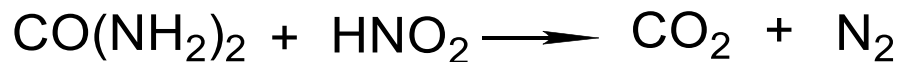
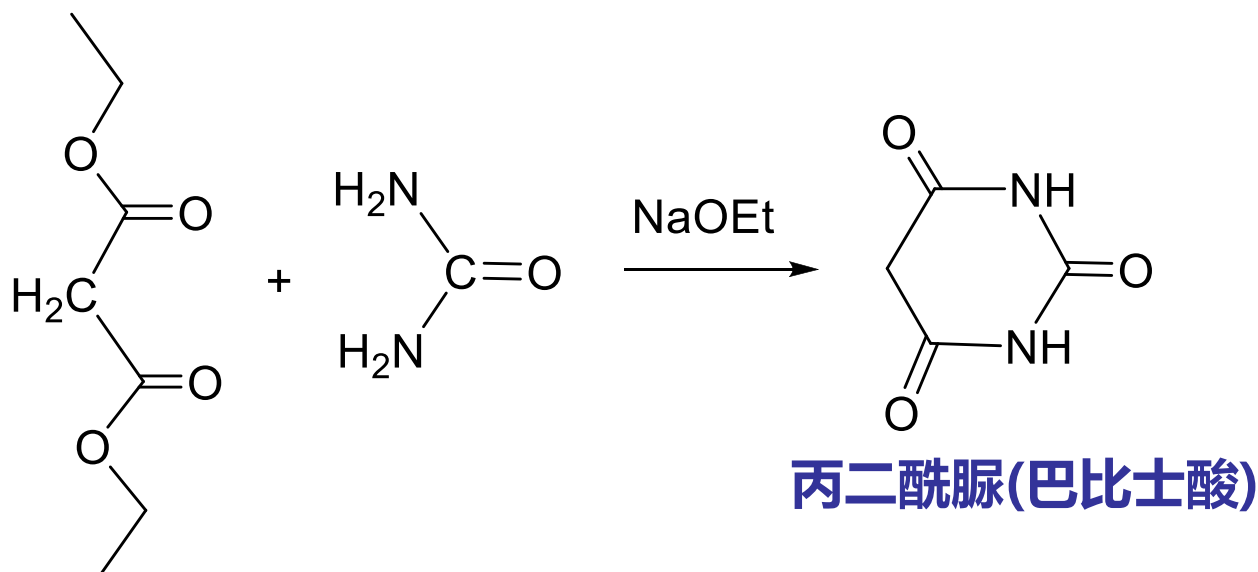


脲在碱或酸催化下水解成氨。

9.9 碳酸衍生物



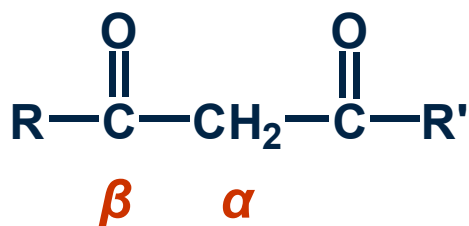
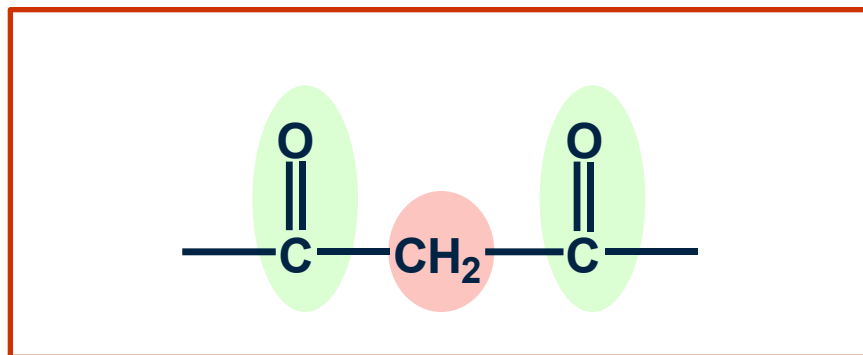
(2) 碳酰胺(脲 or 尿素)



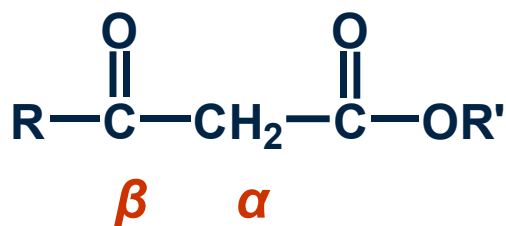
用于除去重氮化过程中过量使用的亚硝酸



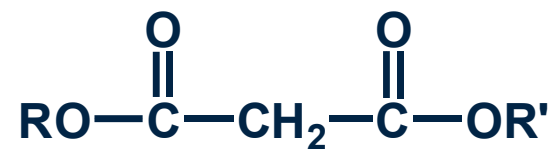
9.10 β -二羰基化合物



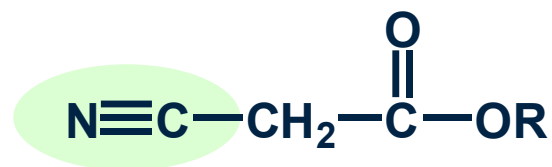
β -二酮



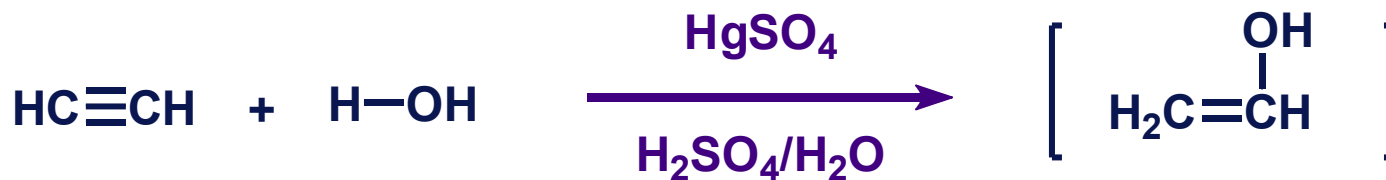
β -酮酸酯



丙二酸二酯



氰乙酸酯



重排

$\longrightarrow$

$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}\text{H}$

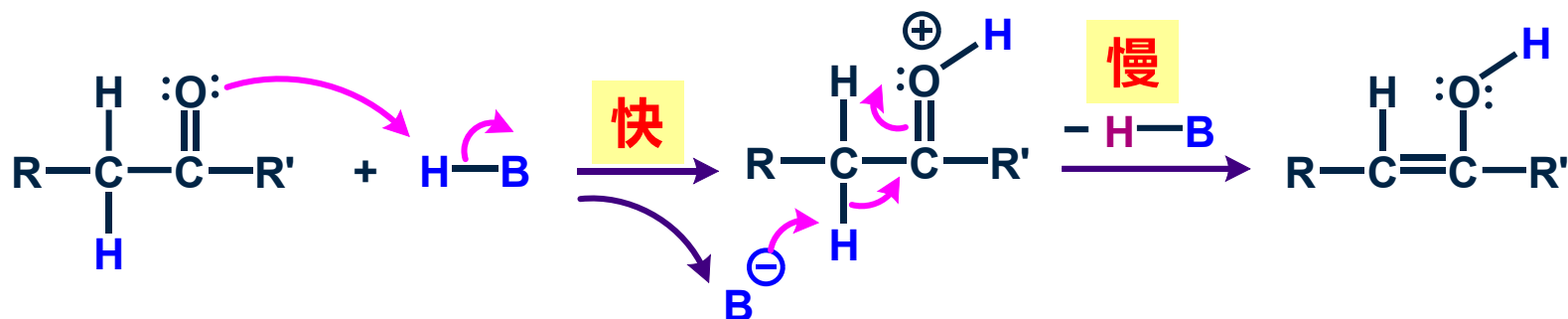
(b) 炔烃水合反应

9.10.1 酮-烯醇互变异构

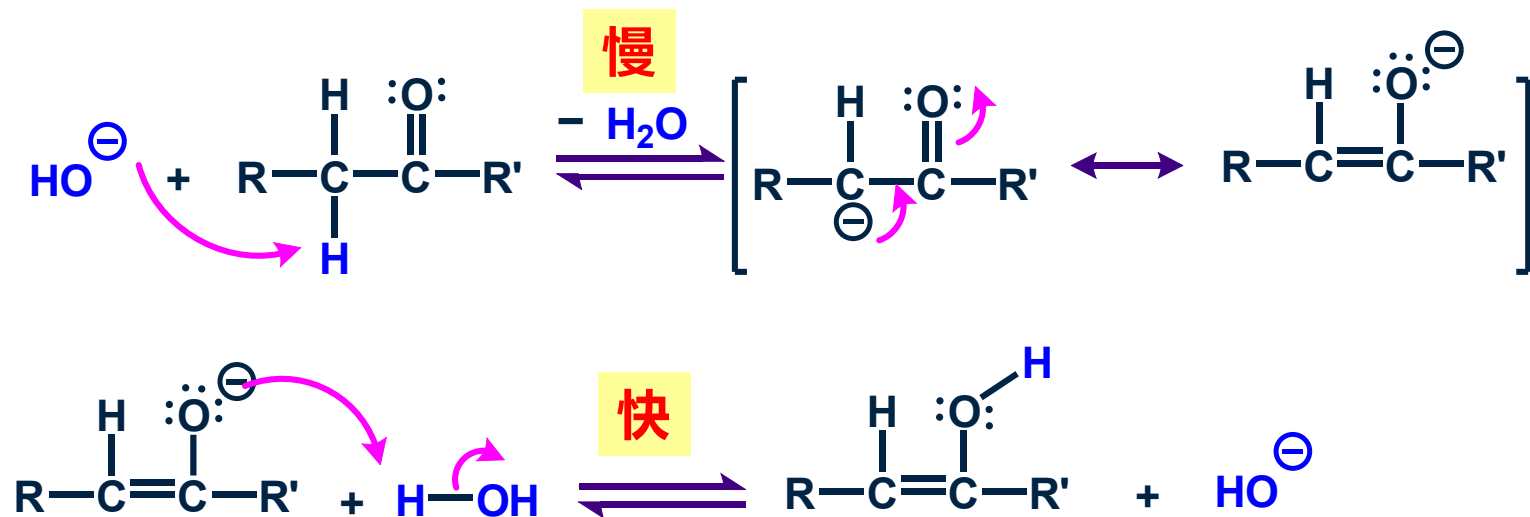


酸催化的酮-烯醇互变异构：

机理与醛酮 α -氢的反应 部分是重叠的



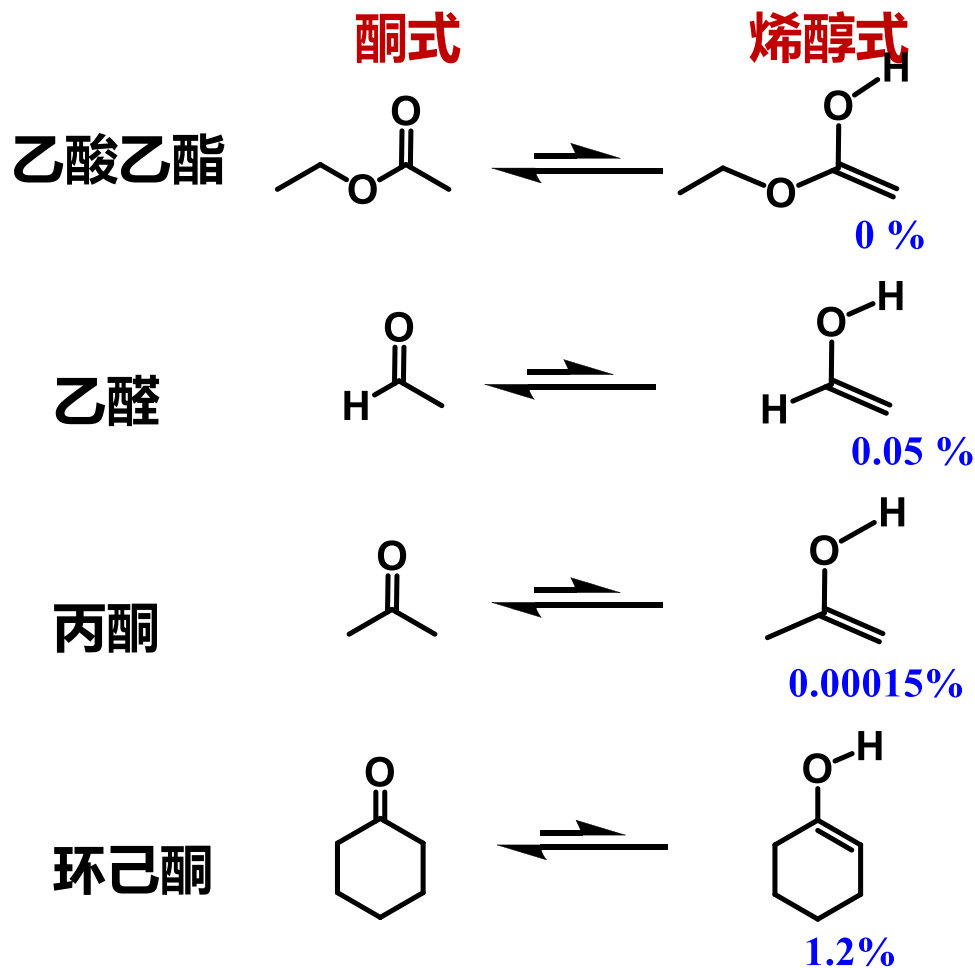
碱催化的酮-烯醇互变异构：



9.10.1 酮-烯醇互变异构



化合物结构对酮-烯醇平衡的影响



单羰基化合物

酮式比烯醇式更稳定。

能差：45 ~ 60 kJ · mol⁻¹

超共轭效应有利于烯醇式生成

破坏酮式的碳氢键和碳氧键所需能量，比破坏烯醇式的氧氢键和碳碳键大

9.10.1 酮-烯醇互变异构



β -二羰基化合物的平衡

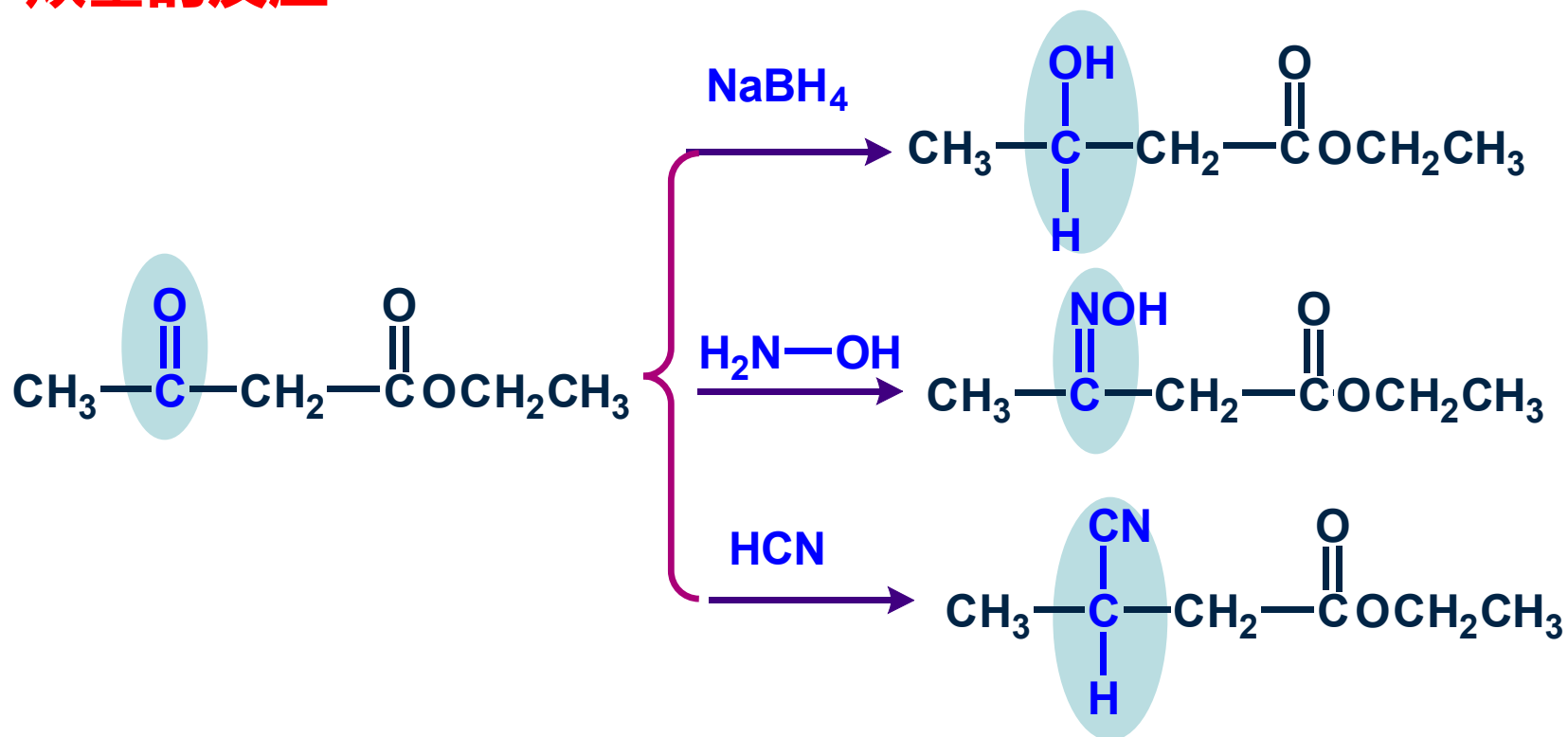


9.10.1 酮-烯醇互变异构



烯醇互变平衡的实验证据 (1)

羰基的反应

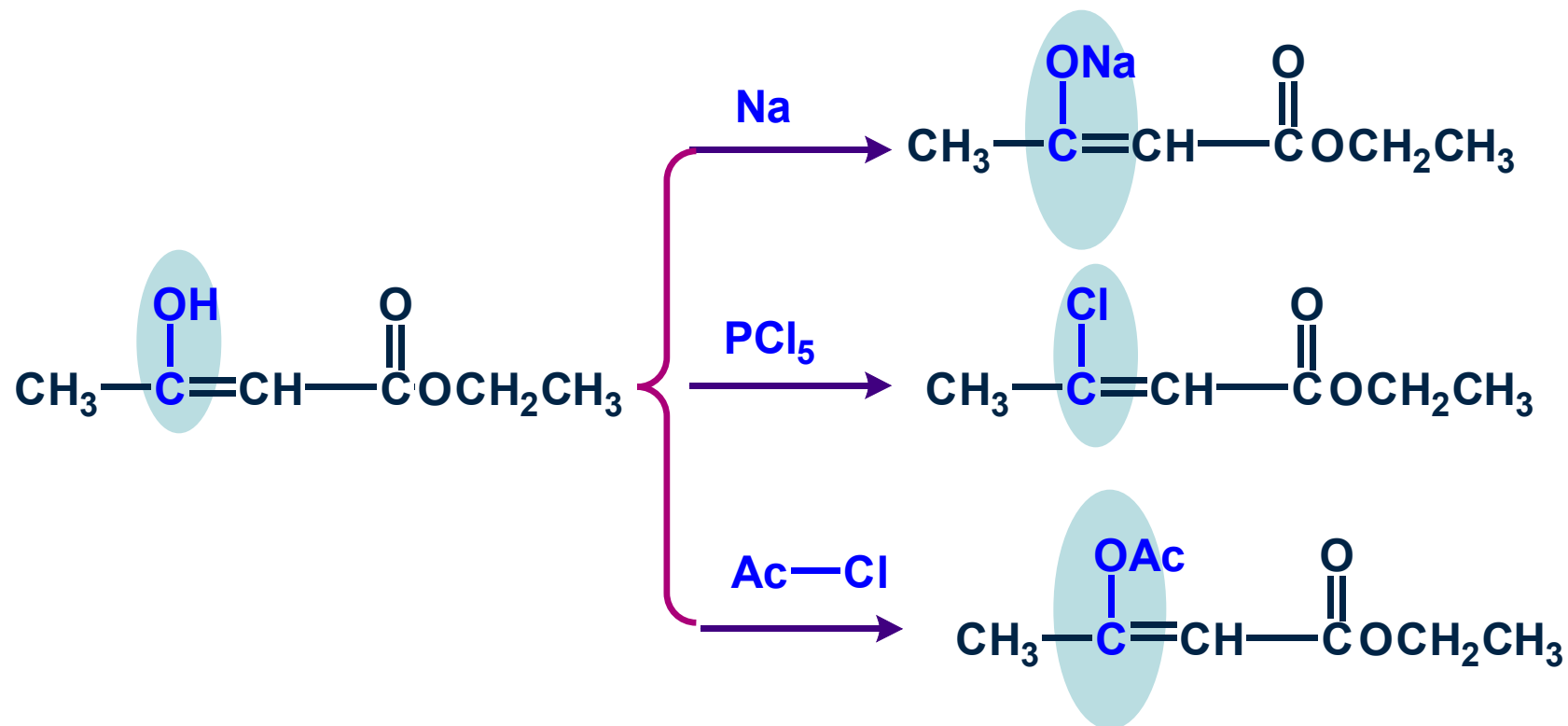


9.10.1 酮-烯醇互变异构



烯醇互变平衡的实验证据 (2)

羟基的反应

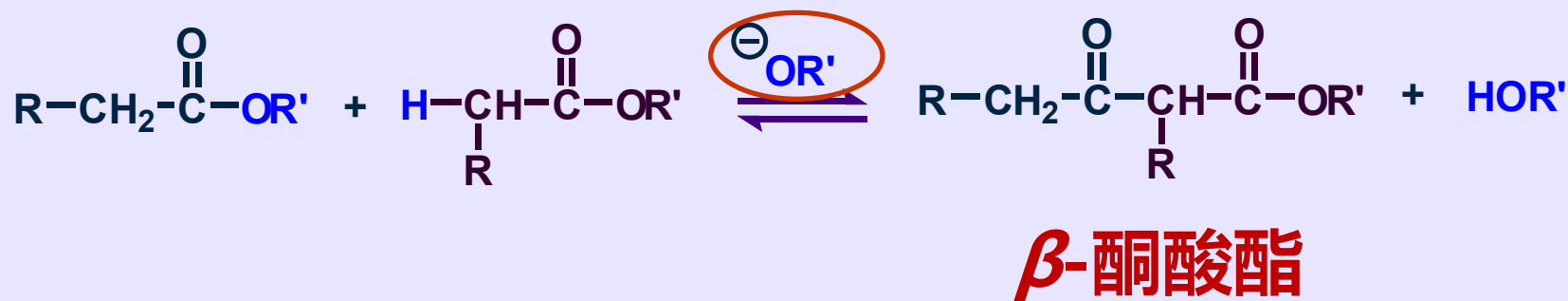




9.10.2 乙酰乙酸乙酯的合成及其应用

乙酰乙酸乙酯的合成

Claisen 酯缩合反应



其它常用碱:

NaH, NaNH₂, LDA,
Ph₃CNa, t-BuOK
(弱亲核性强碱)

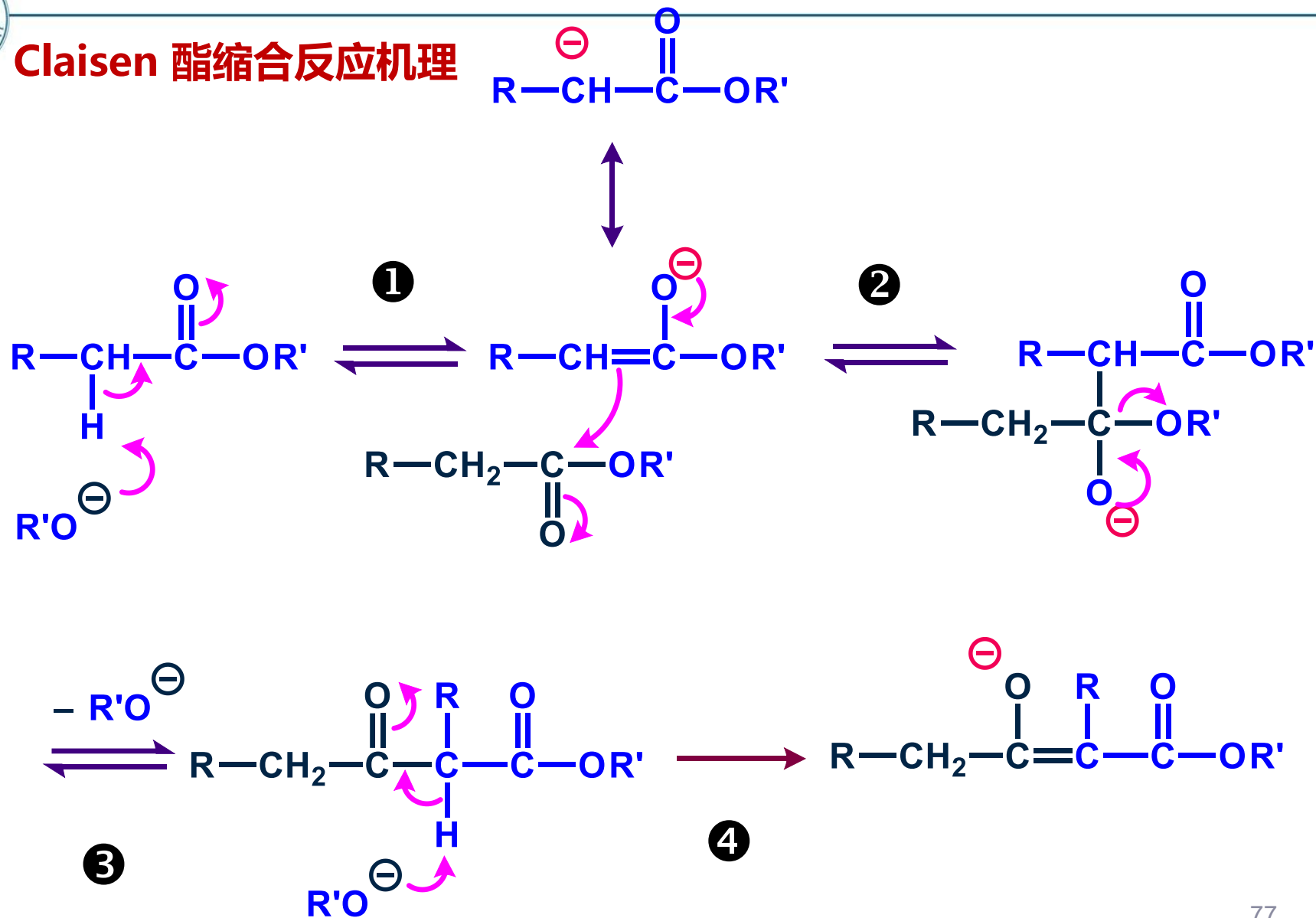
碱用量对反应的影响

催化量: 反应可逆
大于化学计量: 反应完全

9.10.2 乙酰乙酸乙酯的合成及其应用



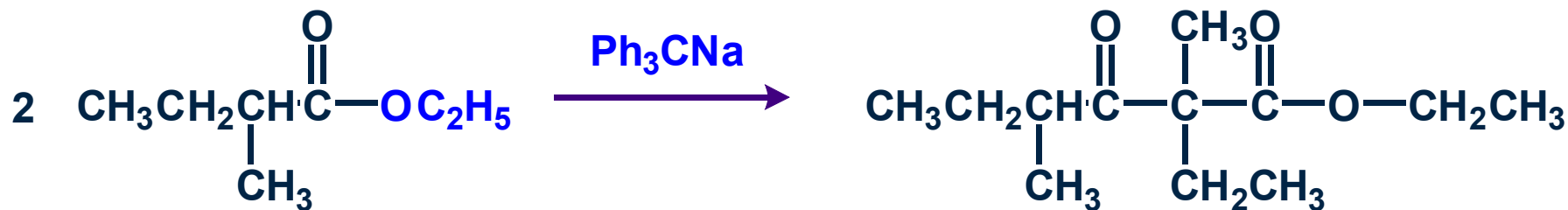
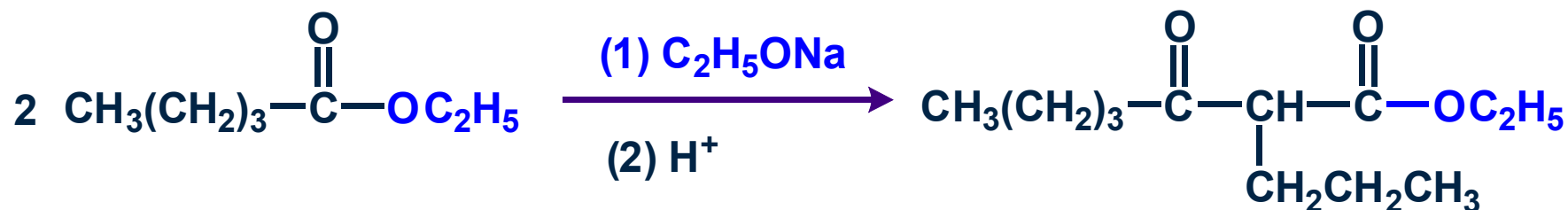
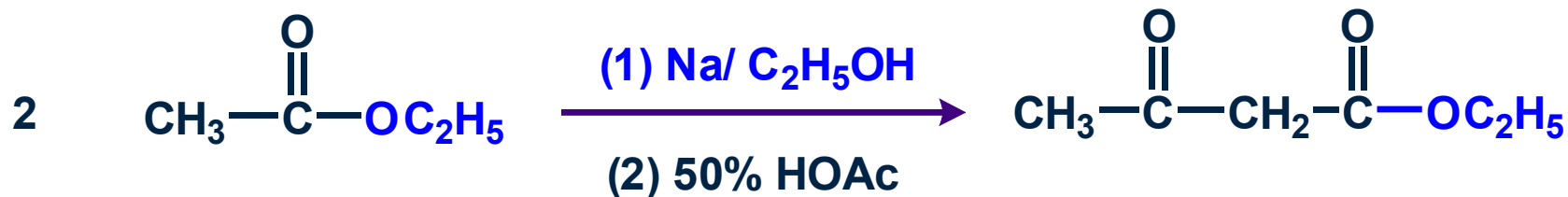
Claisen 酯缩合反应机理





9.10.2 乙酰乙酸乙酯的合成及其应用

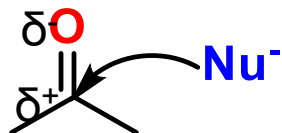
Claisen 酯缩合反应实例



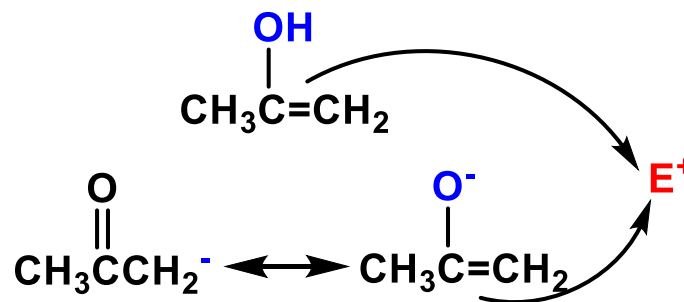
合成1,3-二羰基化合物



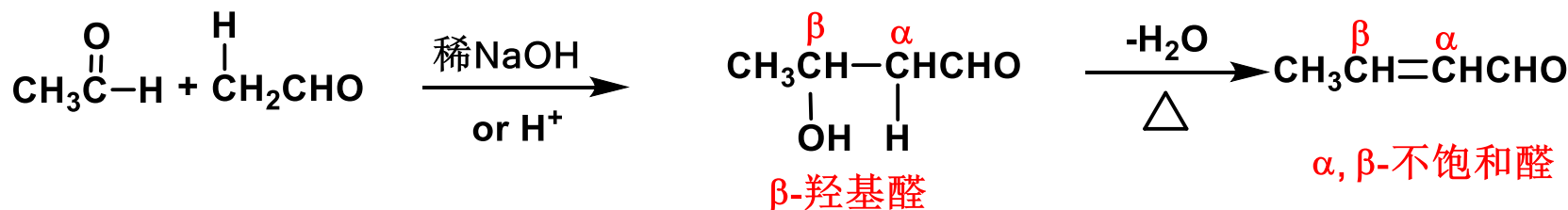
回顾羟醛缩合反应



羰基可以接受
亲核试剂的进攻



烯醇或烯醇负离子可以当做亲核试剂

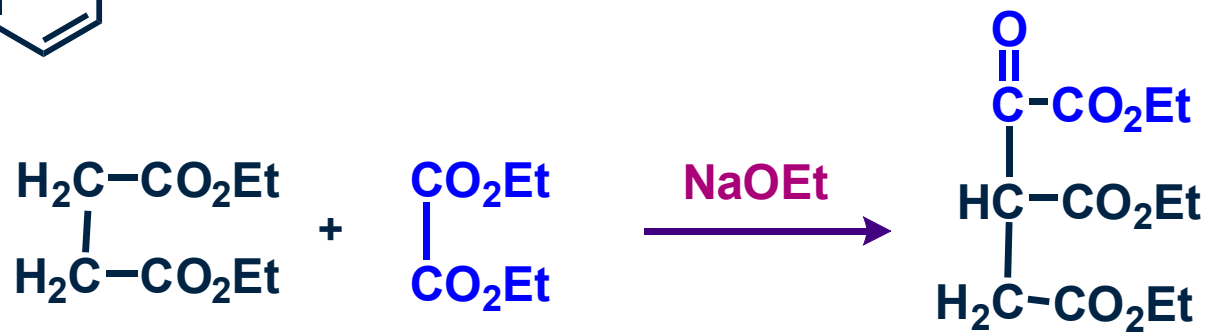
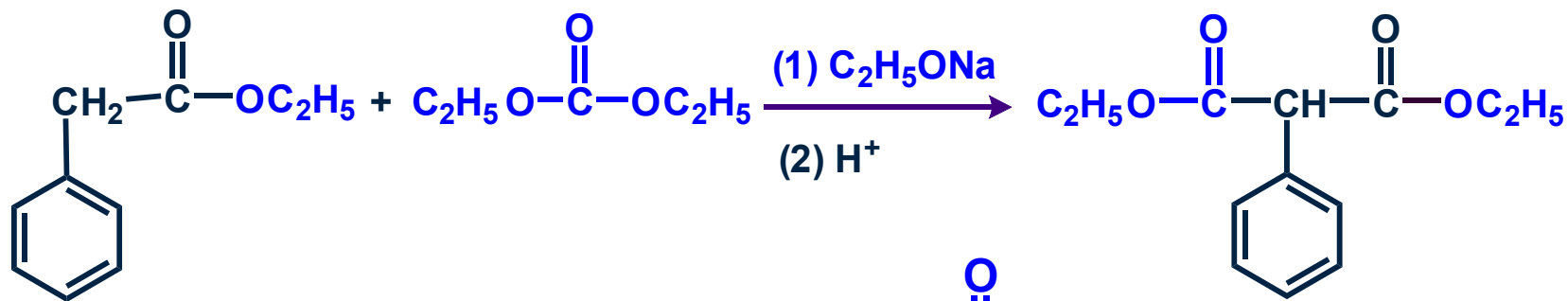
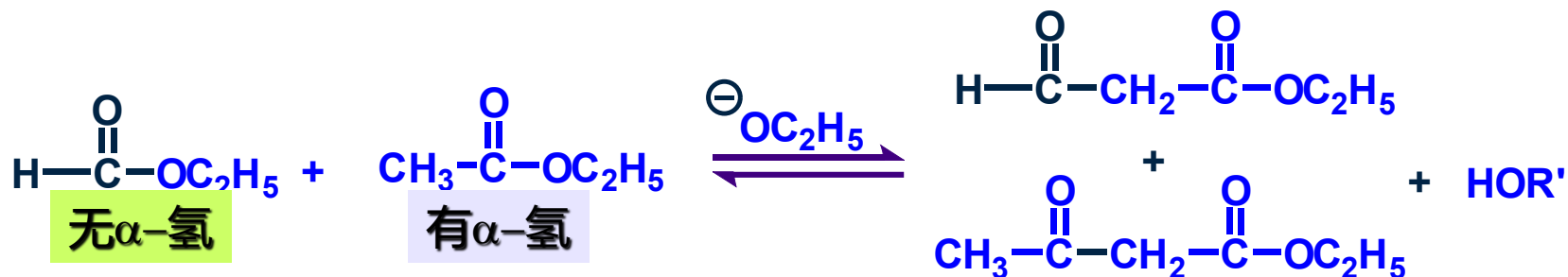


亲核加成-脱水反应



9.10.2 乙酰乙酸乙酯的合成及其应用

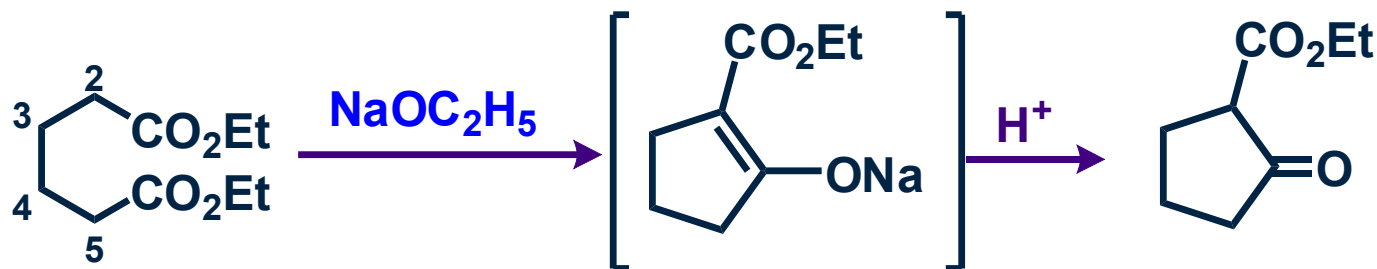
交叉酯缩合 (两个不同酯之间的缩合)



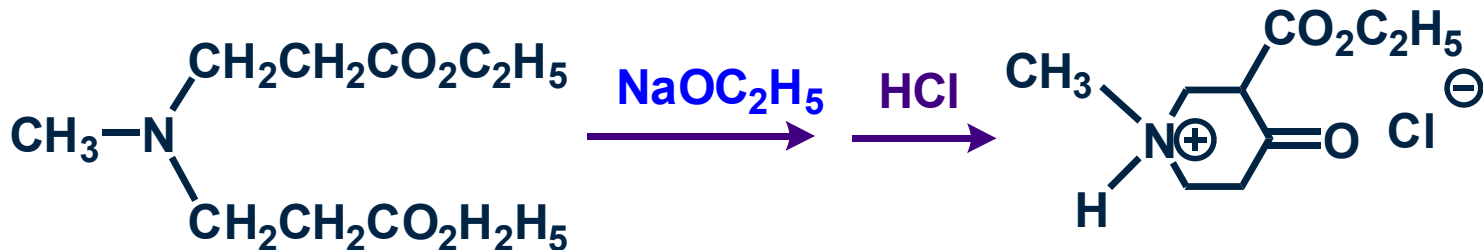
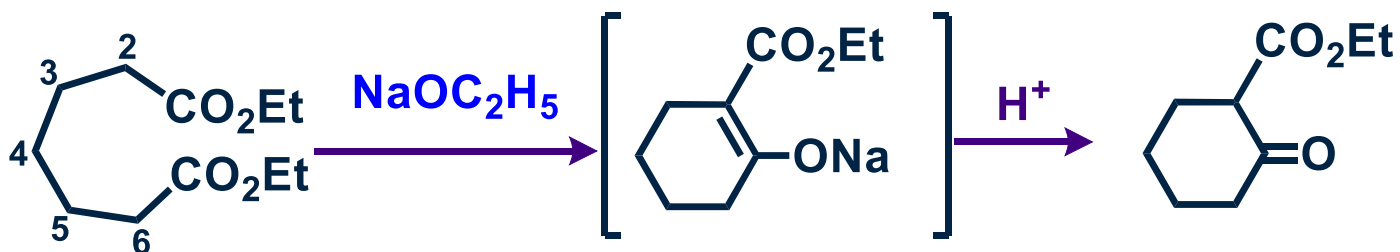


9.10.2 乙酰乙酸乙酯的合成及其应用

分子内酯缩合 - Dieckmann Condensation



更多实例:

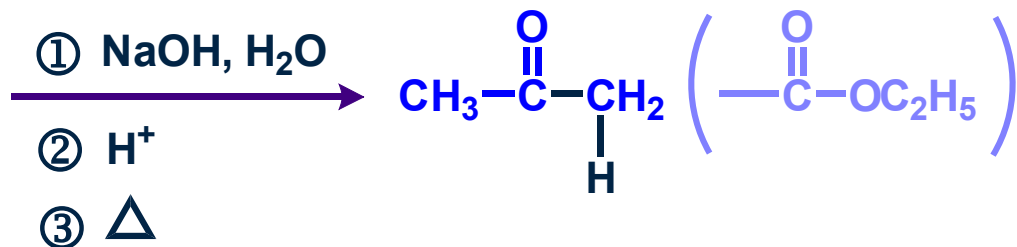
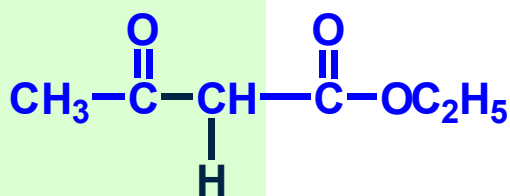
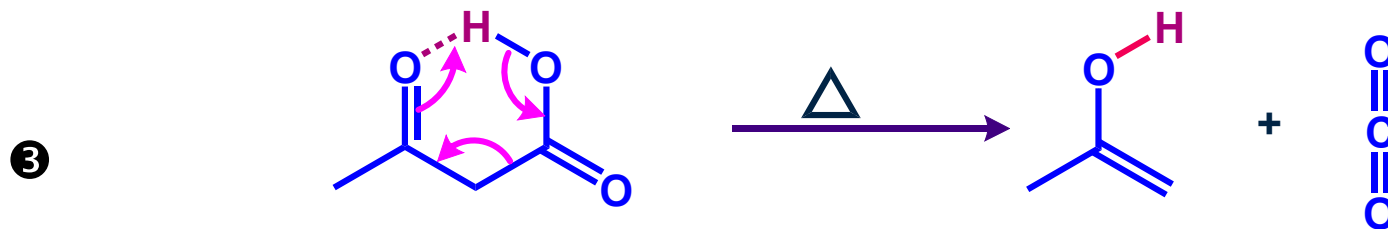




9.10.2 乙酰乙酸乙酯的合成及其应用

乙酰乙酸乙酯的性质

酮式分解： β -酮酸加热 脱羧 生成酮

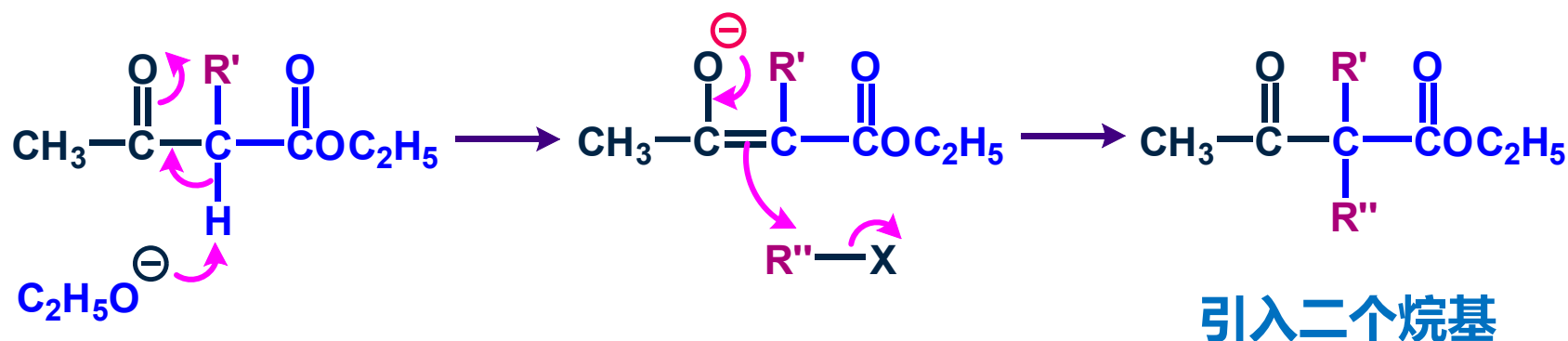
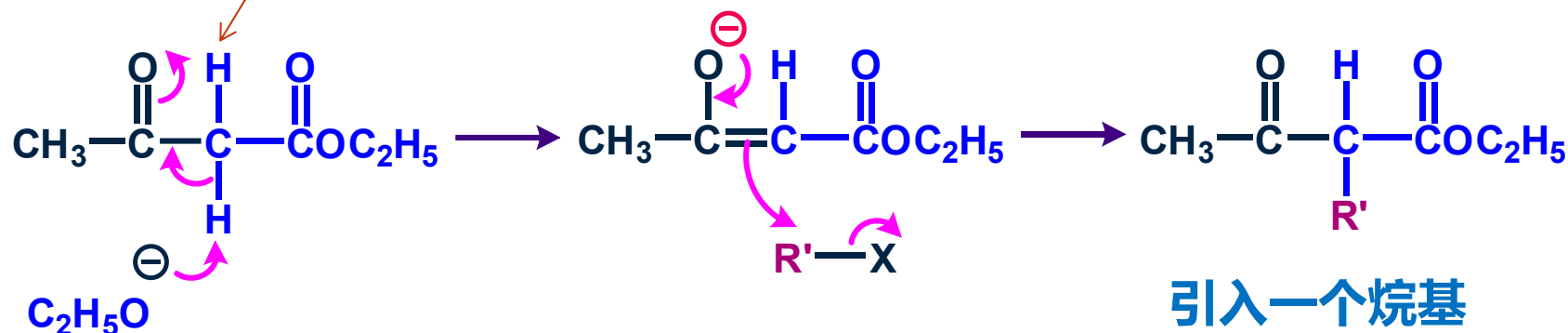


9.10.2 乙酰乙酸乙酯的合成及其应用



乙酰乙酸乙酯的烷基化

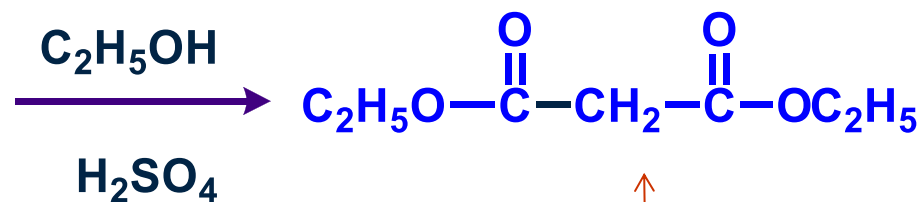
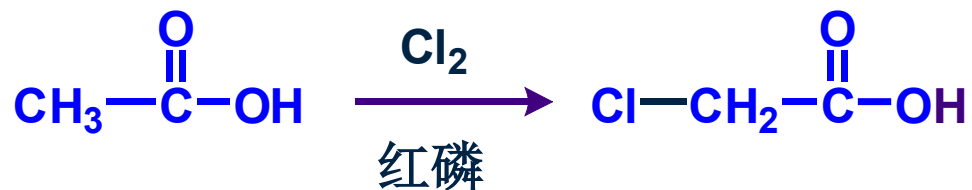
借助于酸性





9.10.3 丙二酸二乙酯的合成及其应用

丙二酸二乙酯的合成

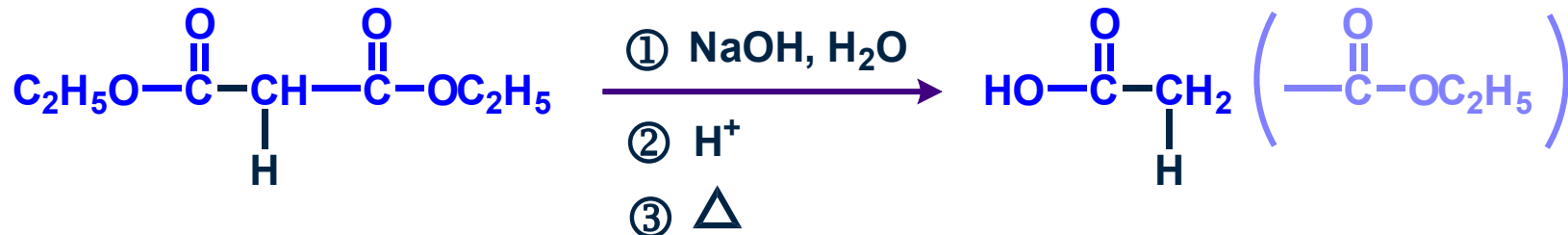
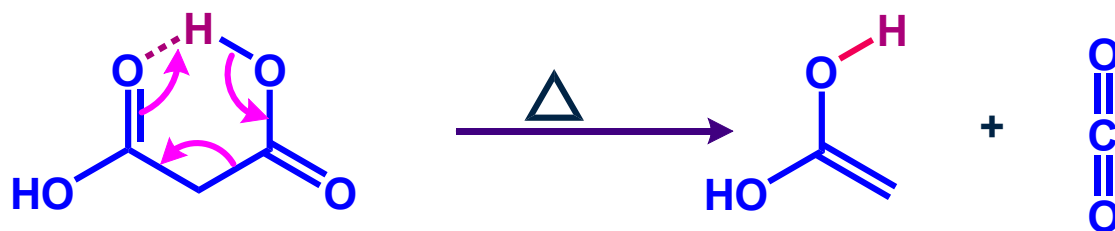
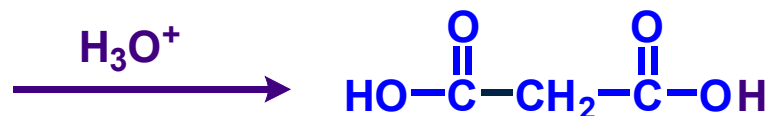


$pK_a = 13.7$



9.10.3 丙二酸二乙酯的合成及其应用

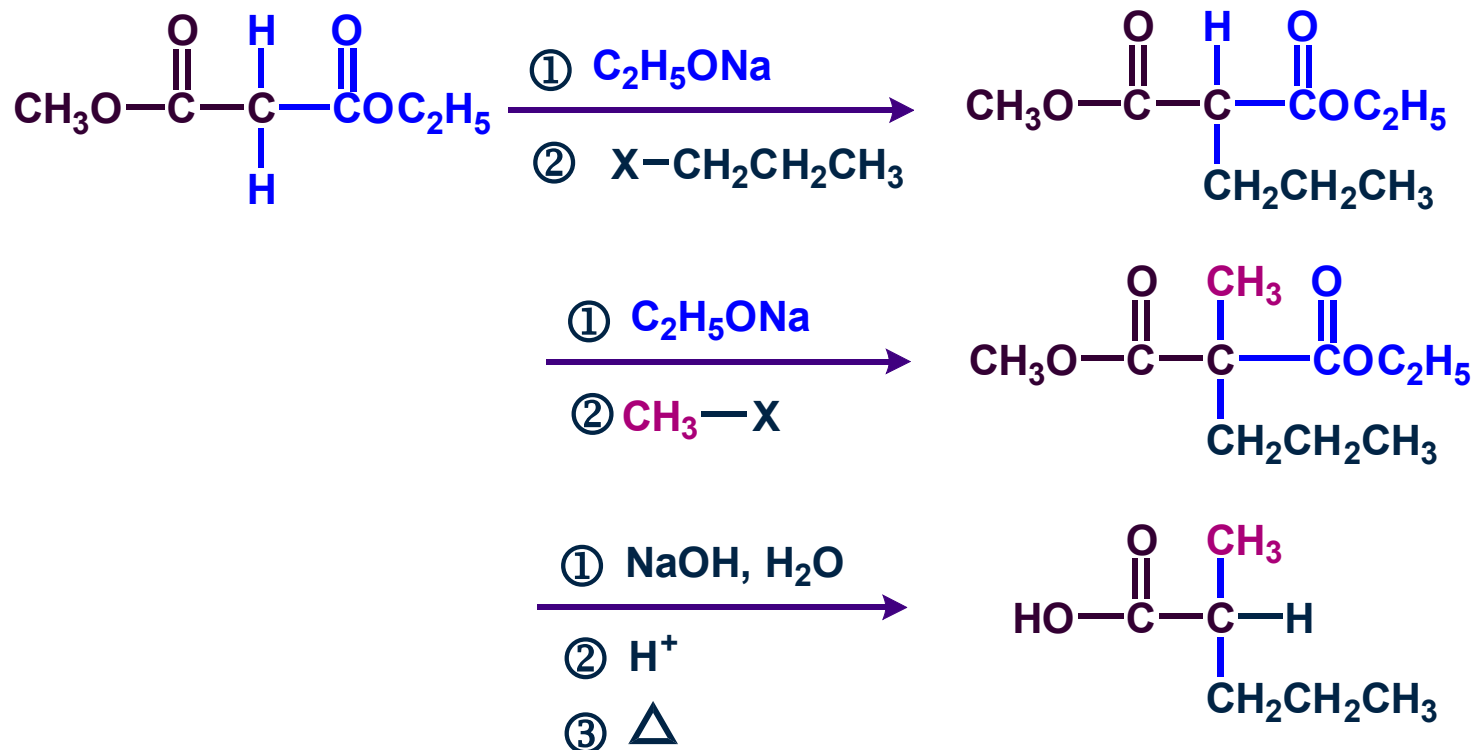
丙二酸二乙酯的性质：





9.10.3 丙二酸二乙酯的合成及其应用

丙二酸二乙酯的烷基化——取代的羧酸

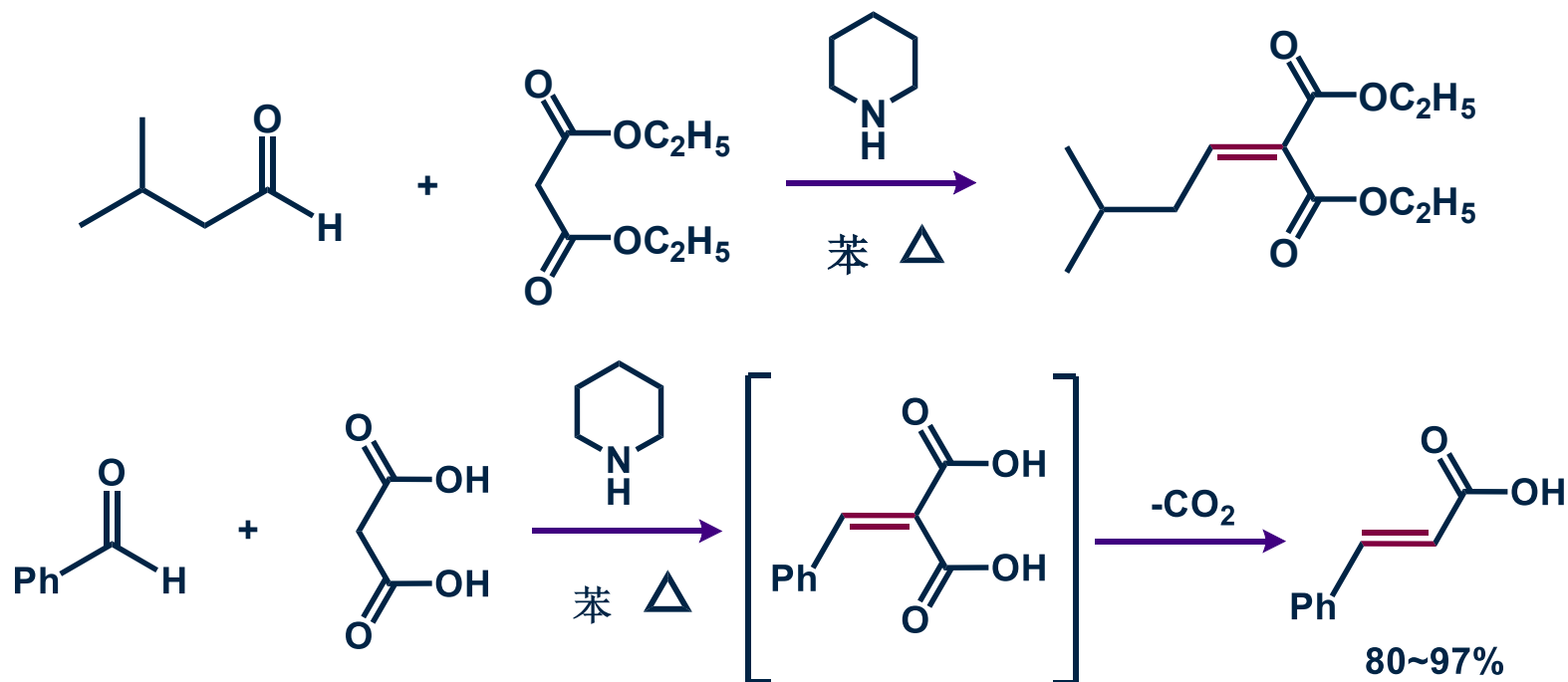


利用丙二酸酯的 α -烃基化反应是制备“ α -烃基取代乙酸”的高效方法。



9.10.4 Knoevenagel缩合

- 醛、酮和含有双重致活的活泼亚甲基化合物在弱碱（胺、吡啶、哌啶）催化下的缩合反应。

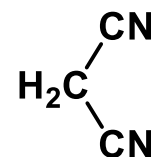
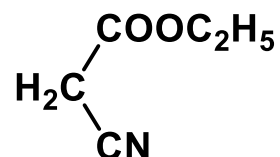
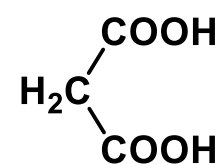
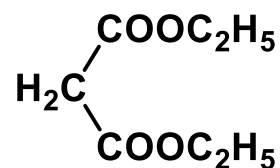
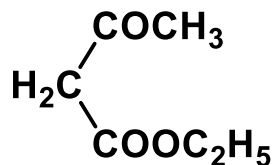


醛酮在弱碱条件下不发生自身间的羟醛缩合反应，提高的反应效率

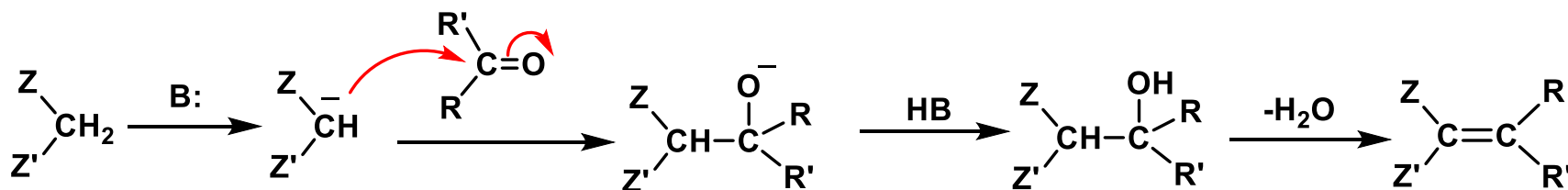


9.10.4 Knoevenagel缩合

双重致活的活泼亚甲基化合物:



反应机理:



类似羟醛缩合反应机理

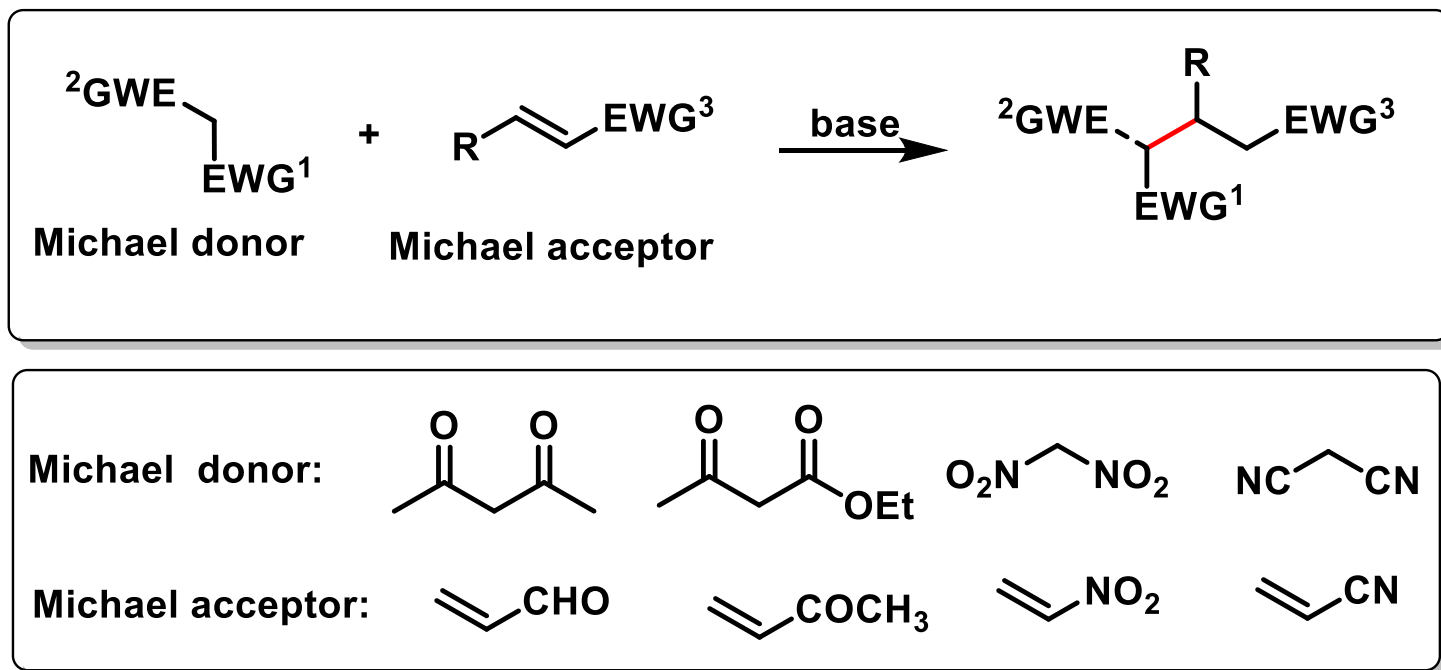


9.10.5 Michael加成反应

α, β -不饱和醛、酮与亲核的碳负离子的1,4-共轭加成反应

亲核的碳负离子一般指活泼的亚甲基化合物

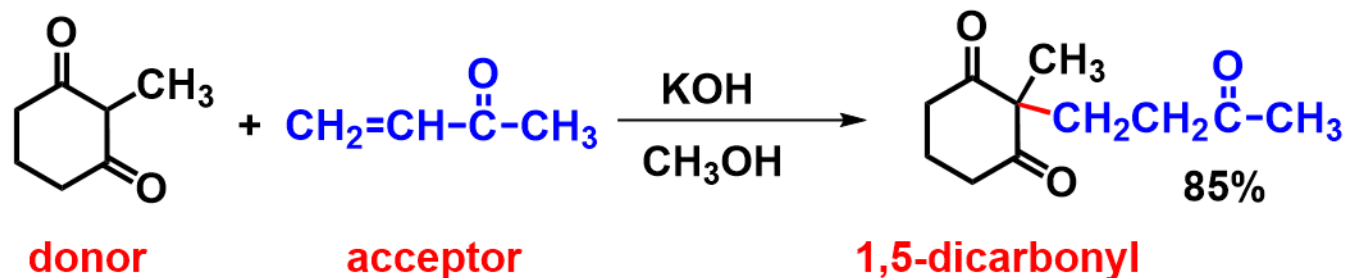
常用催化量的碱: NaOH, KOH, $\text{C}_2\text{H}_5\text{ONa}$, NaNH_2 , Et_3N 等



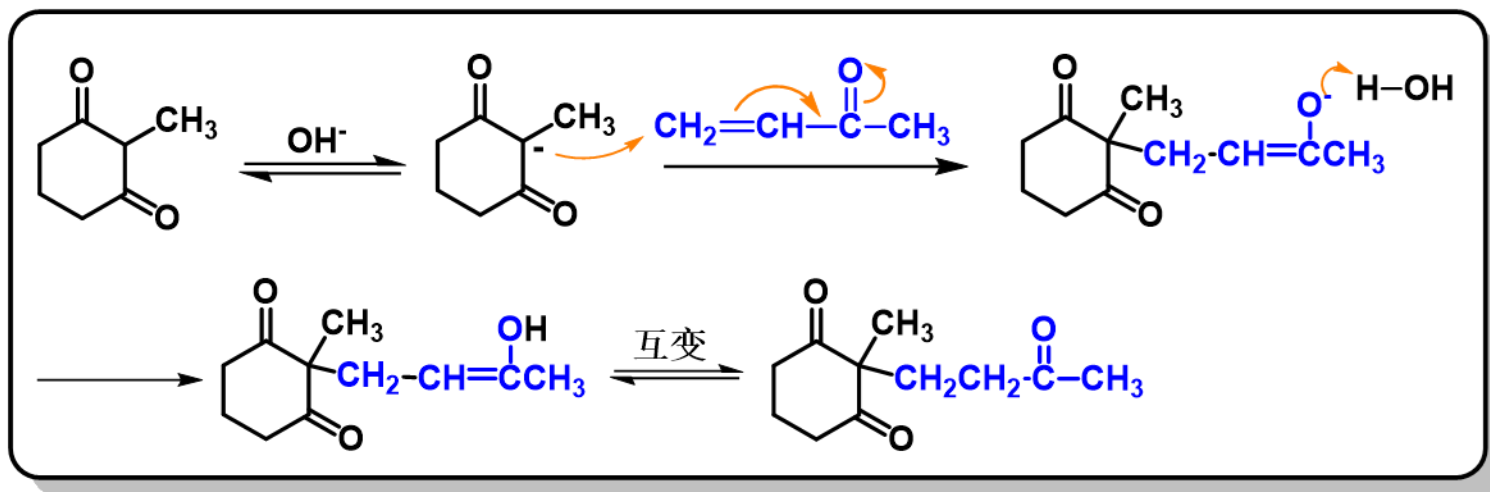


9.10.5 Michael加成反应

α, β -不饱和醛、酮与亲核的碳负离子的1,4-共轭加成反应

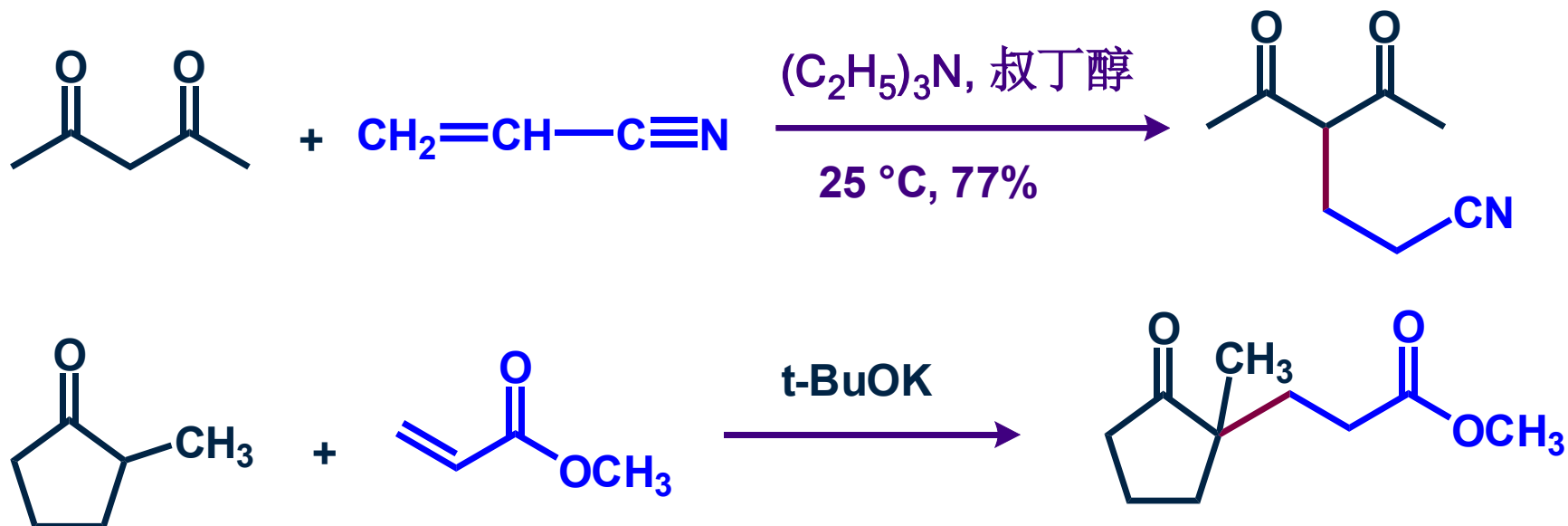


✓ 反应机理:





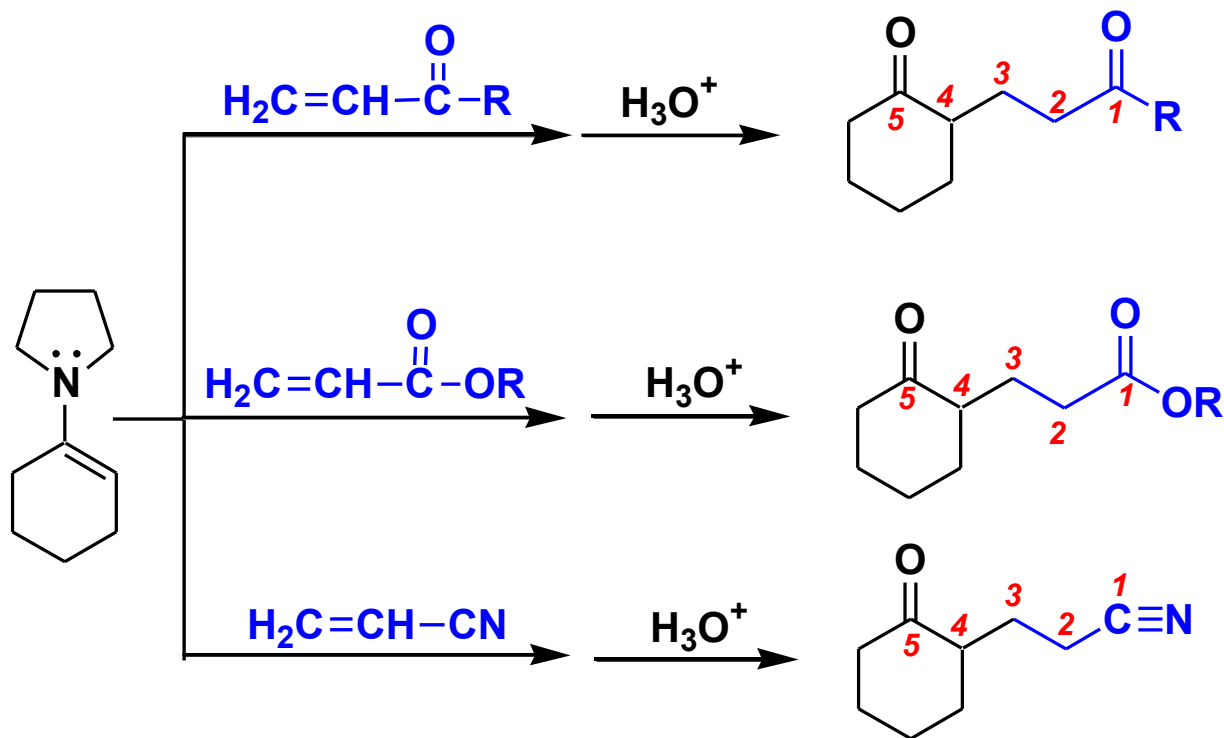
9.10.5 Michael加成反应





9.10.5 Michael加成反应

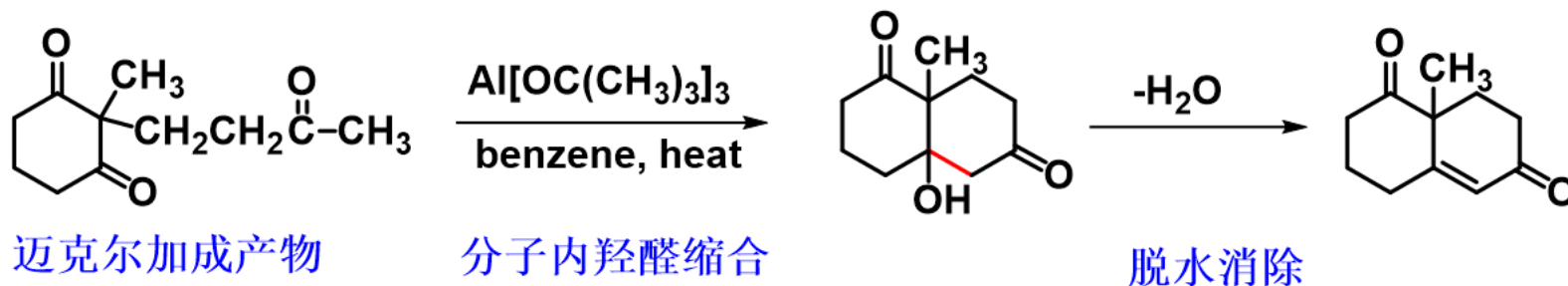
酮与仲胺反应形成的烯胺可与与 α, β -不饱和羰基化合物反应



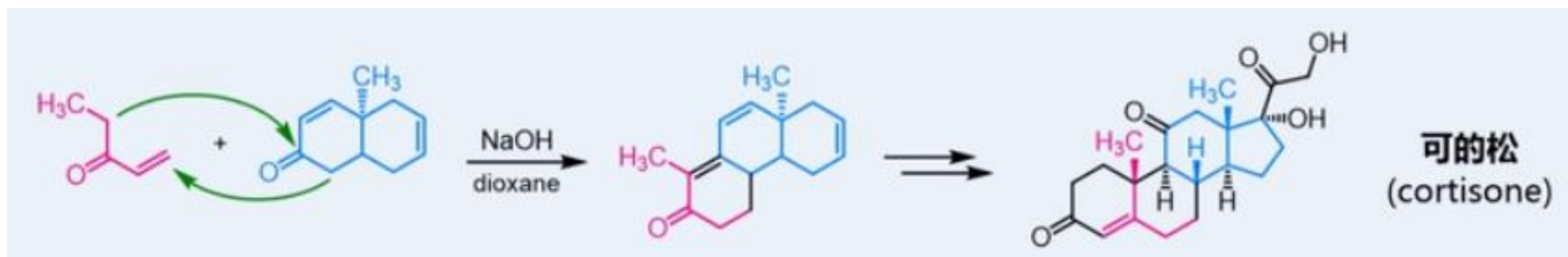


9.10.5 Robinson 环化反应

- 迈克尔加成与分子内羟醛缩合的串联反应
- 构筑六元环（环己烯酮）的重要方法之一



肾上腺皮质激素“可的松”的合成

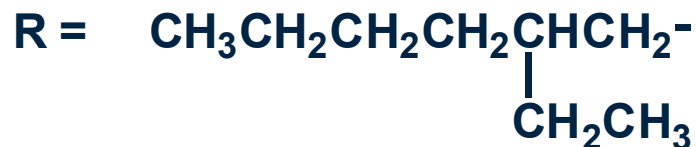
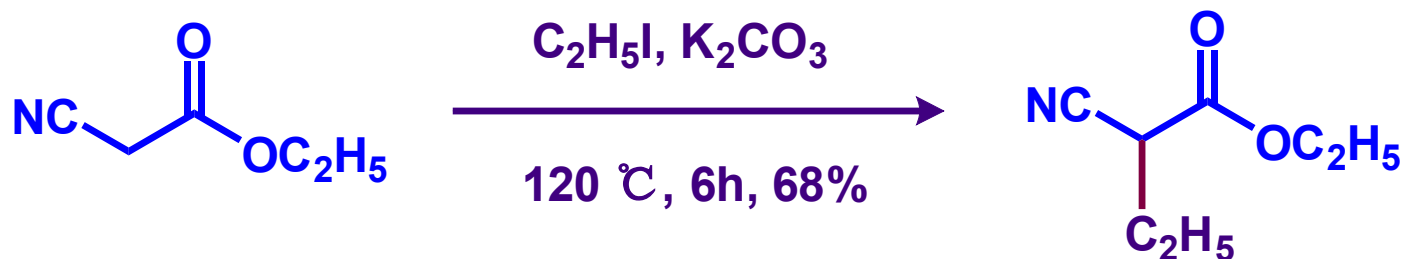


此外，在天然产物合成中的重要原料如紫杉醇和长叶烯的全合成



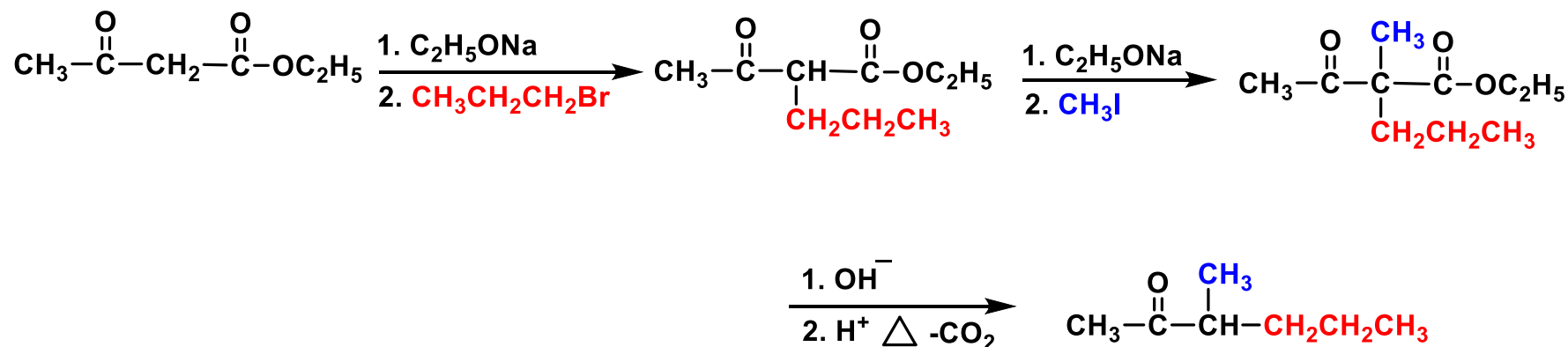
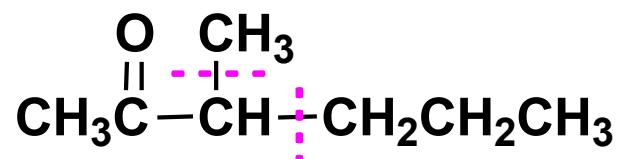
9.10.6 其它含活泼亚甲基的化合物

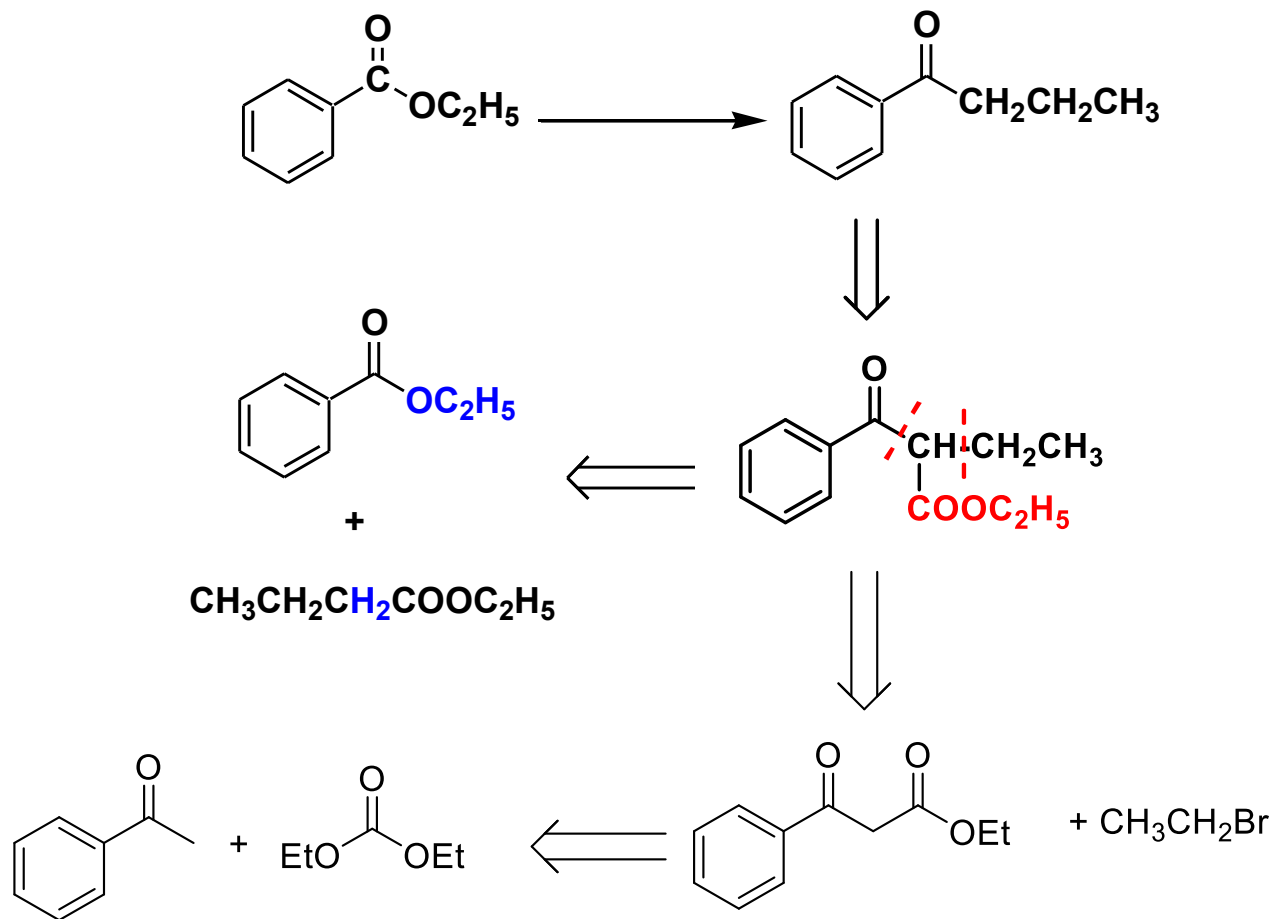
两个吸电基(如 $-\text{CHO}$, $-\text{COR}$, $-\text{COOH}$, $-\text{CONR}_2$, $-\text{CN}$, $-\text{NO}_2$ 等)连接在**同一个碳原子上时**, 其亚甲基的氢原子也具有活泼性, 常称为**活泼亚甲基**。





从乙酰乙酸乙酯及必要试剂合成







课后习题

- **作业：**

- P433 练习 13.8
- P436 练习 13.11
- P449 练习 14.3

- **作业要求：**

- 使用统一的作业本，**先抄一遍题目再作答。**
- 下次课前完成，随机抽查。
- **下次作业提交时统一批改给分后计入平时分。**
- 课代表周六下午前收齐后交至助教